



4.0 Nipped Systems

Timothy J. Walker, TJWalker + Associates Inc.

1620 Edgcumbe Road, Saint Paul, MN 55116

tjwalker@tjwa.com www.webhandling.com 651.686.5400

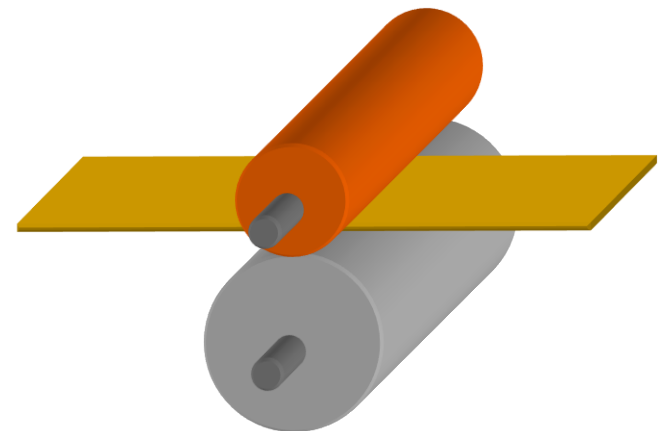
What is a Nip?

nip ¹  (nip)

v. nipped, nip·ping, nips

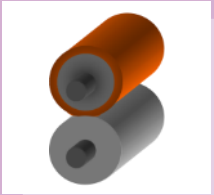
v.tr.

1. To seize and pinch or bite: *The fish nipped the wader's toe.*
2. To remove or sever by pinching or snipping: *nipped off the plant leaf.*
3. To bite or sting with the cold; chill.
4. To check or cut off the growth or development of: *a conspiracy that was nipped in the bud by the police.* See Synonyms at [blast](#).
5. Slang
 - a. To snatch up hastily.
 - b. To take (the property of another) unlawfully; steal.



WH Fundamentals

4: Nipping & Lamination



**Nipped Roller
Systems**

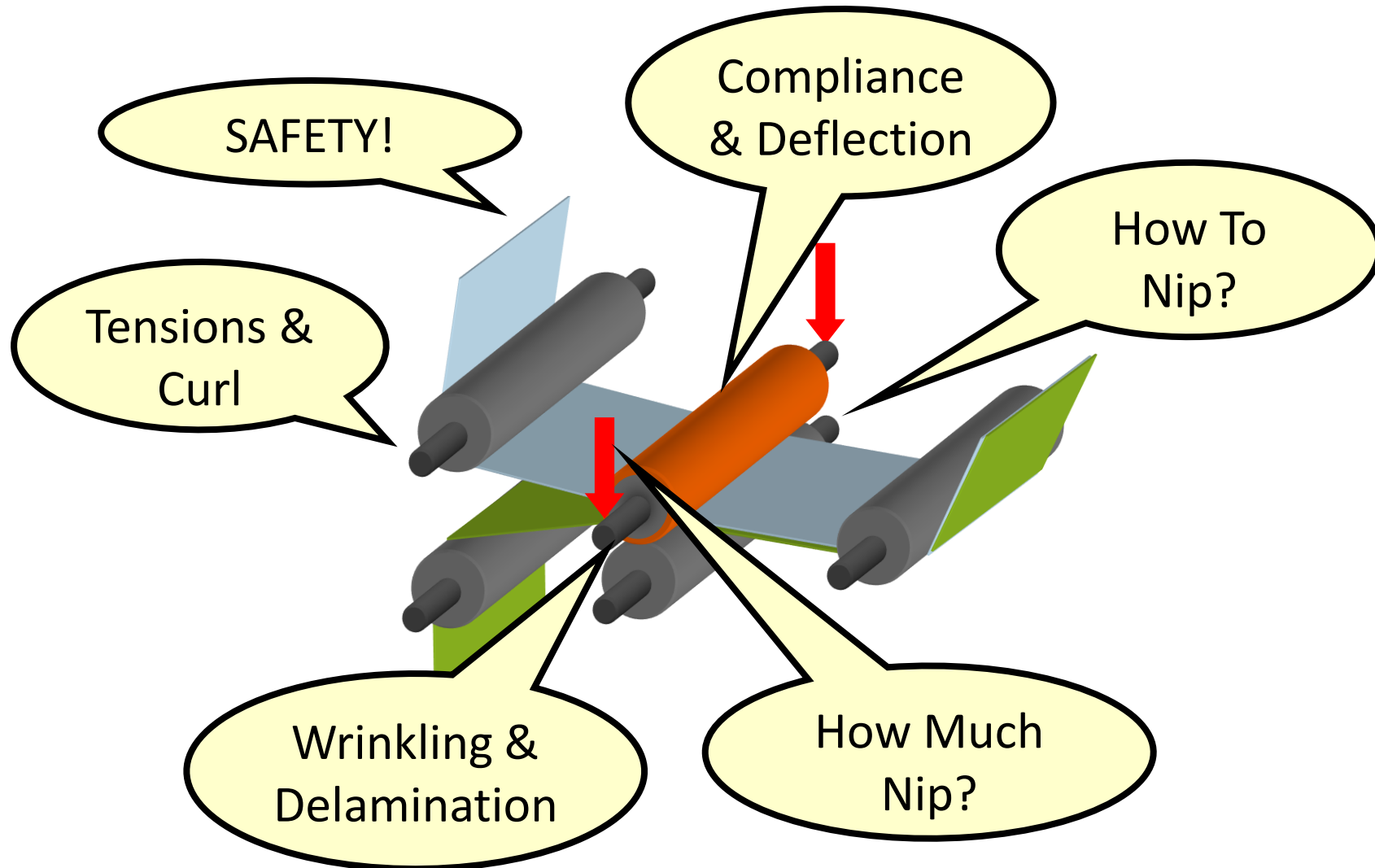


Lamination

+ Nipping and Lamination

- ✓ What is the load and pressure in a nip?
- ✓ What causes or reduced pressure variations?
- ✓ What are the best practices of lamination?

Nipping and Lamination – What to Consider?



4.0 Learning Objectives

- ▶ Where are nipped rollers used?
- ▶ Do you need nipped rollers?
- ▶ What should you think about in any nipped system?
- ▶ What is lamination?
- ▶ What are different types of lamination processes?

Where are Nipped Rollers in Converting?

Nipped rollers are key to many converting processes, including:

Printing

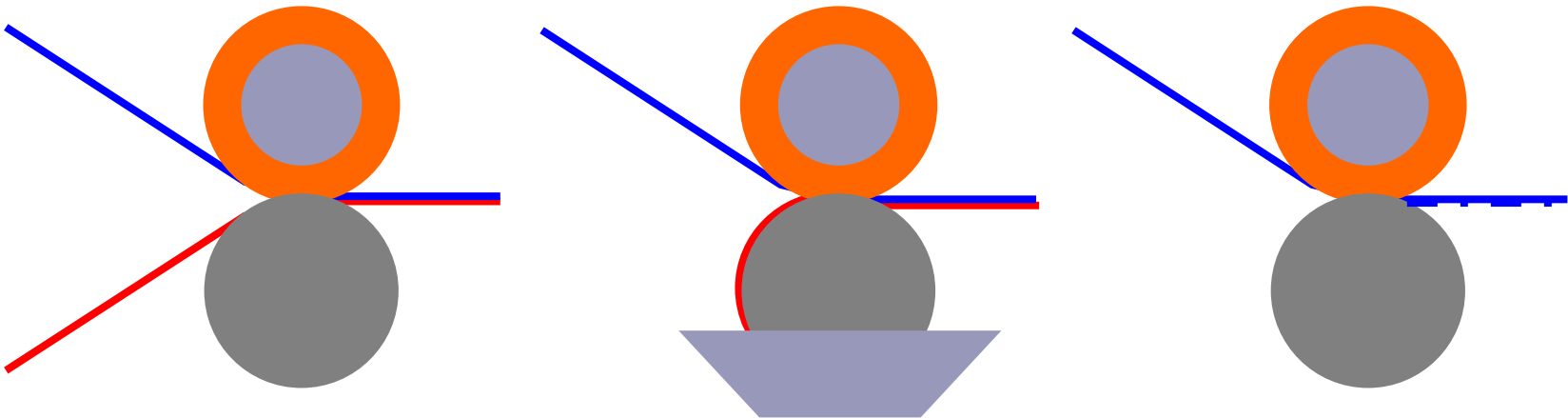
Calendering

Sealing

Laminating

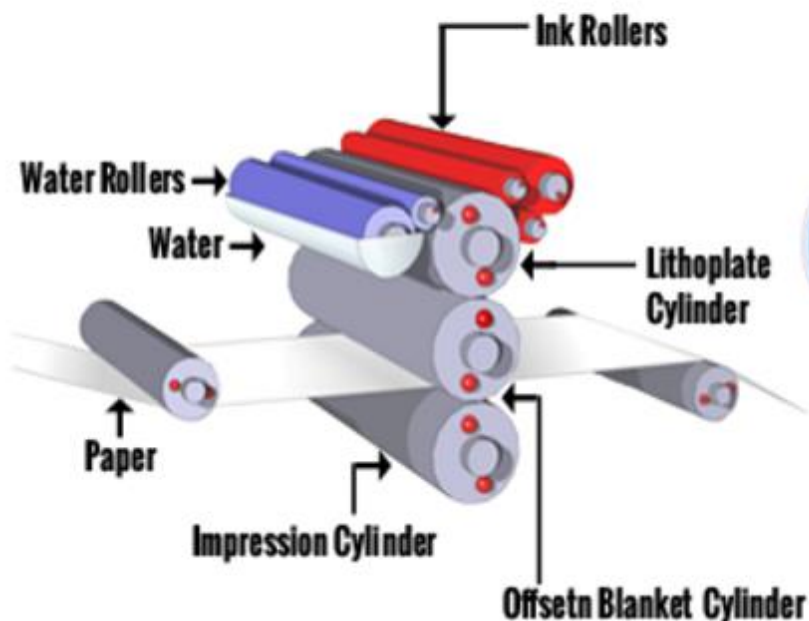
Coating

Embossing



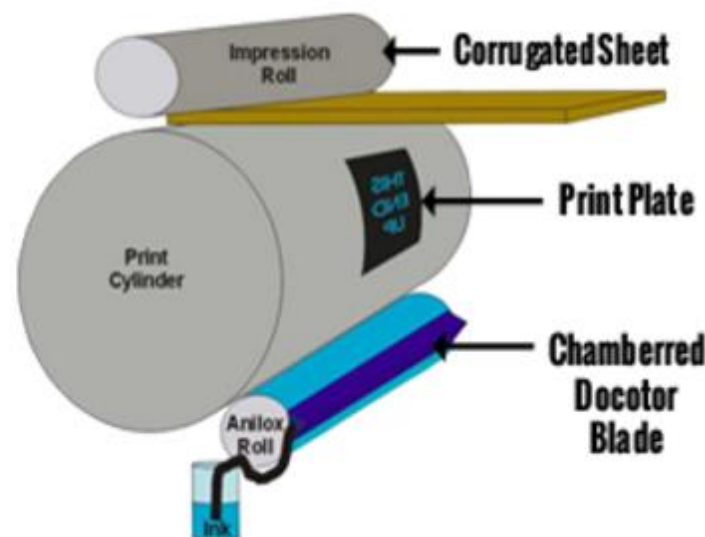
Nipping Examples: Offset and Flexographic Printing

OFFSET PRINTING



VS

FLEXOGRAPHY PRINTING



<http://www.leodesignpackaging.com/>

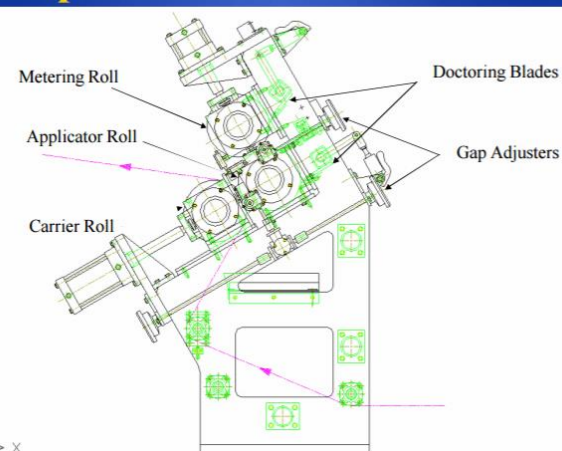
Nipping Examples: Roll Coating



- Direct Gravure
- Reverse Roll Gravure



Nip Fed Reverse Roll Coater



NEW ERA CONVERTING MACHINERY

🏠 235 McLean Blvd.
Paterson, NJ 07504

☎ [\(201\) 670-4848](tel:(201)670-4848)

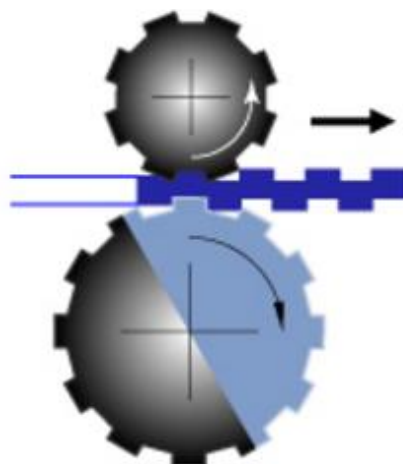
✉ info@neweraconverting.com

Nipping Examples: Embossing Rollers

One-side embossing



Both-side embossing



Example Embossed Product:
lidding for yogurt, coffee pods,
creamer



YURI ROLL Co., Ltd.
technical innovation

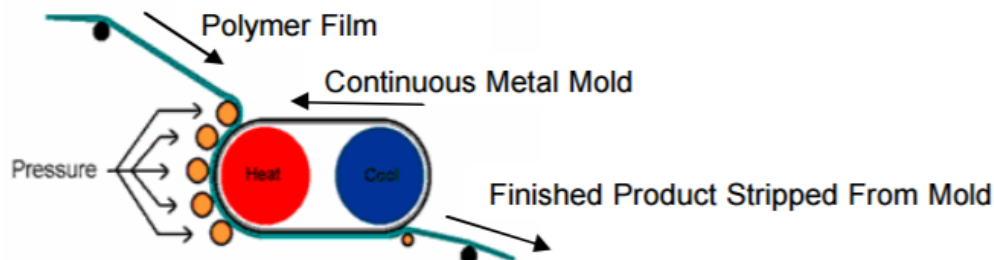
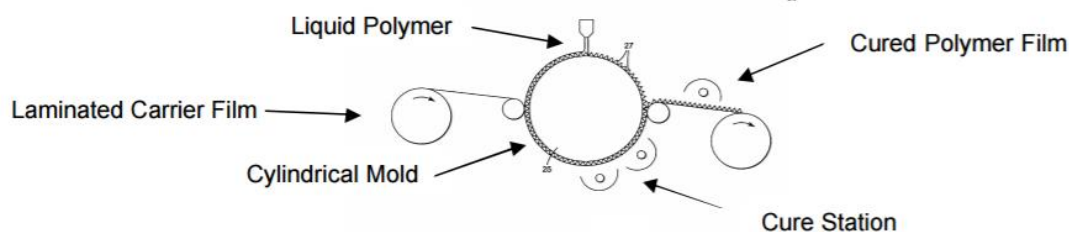
Nipping Examples: Calender for Lithium Foil



<https://en.wikipedia.org/wiki/Calender>

Nipping Exaples: Microreplication

Example of
microreplicated processes

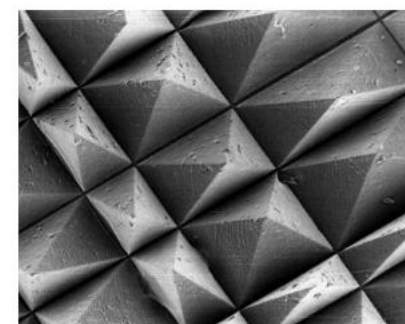
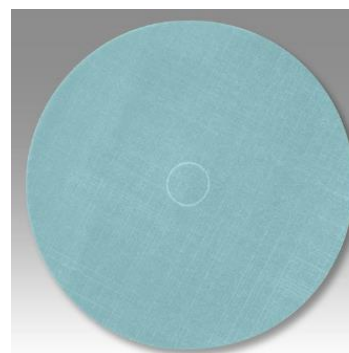
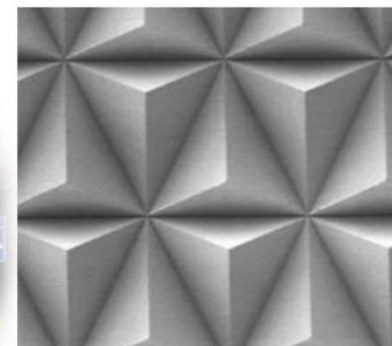


Courtesy of:

robert@10xtechnology.com

www.10xtechnology.com

Examples of
microreplicated product

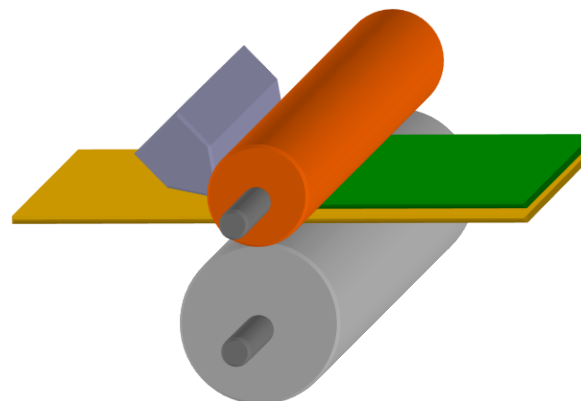


3M™ Trizact™ Abrasive surface, 50x magnification

Courtesy of 3M Co.

When to Use Nipped Rollers

Though many processes require high pressure nipped rollers:

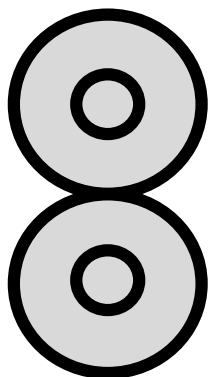


Avoid nipped rollers
wherever possible.

Use unnipped or vacuum
pull rollers.



Thickness vs Pressure Control

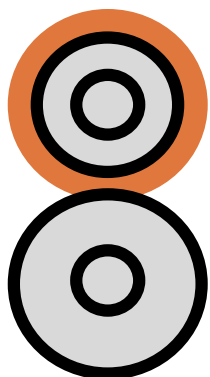


Steel-Steel Nipped Rollers



Is your nip systems for thickness control?

- Coating thickness
- Rubber calendar
- Steel mill



Rubber-Steel Nipped Rollers



Is your nip system for pressure control?

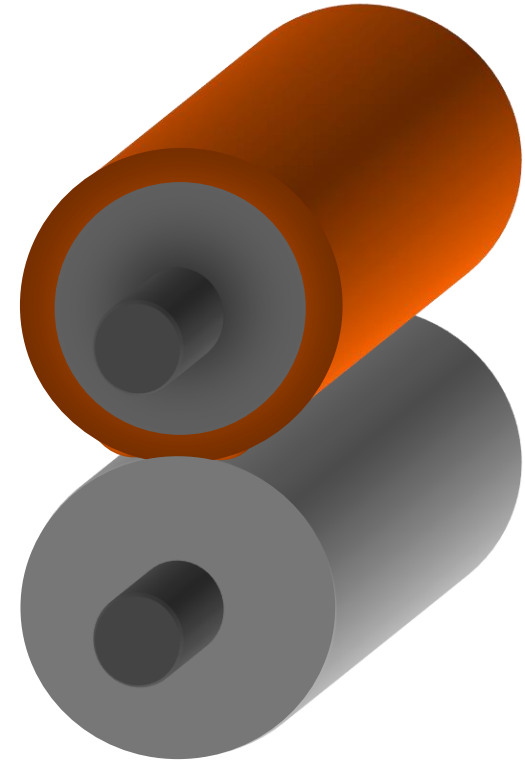
- Squeegee gas or fluid
- Transfer coat/printing
- Emboss solid
- Calendar paper

Compliant Roller Benefits

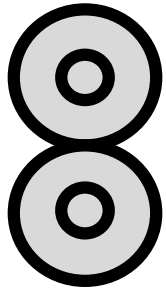
Pressure controlling nip systems usually include at least one compliant roller.

Compliance is needed to compensate for:

- Roller deflection
- Roller diameter variations
- Web thickness variations



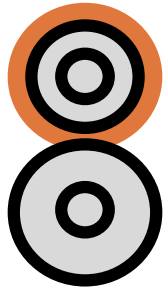
Rubber vs. Steel Nipped Rollers



Steel-Steel Nipped Rollers



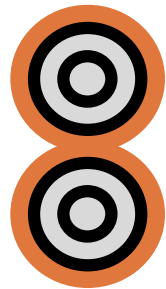
- Rollers have equal compliance
- Speed $\neq f(\text{load})$
- Gapped for thickness control
- Contact for high pressure control
- Good for high temperatures
- Minimal compliance



Rubber-Steel Nipped Rollers



- Rollers have differing compliance
- Rubber speed = $f(\text{load})$
- Good for pressure control
- Limited temperatures
- Limited life due to rubber wear and aging
- Good compliance

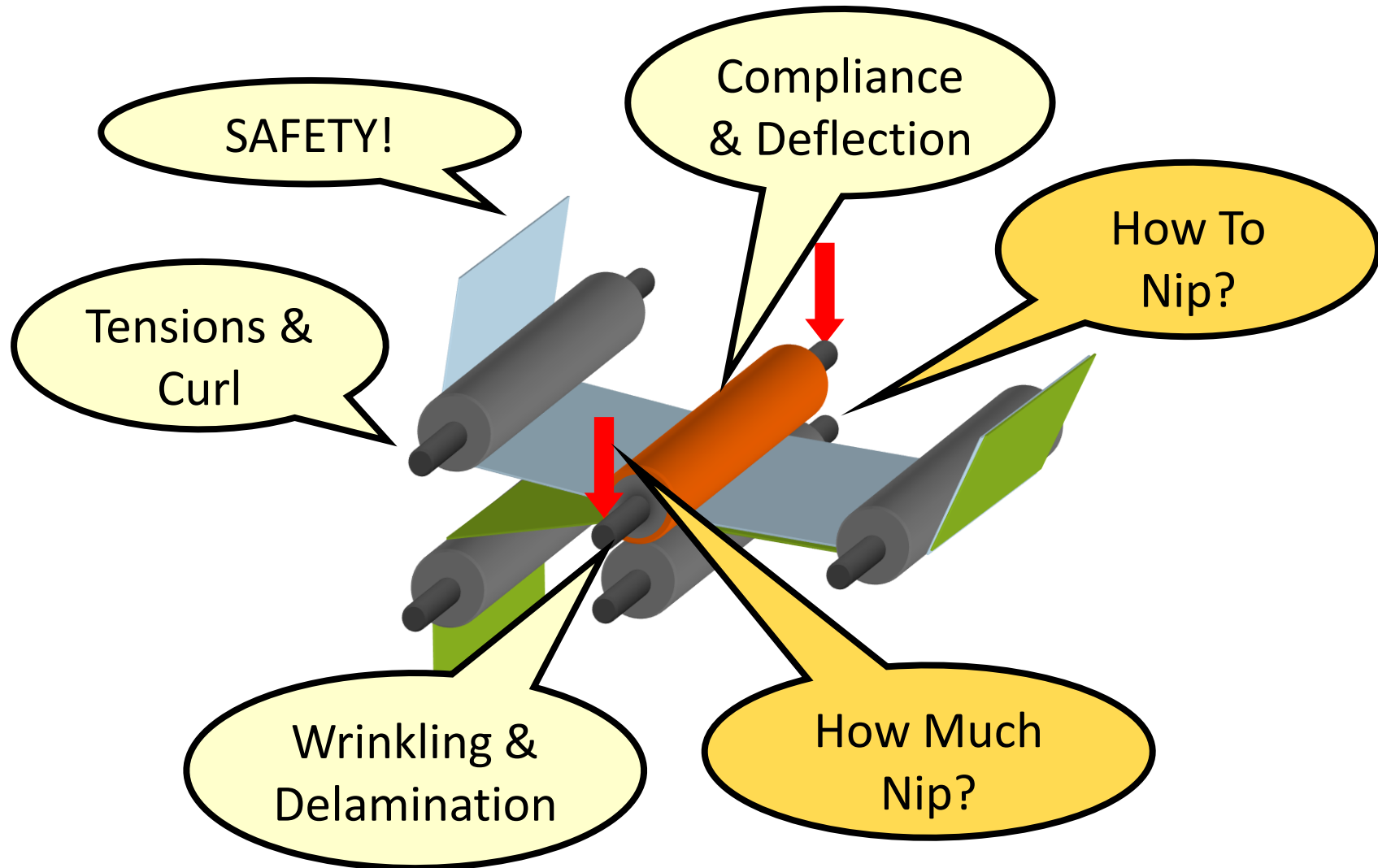


Rubber-Rubber Nipped



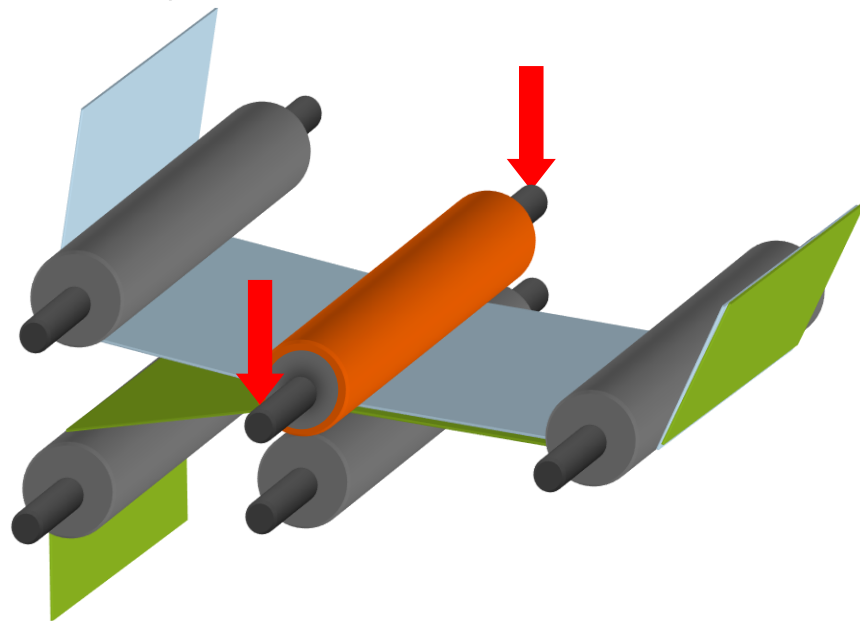
- Rollers are speed matched
- Speed = $f(\text{load})$
- Pressure control
- Limited temperatures
- Limited life due to rubber wear and aging
- Maximum compliance

Nipping and Lamination – What to Consider?



Optimize a Nipped Process

- ▶ Nip force from pressure and geometry
- ▶ Effective width
- ▶ Nip footprint
- ▶ Nip average pressure
- ▶ Nip residence time
- ▶ Rubber covering indentation
- ▶ Roller deflection
- ▶ Compare indentation to deflection
- ▶ Compensate for deflection, if needed



Nipping and Lamination Best Practices

How Much Nip?:

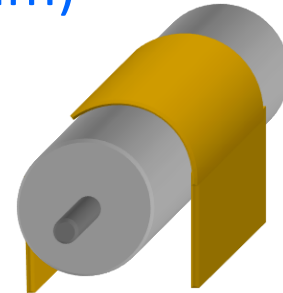
- Calculate the load, $F(\text{nip})$.
- Determine effective width (web width, roller width), $w(\text{eff})$.
- Calculate nip load per width, $N = F(\text{nip})/w(\text{eff})$
- Roller width should be greater than web width, but not excessively wider than the web (e.g. roller width is $< 105\%$ of web width)
- Create a table or chart of N or $F(\text{nip})$ vs. supply pressure
- Measure the footprint and CD footprint variations, b .
- Estimate average process pressure from load per width and footprint $P=N/b$. ($P_{\text{max}} = 4/\pi * P_{\text{avg}}$, 27% higher than average.)
- Avoid gapped or indentation controlled nipping.
- For gapped or indentation nips, measure the gap or indentation.

Nips Can Create High Pressure

Unnipped systems are dependent on web tension and roller radius. **Pressure (kPa) = Tension (N/m) / Radius (mm)**

$$P = \frac{T}{r}$$

Example:
1 PLI, 2" radius
= 0.5 psi



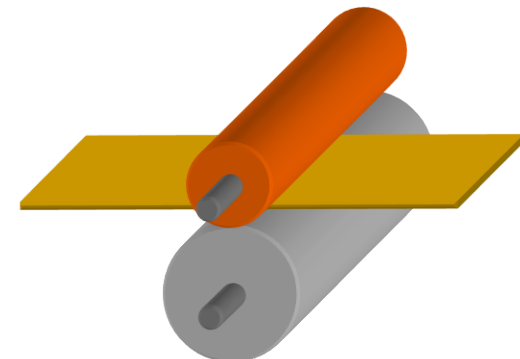
Nipped systems can focus high load over a small area, creating high pressure, independent of tension and radius.

Average Pressure (kPa) = Nip Load (N/m) / Footprint (mm)

**Nips create
10-1000x more
pressure.**

$$P_{avg} = \frac{N}{b}$$

Example:
10 lbs/in.
0.4" footprint
= 25 psi



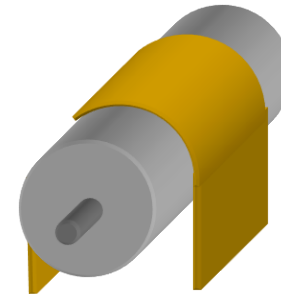
Nips Can Create High Pressure

Unnipped systems are dependent on web tension and roller radius. **Pressure (psi) = Tension (pli) / Radius (in.)**

$$P = \frac{T}{r}$$

Example:

200 N/m, 50mm radius
= 4 kPa (0.04 bar)



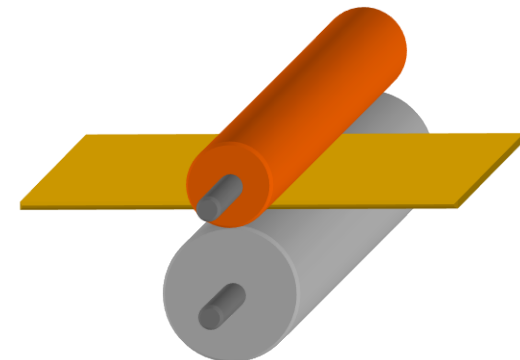
Nipped systems can focus high load over a small area, creating high pressure, independent of tension and radius. **Average Pressure (psi) = Nip Load (pli) / Footprint (in)**

**Nips create
10-1000x more
pressure.**

$$P_{avg} = \frac{N}{b}$$

Example:

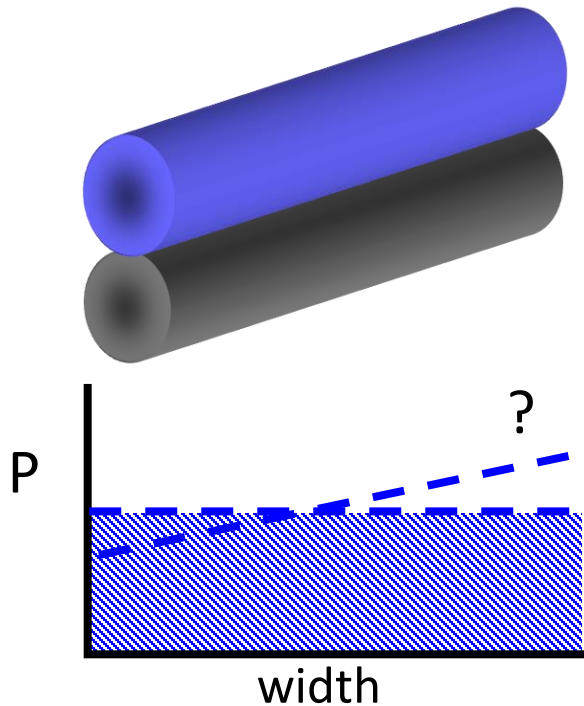
2000 N/m
10 mm footprint
= 200 kPa (2 bar)



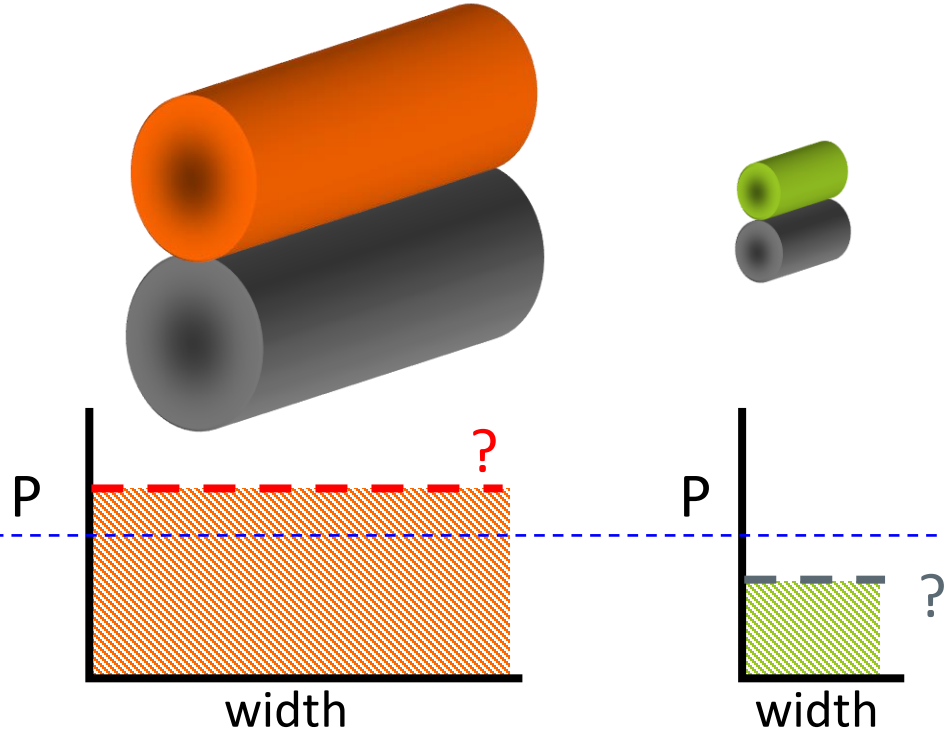
Why Measure Pressure Variations?

For consistent product quality, you need:

Uniform pressure across a nip.



Uniform pressure across nips of sister process lines.

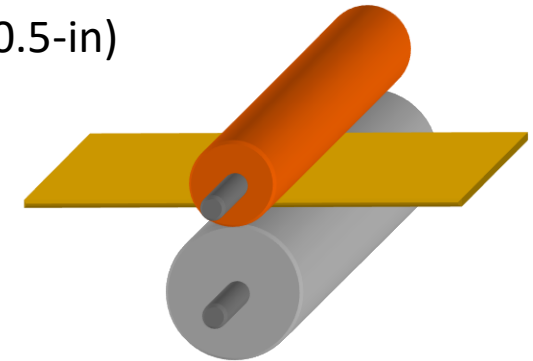


Typical Nip Loads and Pressures?

Process nip load and pressure will be a function of:

- Nip hardness (Durometer, diameter, thickness)
- Material (gas, liquid, viscous liquid, low modulus solid, high modulus solid)
- Rate

PROCESS	NIP LOAD	PRESSURE (b = 0.5-in)
Soft winding nips	< 1 lb/in	2 psi
Nipped drive rolls (usually >5x tension differential)	2-10 lbs/in	4-20 psi
High speed winding nips	1-10 lbs/in	2-20 psi
2-drum paper winding	2-30 lbs/in	4-60 psi
Laminating nips	1-100 lbs/in	2-200 psi
Roll coating nips	2-100 lbs/in	4-200 psi
Extrusion nips	2-200 lbs/in	4-400 psi
Microreplication/embossing	10-200 lbs/in	20-400 psi
Calendar nips (according to online refs.)	200-4000 lbs/in	400-8000 psi



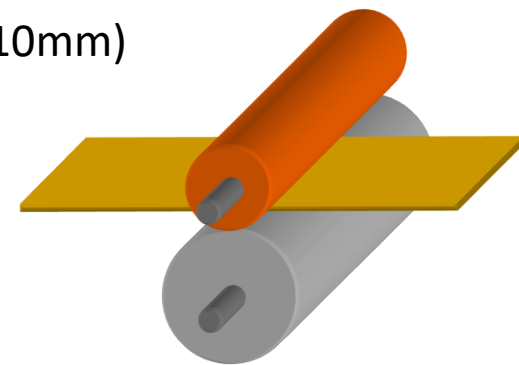
Required nip loads and pressures by material and process needs. These examples should not be considered recommended values.

Typical Nip Loads and Pressures?

Process nip load and pressure will be a function of:

- Nip hardness (Durometer, diameter, thickness)
- Material (gas, liquid, viscous liquid, low modulus solid, high modulus solid)
- Rate (faster processing requires more load)

PROCESS	NIP LOAD	PRESSURE (b = 10mm)
Soft winding nips	< 200 N/m	20 kPa
Nipped drive rolls (usually >5x tension differential)	400-2000 N/m	40-200 kPa
High speed winding nips	200-2000 N/m	20-200 kPa
2-drum paper winding	400-6000 N/m	40-600 kPa
Laminating nips	0.2-20 kN/m	20-2000 kPa
Roll coating nips	0.4-20 kN/m	40-2000 kPa
Extrusion nips	0.4-40 kN/m	40-4000 kPa
Microreplication/embossing	2-40 kN/m	0.2-4 MPa
Calendar nips (according to online refs.)	40-800 kN/m	4-800 MPa



Required nip loads and pressures by material and process needs. These examples should not be considered recommended values.

Nip Force per Width (N), Average Nip Pressure

Total Nip Load Force

$$F_N = F_1 + F_2 (\pm W_{eff})$$

Nip Load in Force per Width

$$N = F_N / w$$

Average Pressure in Nip

$$P_{avg} = N / b = F_N / wb$$

F_1 = Force applied to side
1 of nipping roller

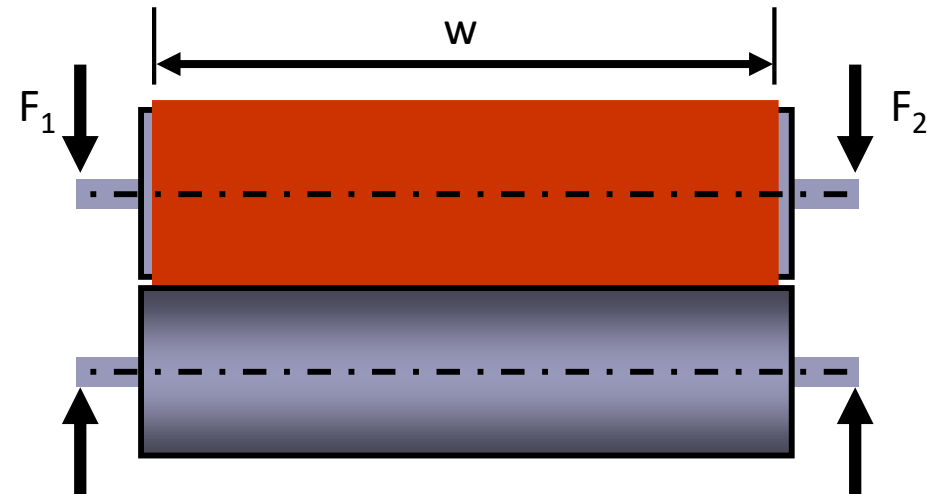
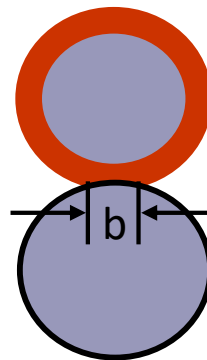
F_2 = Force applied to side
2 of nipping roller

W_{eff} = Effective roller weight
on nipping load

N = Nip load

P = Nip pressure, average

b = Nip footprint
(machine direction
contact length)



Controlling Nip Load or Indentation

Indentation Controlled Nip



Courtesy of Delta Industrial

Pressure Controlled Nip



Pressure Controlled Nip



Courtesy of Converter Assy. Corp.

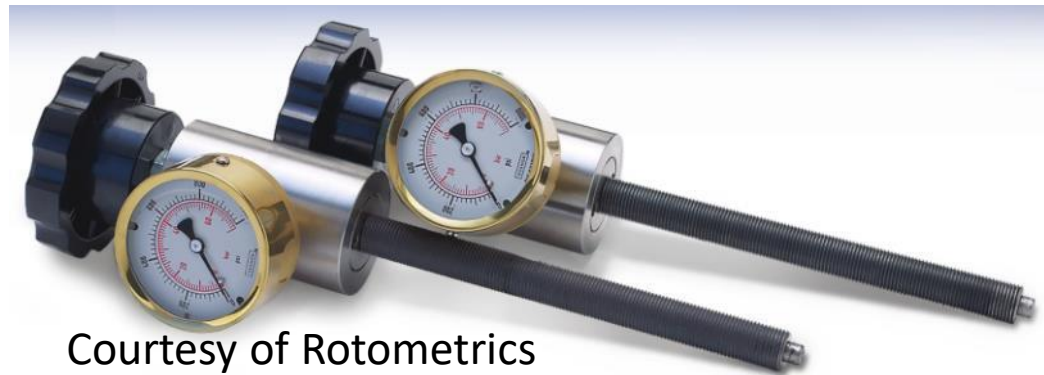
Air Cylinders and Hydra Jacks

Most pressure controlled nipped rollers systems use double acting air cylinders (allow pressure control to extend or retract the cylinder shaft).



Courtesy of Parker

A Hydra Jack (Rotometrics) replaces a screw with a pressure controlled screw.



Courtesy of Rotometrics

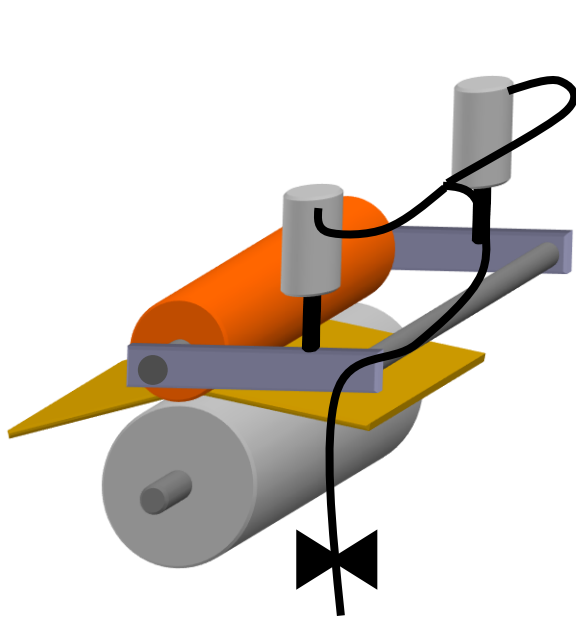
Nipped System with Gap or Indentation Control



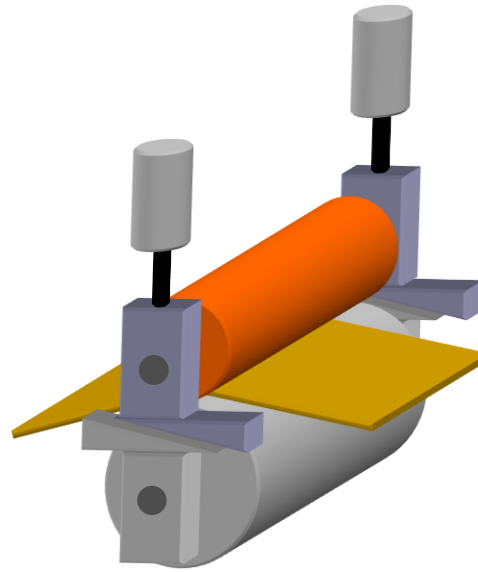
1. With rollers touching with no load, dial indicator is set to zero.
2. For gap mode, stops are set to hold the nip open a fixed amount.
3. For negative gap mode, nip load is turned up, creating a large indentation into the rubber roller, then stops are set to create the target indentation (negative gap).

Nipped Rollers Load Options

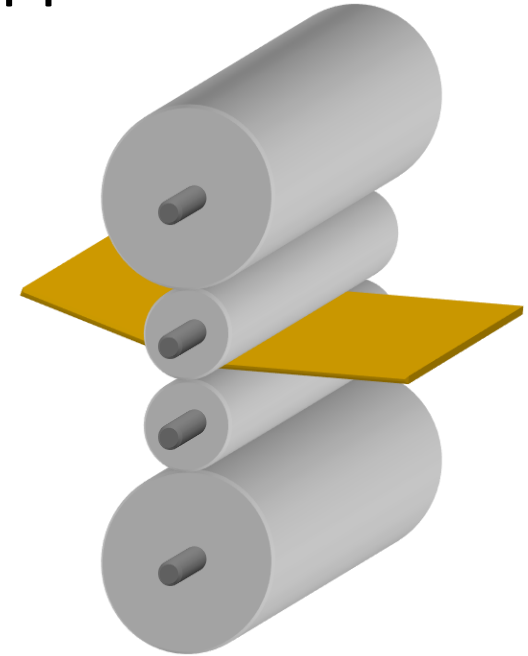
Many processes require high pressure nipped rollers:



Pneumatic Loaded



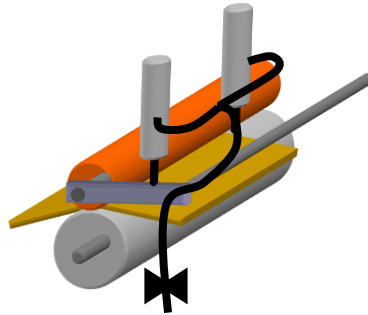
Engagement Loaded



Stacked / Gravity Loaded

Nip System Comparison

Pneumatic



Advantages

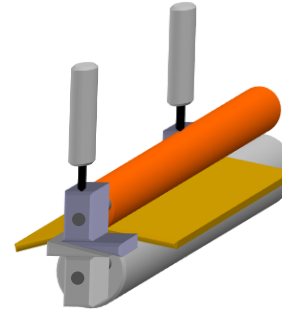
Pressure is independent of web thickness

Direct relation of $N = f(P, \text{psi})$

Disadvantages

Added complexity

Engagement

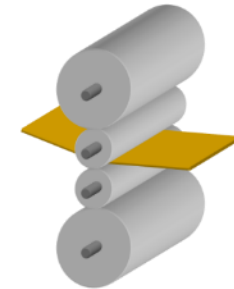


Simple

Can control thickness (gap mode)

P is NOT independent of rubber, web thickness, speed

Stacked / Gravity



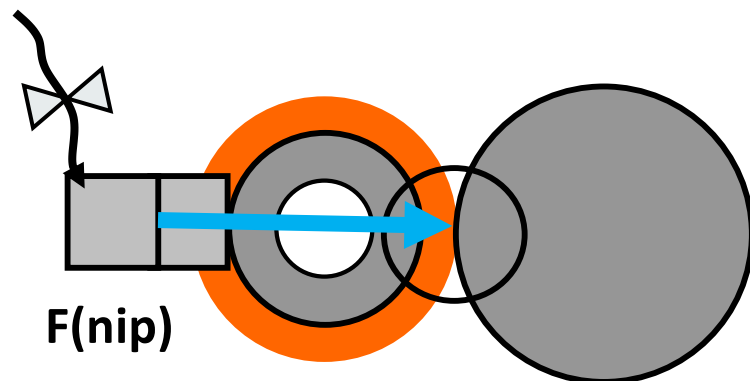
Simple

Load is applied uniformly and inline with process, greatly reducing deflection

Added complexity

Turning Air Pressure into Nip Force

Force Controlled Nip



$F(nip) = \text{Force}$

$P = \text{Pressure}$

$A = \text{Area of cylinder}$

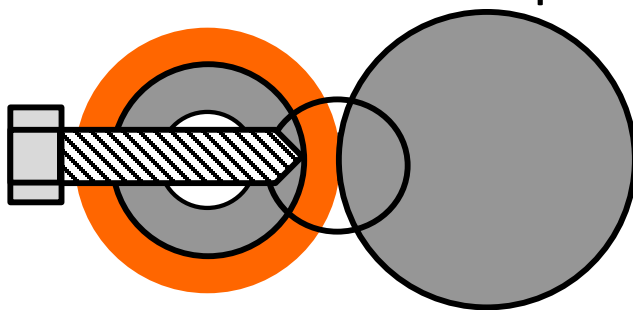
$N = \text{No. of cylinders}$

$F(nip) = N \times P \times A$

$F(nip) = COF \times F(nip)$

- or -

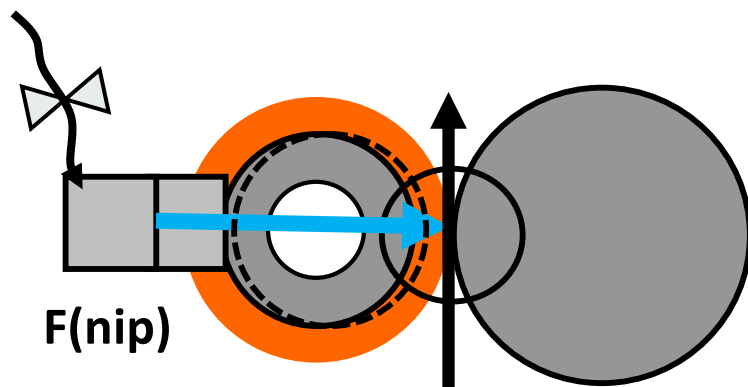
Indentation Controlled Nip



$\delta = \text{Rubber indentation, the difference in covering thickness in the unnipped vs. nipped zone.}$

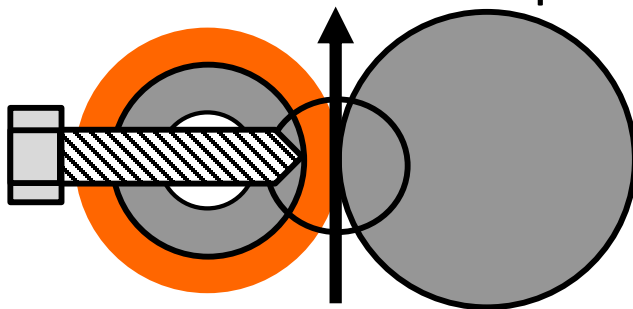
Turning Air Pressure into Nip Force

Force Controlled Nip



$$F(\text{web}) = F(\text{no web})$$

Indentation Controlled Nip



$$F(\text{web}) > F(\text{no web})$$

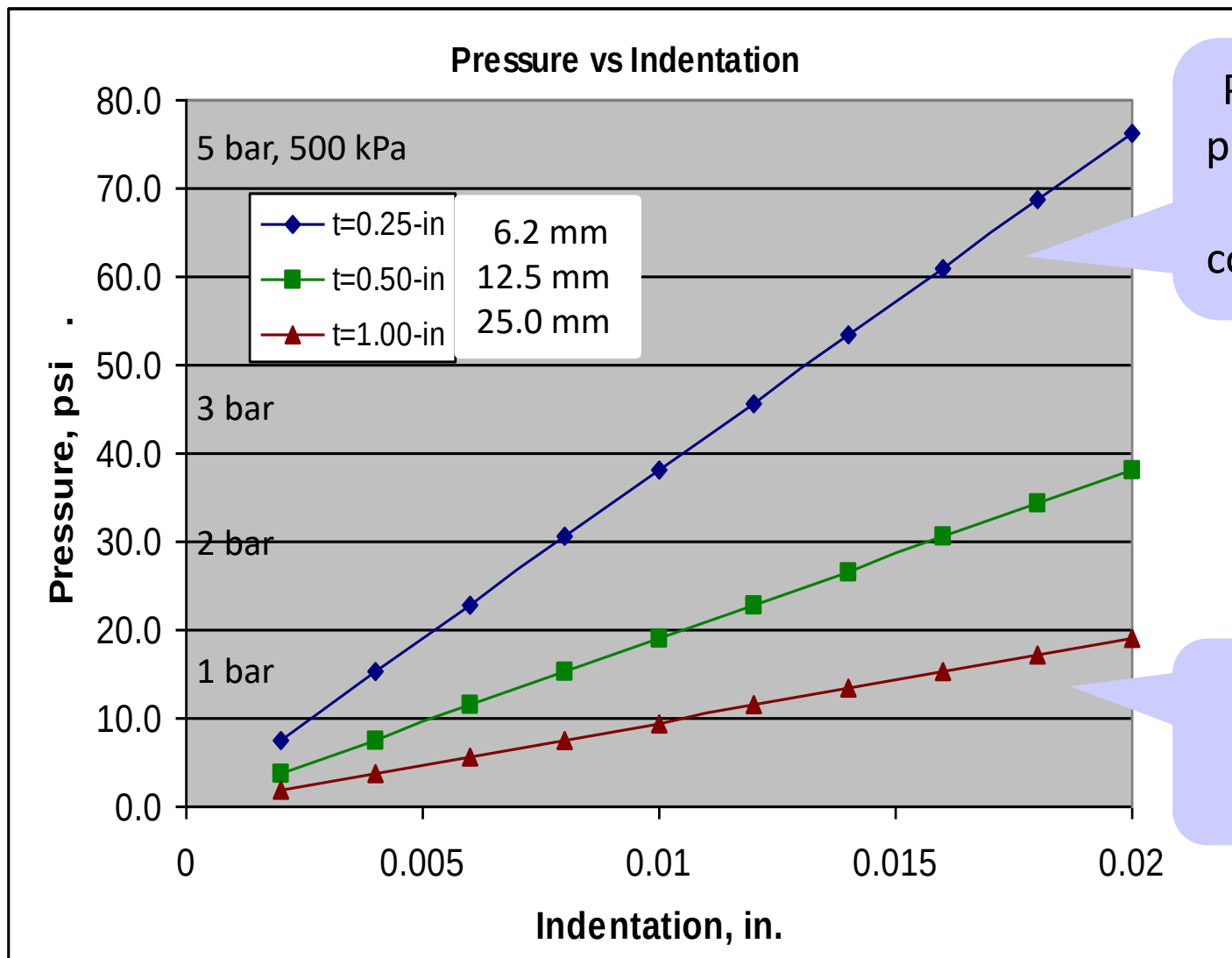
Compare these two system:

If each indents the same rubber covering by the same amount, what is the difference between force or indentation controlled nipping?

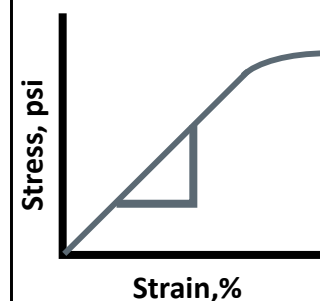
There is no difference without the web in the nip.

However, adding the web will make little change in the force loaded nip pressure, but will increase the indentation loaded nip pressure by the force required to indent the rubber covering an additional amount equal to the thickness of the compressed web.

Average Pressure vs. Indentation



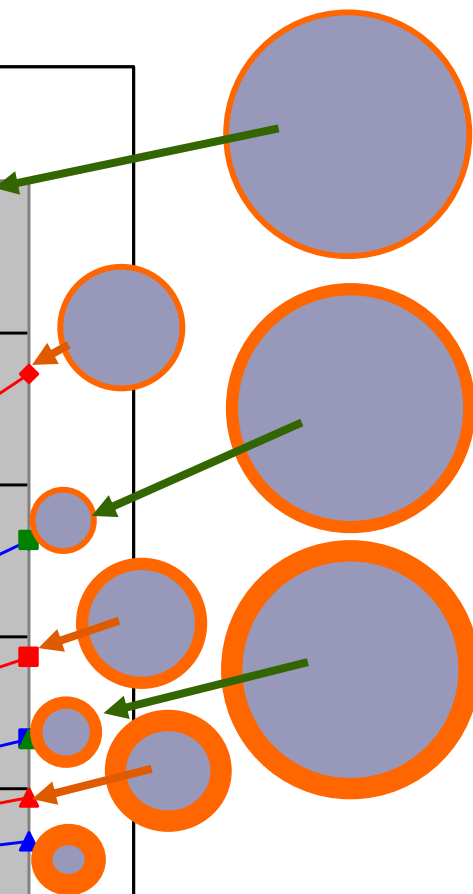
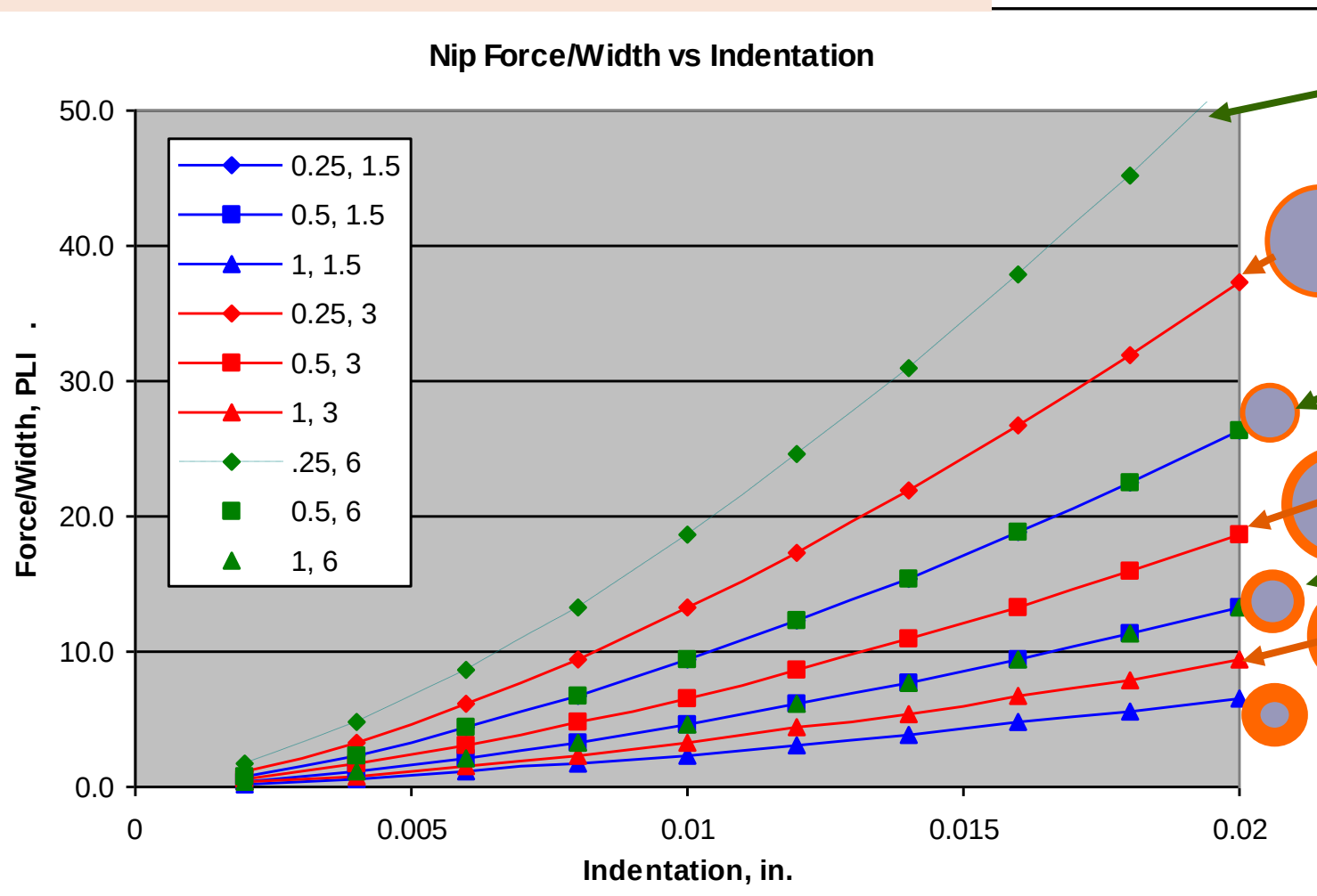
Pressure is directly proportional to ratio of indentation to cover thickness (δ/t).



Pressure is independent of radius!

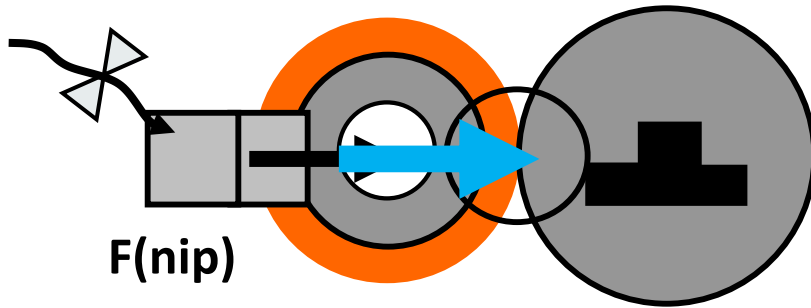
Nip Load (PLI) vs. Indentation

Larger diameters and thinner coverings require more nip load to create indentation.



Effective Nip Force – In-Line Cylinders

Calculate cylinder force from pressure and area



$F_{CYL\&NIP}$ = Total force of cylinder and nip.

n = Number of cylinders

P = Pressure applied to cylinders

A = Area of individual cylinder

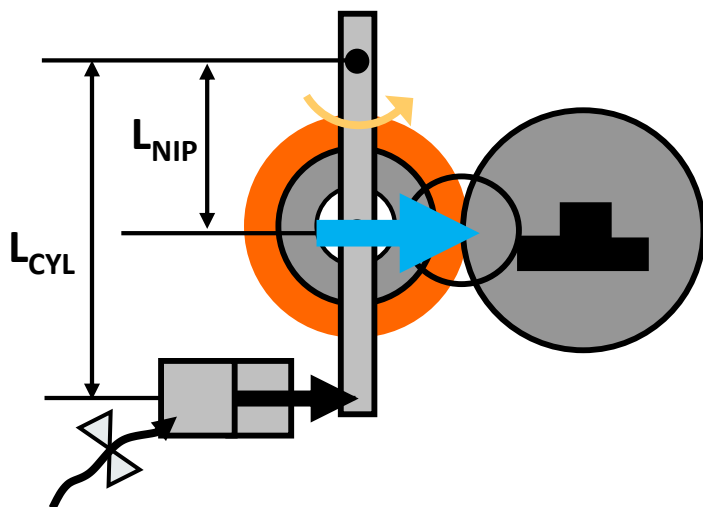
r_{CYL} = Radius of cylinder bore

$$F_{CYL\&NIP} = nPA = nP(\pi r_{cyl}^2)$$

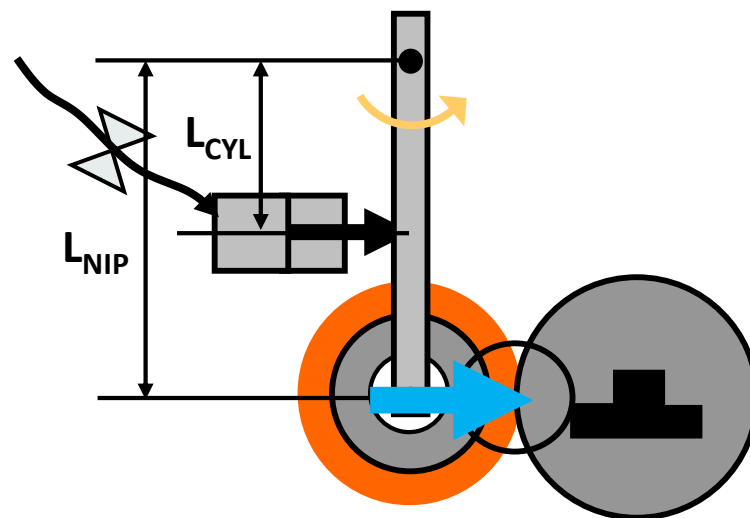
Force by Cylinder Size (per two cylinders)

Cylinder Diam.	Area of 2 Cyls.	Force (10psi)	Force (80psi)
1-inch	1.6-inch	16-lbs	128-lbs
2	6.3	63	500
3	14	140	1100
4	25	250	2000
5	39	390	3100
6	56	560	4500
8	100	1000	8000
10	157	1570	12500
12	226	2260	18000

Effective Nip Force – Pivoting Nip



Cylinder with Mechanical Advantage



Cylinder with Mechanical Disadvantage

Calc. cylinder force from pressure and area

$$F_{CYL} = nPA = nP(\pi r_{cyl}^2)$$

Calc. leveraged nip force

$$F_{NIP} = F_{CYL} (L_{CYL} / L_{NIP})$$

For pivoting system:

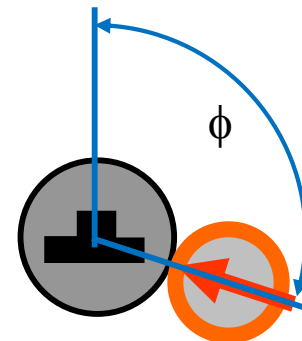
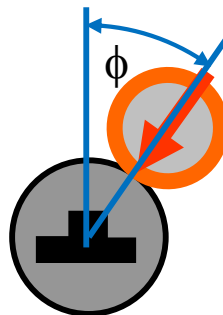
L_{CYL} = Length from pivot to cylinder

L_{NIP} = Length from pivot to nip

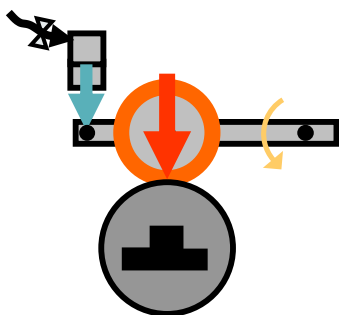
Calculate Nip Force (Considering Gravity)

Depending on angle of nipping, roller weight can be additive, neutral, or subtractive.

$$F_{NIP} = F_{CYL,eff} + W_{NIP} \cos \phi$$

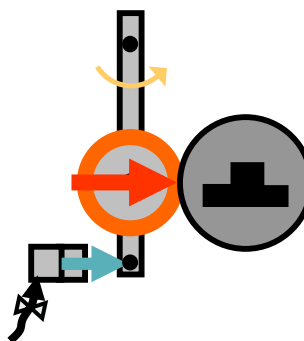


Roller weight
adds to nip force



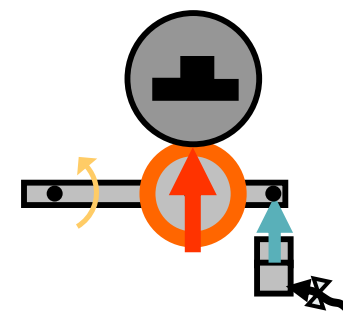
$$F_{NIP} = F_{CYL,eff} + W_{NIP}$$

Nip force is independent
of roller weight



$$F_{NIP} = F_{CYL,eff}$$

Roller weight subtracts
from nip force

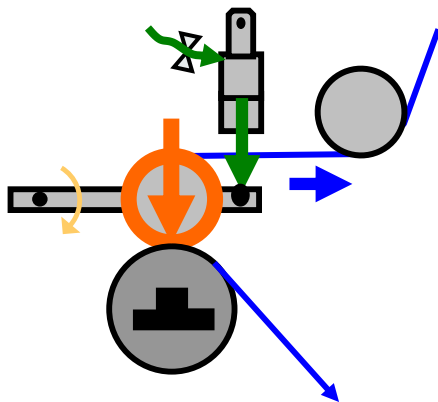


$$F_{NIP} = F_{CYL,eff} - W_{NIP}$$

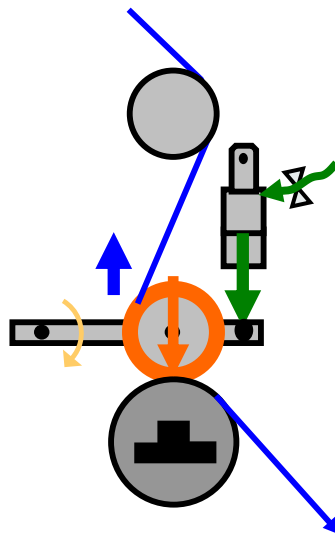
Nips and Tension: Neutral, Subtractive, or Additive

If loaded nip roller is wrapped with anything other than 180-degrees, the web tension will subtract from (or add to) the air cylinder forces, reducing the effective nip load.

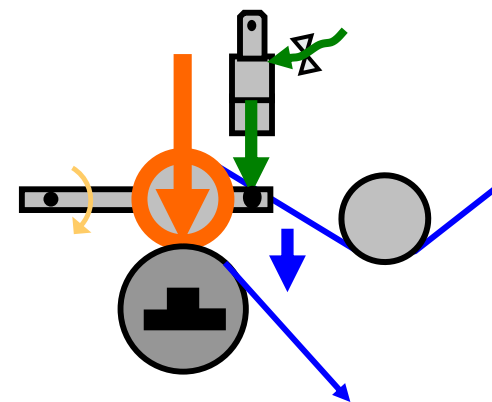
Tension Neutral:
Tension is perpendicular
to nipping direction



Tension **Subtracts** from Nip:
High tension force may pull
open a nip with low loads.

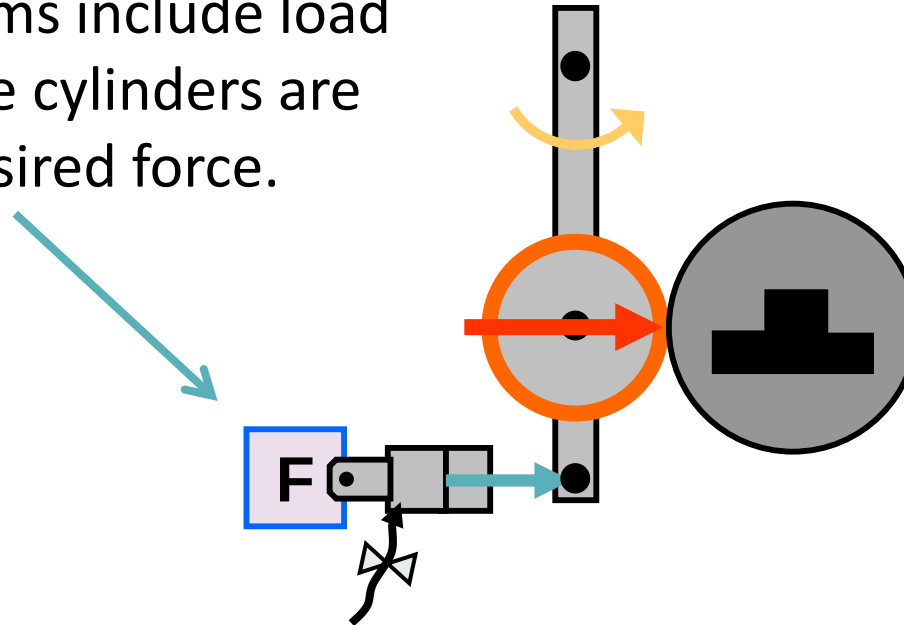


Tension **Adds** to Nip:
Tension adds to force
closing the nip.



Monitoring Nip Load

Some nipping systems include load cells to confirm the cylinders are creating the desired force.





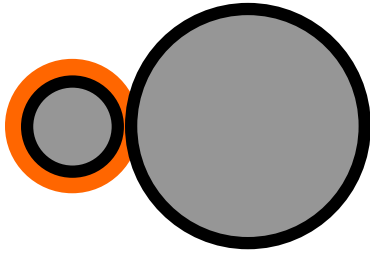
4.0 Nipped Systems

Timothy J. Walker, TJWalker + Associates Inc.

1620 Edgcumbe Road, Saint Paul, MN 55116

tjwalker@tjwa.com www.webhandling.com 651.686.5400

Nip Orientation Options



Advantages:

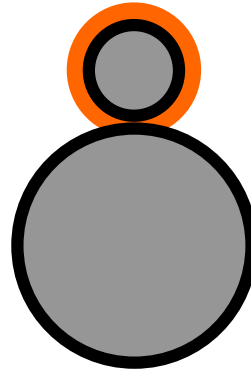
Nip load independent of roller weight.

Roller deflection spreads web with top vertical entry.

Easy to align (level)

Disadvantages:

Top entry may be less safe (easy to fall in)

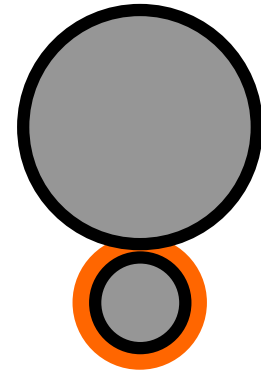


Advantages:

Minimum deflection if roller weight is target nip load.

Disadvantages:

More difficult to prevent twist (tram)



Advantages:

Safest? Nip open is default position.

Disadvantages:

Gravity and nip load are additive making this the highest deflection case.

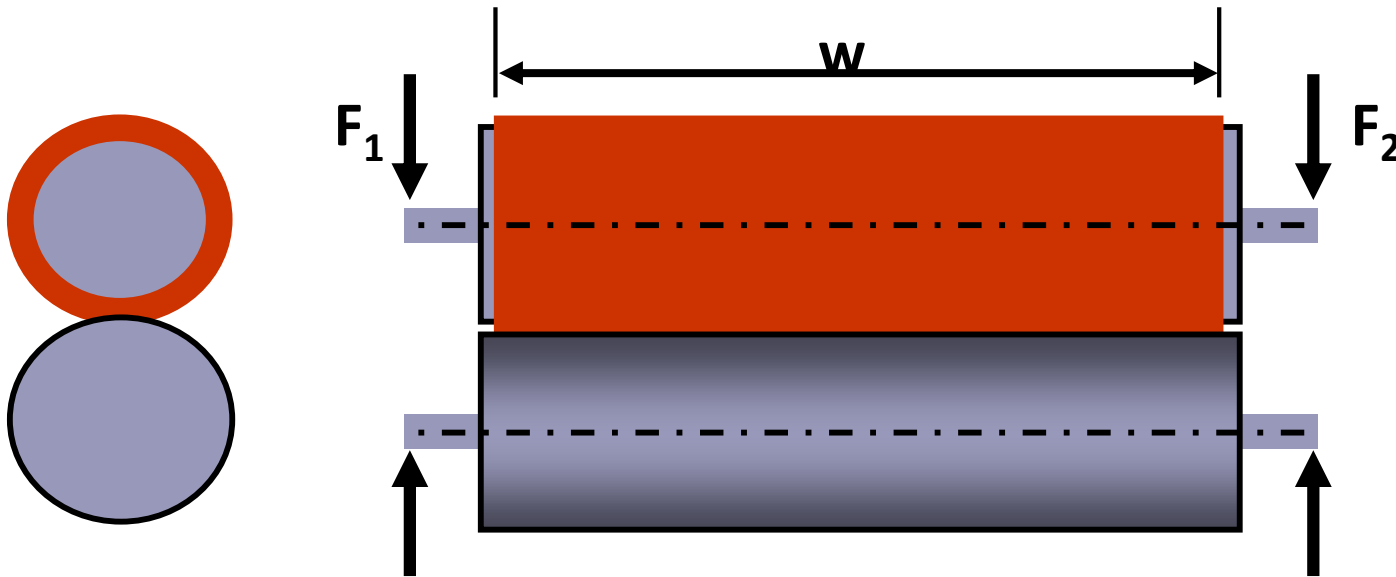
Nip Force per Width (N)

Total Nip Load Force

$$F_N = F_1 + F_2 (\pm W_{eff})$$

Nip Load in Force per Width

$$N = F_N / w$$

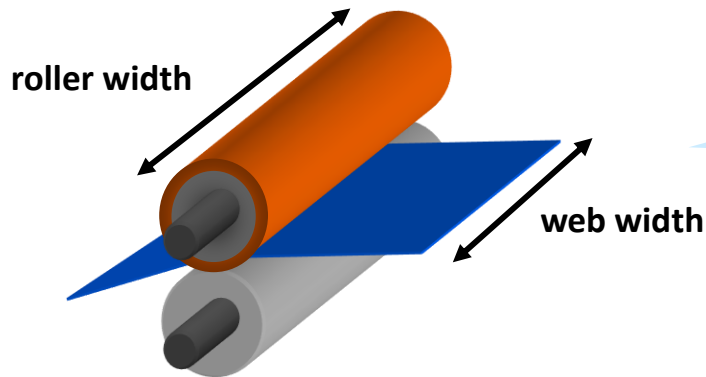


F_1 = Force applied to side 1 of nipping roller

F_2 = Force applied to side 2 of nipping roller

W_{eff} = Effective roller weight on nipping load

What is the Nip Width?



Is the nip width the web or roller width?

t_{web} = web thickness

δ_{rubber} = rubber indentation

$$N = F_N / w_{\text{roller}}$$

$$N = F_N / w_{\text{web}}$$



If $t_{\text{web}} < \delta_{\text{rubber}}$

Nip width is roller width.



If $t_{\text{web}} > \delta_{\text{rubber}}$

Nip width is web width.

Calculate Nip Load - Example

Calc. cylinder force from pressure and area

$$F_{CYL} = nPA = nP(\pi r_{cyl}^2)$$

Calc. leveraged nip force

$$F_{NIP} = F_{CYL} (L_{CYL} / L_{NIP})$$

For vertical nips, include roller weight effect

$$F_{NIP} = F_{CYL} (L_{CYL} / L_{NIP}) \pm W_{NIP}$$

Calculate Nip Load:

$$N = F_{NIP} / w$$

Examples:

n = 2 cylinders

P = 60 psi

R(cyl bore) = 2-in

A = $\pi r^2 = 12.5 \text{ in}^2$

F(cyl) = (2)(60)(12.5)
= 1500 lbs

L(cyl)/L(nip) = 3/2

F(nip) = (1500)(3/2)
= 2250 lbs

w = 50-in

N = 2250-lbs/50-in
= 45 lbs/in

n = 2 cylinders

P = 4 bar = 0.4 MPa

R(cyl bore) = 50mm

A = $\pi r^2 = 7850 \text{ mm}^2$

F(cyl) = (2)(0.4)(7850)
= 6280 N

L(cyl)/L(nip) = 3/2

F(nip) = (6280)(3/2)
= 9420 N

w = 1.2m

N = 9420N/1.2m
= 7850 N/m

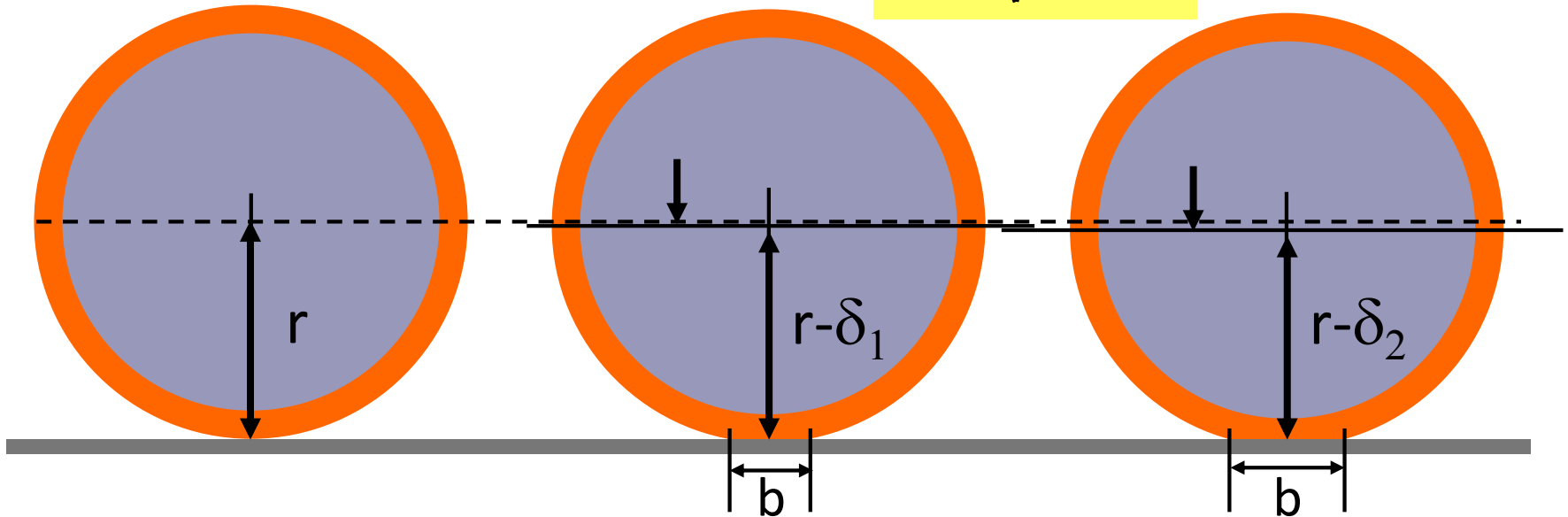
Optimize a Nipped Process

- ▶ Nip force from Pressure and Geometry
- ▶ Effective width
- ▶ Nip footprint
- ▶ Nip average pressure
- ▶ Nip residence time
- ▶ Rubber covering indentation
- ▶ Roller deflection
- ▶ Compare indentation to deflection
- ▶ Compensate for deflection, if needed

Footprint vs. Indentation

$$b^2 = 8r_{eq}\delta$$

$$b = \sqrt{8r_{eq}\delta}$$



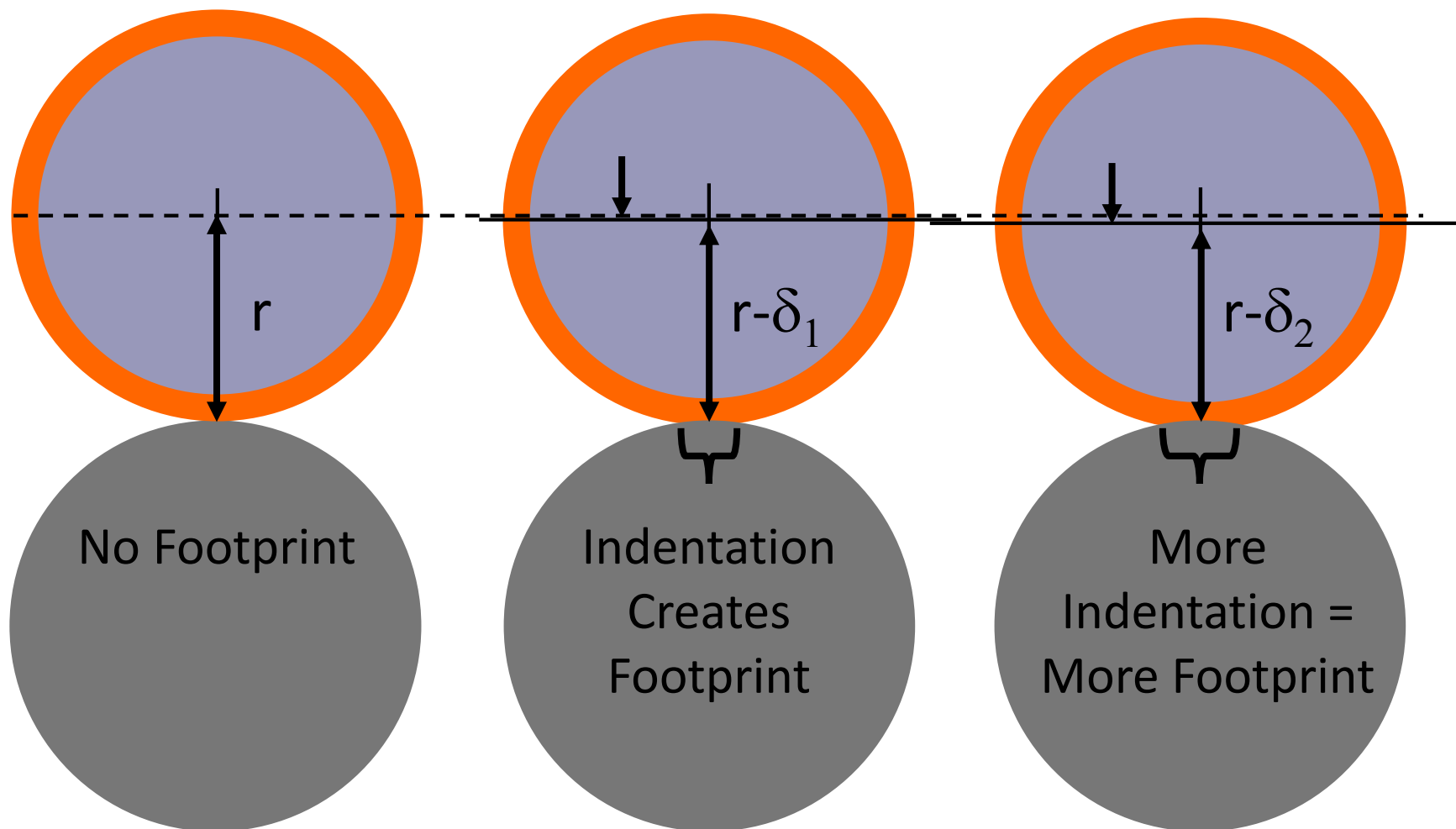
No Footprint

Indentation
Creates
Footprint

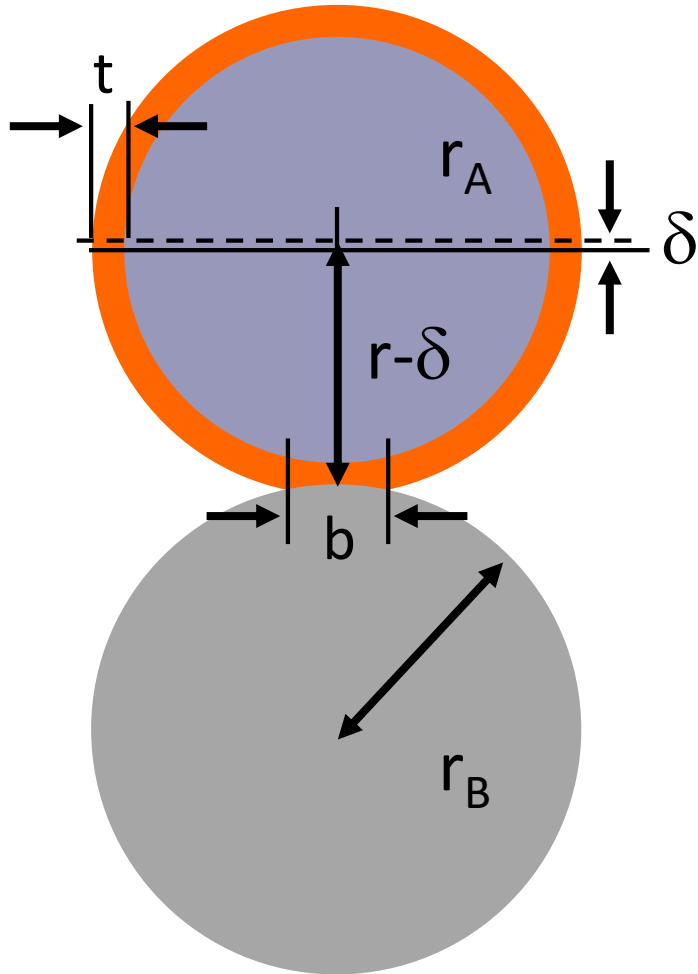
More
Indentation =
More Footprint

Footprint vs. Indentation

Smaller diameter = Smaller footprint



Footprint vs. Indentation



$$b = \sqrt{8r_{eq}\delta}$$

$$r_{eq} = \frac{r_A r_B}{(r_A + r_B)}$$

δ = Indentation

b = Footprint

r_A = Rubber Roller Radius

r_B = Steel Roller Radius

$$\delta = \frac{b^2}{8r_{eq}}$$

This relationship is accurate for small indentations, $\delta/t < 0.07$

Recommendation #7

Rubber rollers should have 10-30 mils indentation under anticipated process pressures.

Why? To forgive sins of diameter, hardness, deflection, alignment, and loading variations.



Recommendation #14

Use nip footprint measurements to verify crossweb nip uniformity.

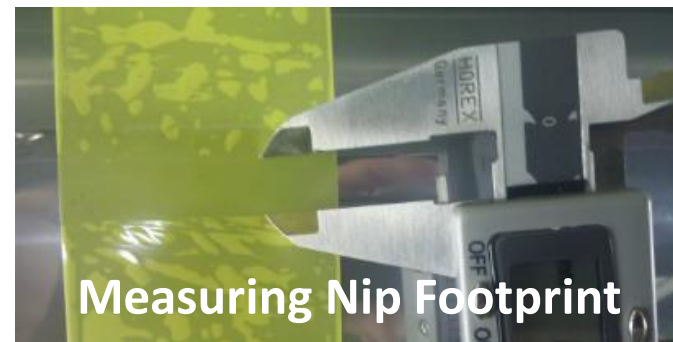
Why? Footprint is a easy measurement and a good indicator of crossweb pressure.



Tape Bubble Footprint Measurement



Tape Before Nipping



Measuring Nip Footprint

Colored adhesive polypropylene splicing tape is a good candidate for this measurement.

Operator Side

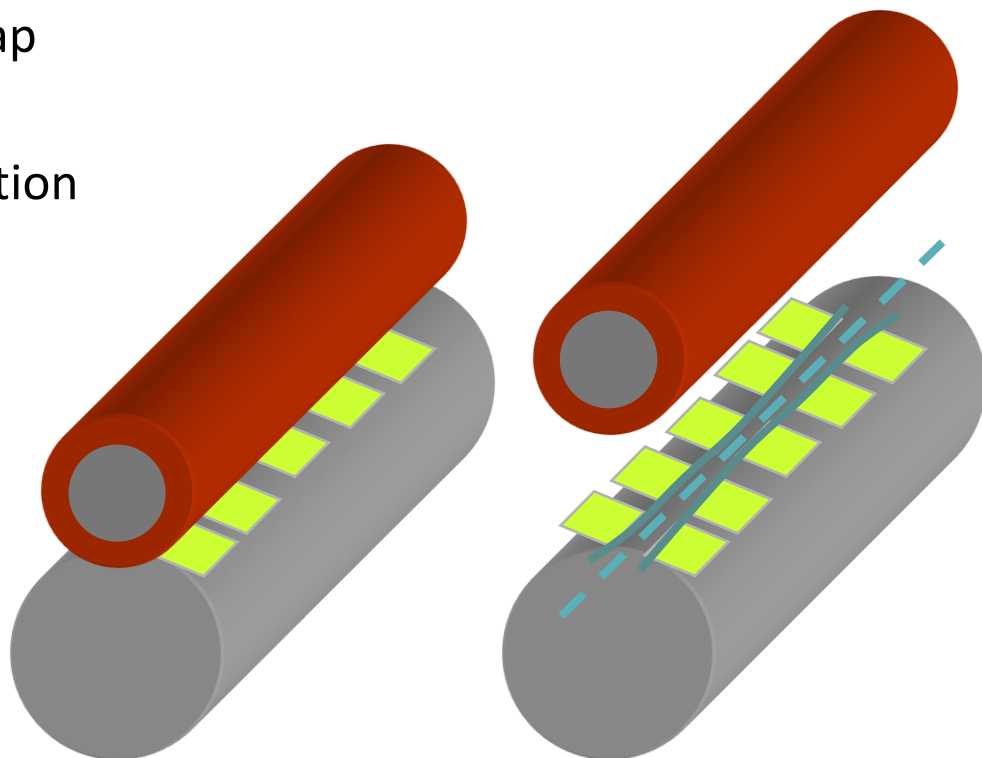
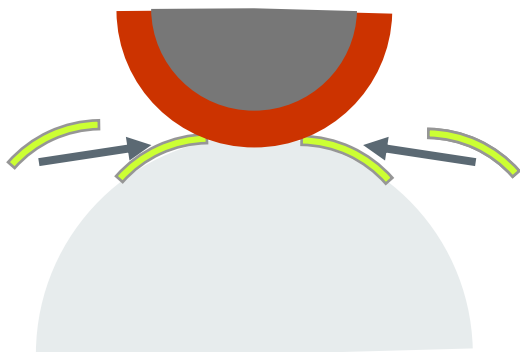
Drive Side



Nip Footprint Measurement 1

Measure nip contact area using a series of sticky notes.

1. Close the nipping roller and apply full load (with nip not rotating).
2. Place a series of Post It™ notes on one roller, pushing them as far into the nip point as possible and stick them in place.
3. Open the nip and measure the gap between opposing notes.
4. Look for MD contact length variation as a sign of non-uniform nip pressure.



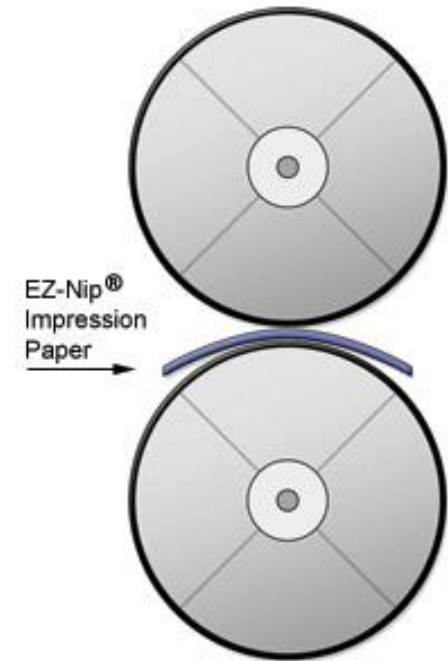
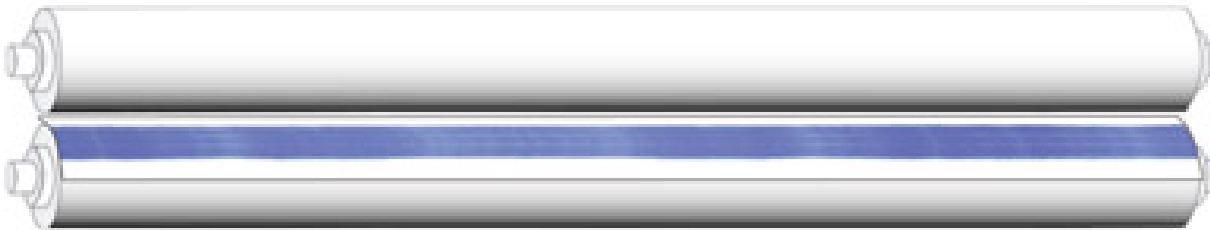
Nip Footprint Measurement 3

EZ-Nip[®]

EZ-Nip - \$300 / 200 ft.

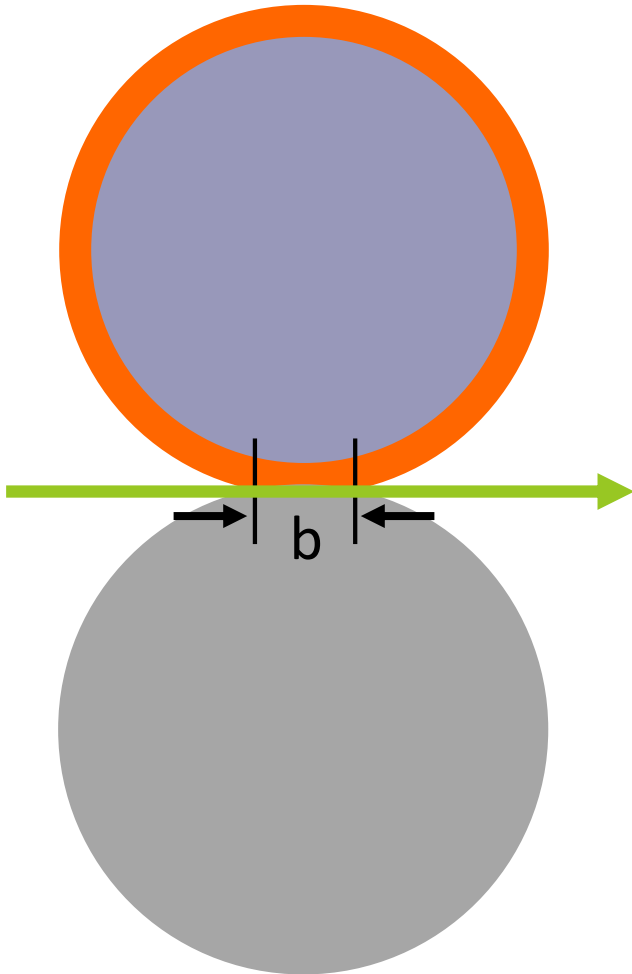
A single sheet pressure sensing film.

Thickness	15 mils (0.076 mm)
Humidity Range	10-90% RH
Pressure Minimum	300 PSI (20.8 kg/cm ²)



www.sensorprod.com/nip.php

Footprint vs. Indentation



$$t_{nip} = \frac{b}{V}$$

b = Footprint

V = Process Speed

t = Time in nip

Example:

B = 0.5-inch, V = 100 in/s = 500 fpm

T = 0.5/100 = 0.005s = 5ms

Optimize a Nipped Process

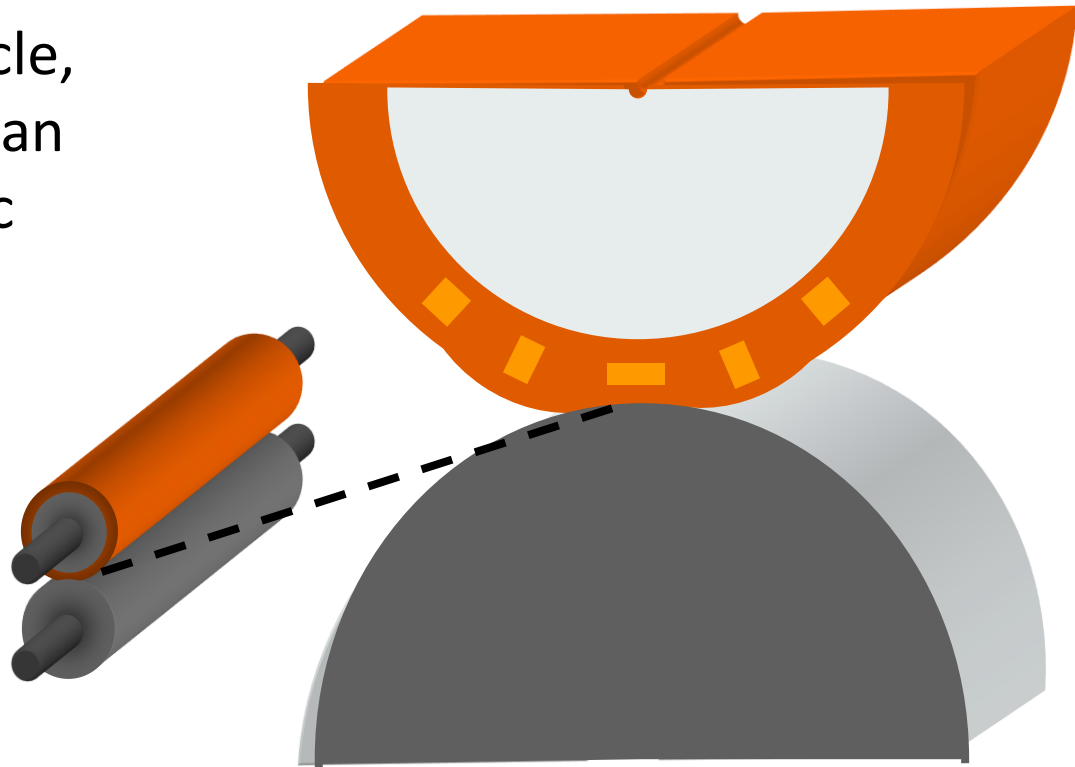
- ▶ Nip force from Pressure and Geometry
- ▶ Effective width
- ▶ Nip footprint
- ▶ Nip average pressure
- ▶ Nip residence time
- ▶ Rubber covering indentation
- ▶ Roller deflection
- ▶ Compare indentation to deflection
- ▶ Compensate for deflection, if needed

Elastomer Nip Deformation

Elastomeric roller covers deform, elongating near the nip point and bulging before and after.

Hysteresis, a differing response in the compression-recovery cycle, creates internal heat, which can lead to significant elastomeric degradation or melting.

The surface speed differential can also lead to wear if rubber nip roller have large wrap angles.



Example: Calculating Rubber Indentation

INPUTS

WEB

poisson's	nu	0.46 --
Shore A	H(SA)	80 --
width	w(web)	30 in

ROLLERS

base roller radius steel	R2	12.0 in
rubber roller outer radius	R1	2.00 in
rubber roller core outer radius	RCO	1.75 in
rubber roller core inner radius	RCI	1.5 in
roller width	w(roller)	30 in

NIPPING

air pressure	P(nip)	40 psi
radius cylinders	r(cyl)	1 in
no. of cylinders	n	4 --
nip/cyl leverage	L(cyl)/L(nip)	1 --
speed	V	300 fpm

OUTPUTS

NIP FORCE / LOAD

Nip force	F(nip)	503 lbs
Nip load	N(force/width)	16.8 lbs/in

INDENTATION / DEFLECTION

thickness	t(r)	0.25 in
Rubber modulus	Eo	1910 psi
radius equiv	req	1.7 in
indentation	delta	0.0053 in
footprint	b	0.27 in
%indent	$\Delta r/t(\text{rubber})$	2.1%
deflection	y	0.0017 in
deflect/indent	$=y/\Delta r$	33%

NIP PRESSURE / TIME

Pressure	Pavg	62 psi
Nip Time	t(nip)	0.004 s
overspeed	$\varepsilon(\text{theta})$	0.528% --

Nip Pressure Equations

$$N = \frac{F_N}{w} = \frac{2}{3} \frac{(1-\nu)^2}{1-2\nu} \frac{E_o}{1-\nu^2} \frac{\sqrt{2r_{eq}}}{t} \delta^{\frac{3}{2}}$$

Contact Mechanics by K.L. Johnson, Cambridge University Press, 1989, pp 139-140.
(According to Dr. J.K. Good, this can be used for rubber strains up to 7 percent.)

$$b = \sqrt{8r_{eq}} \delta$$

$$P_{avg} = \frac{N}{b} = \frac{1}{3} \frac{(1-\nu)^2}{1-2\nu} \frac{E_o}{1-\nu^2} \frac{\delta}{t}$$

Pressure in rubber nip rollers is independent of roller diameter!

Nip Calculation Variables

N = Nip load, N/m or lb/in (PLI)

F_N = Nip force, N or lbs

b = Footprint lengths, m or in (MD length of contact zone)

ν = Rubber covering Poisson's ratio (use 0.46 per OSU WHRC)

E_0 = Young's elastic modulus of rubber covering, kPa or psi
[from Shore A and OSU WHRC equation of $E=f(\text{Shore A})$]

r_{eq} = Equivalent radius of two rollers, m or in = $(R_1 R_2) / (R_1 + R_2)$

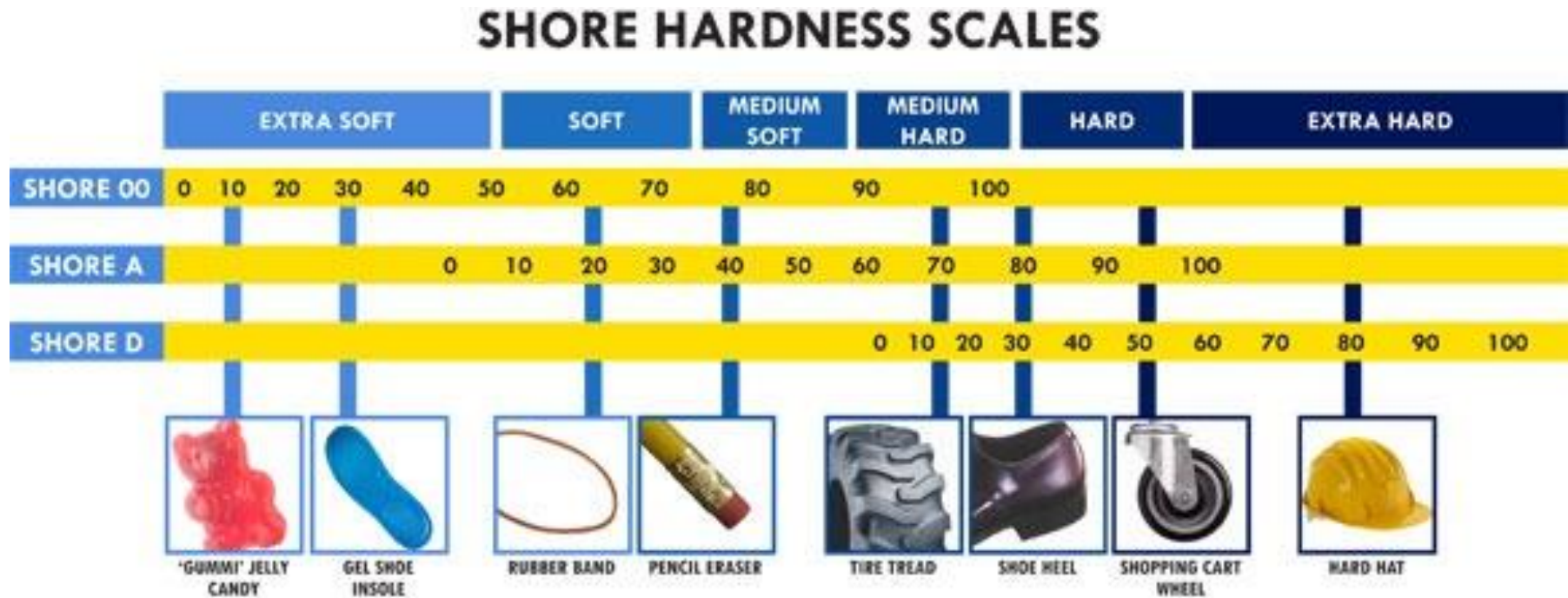
t = Covering thickness, m or in

δ = Rubber radial indentation, m or in

H_{SA} = Hardness per Shore A Durometer tester

Comparing Hardness Scales (with examples)

Courtesy of smooth-on.com



How is Rubber Hardness Measured?

Shore A or D Durometer



The three main suppliers of Durometers (alphabetically)...
Asker (Japan) - **Bareiss** (Germany) - **Shore/Instron** (USA)

shore-durometer-history.blogspot.com

Scale of Durometer

Material To Be Used

A

Frustum cone



Rubber: soft vulcanized (ie tire), natural nitrile.

Elastomeric Materials (rubber and rubber-like):

GR-S, GR-1, neoprene, thiokol, flexible polyacrylic esters. **Other:** wax, felt, leather, etc. (materials that normally yield under fingernail pressure such as the heel on your shoe).

D

Sharp 30° angle

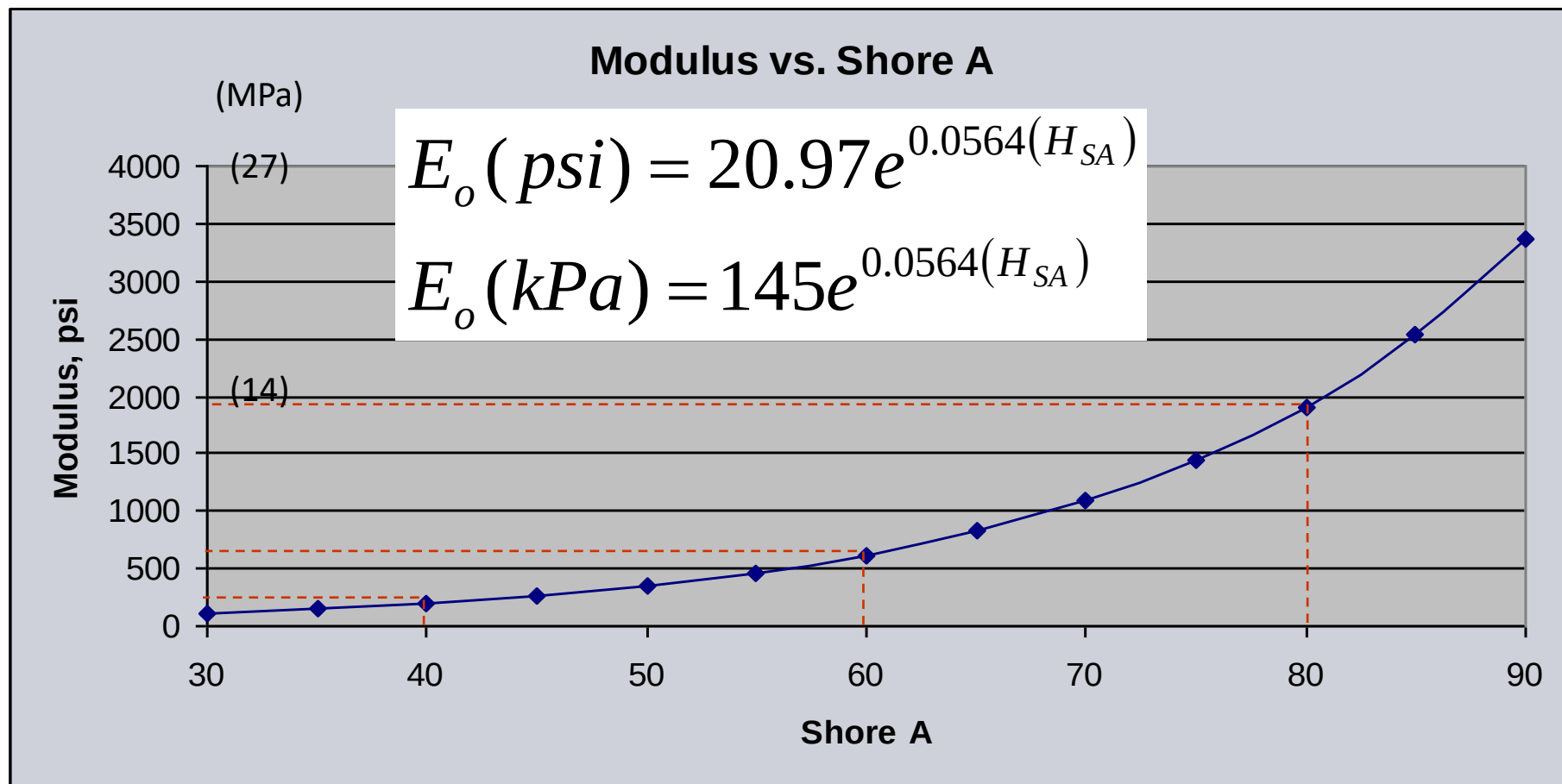


Rubber, Hard Plastics: harder grades such as rigid thermoplastics, plexiglas, thermopolystyrene, vinyl, sheet, cellulose acetate, thermosetting laminates (ie formica) **Other:** paper-filled calendar rolls, calendar bowls, etc. (materials that would not normally indent under fingernail pressure, such as a pocket comb).



▲ Rubber test block kit

Rubber Modulus vs. Shore A Hardness (H_{SA})



Modeling Rubber Covered Nip Rollers in Web Lines, J.K. Good, Oklahoma State University, Stillwater, OK, Proceedings of the Sixth Annual Conference on Web Handling, 2001 pp. 159-177.



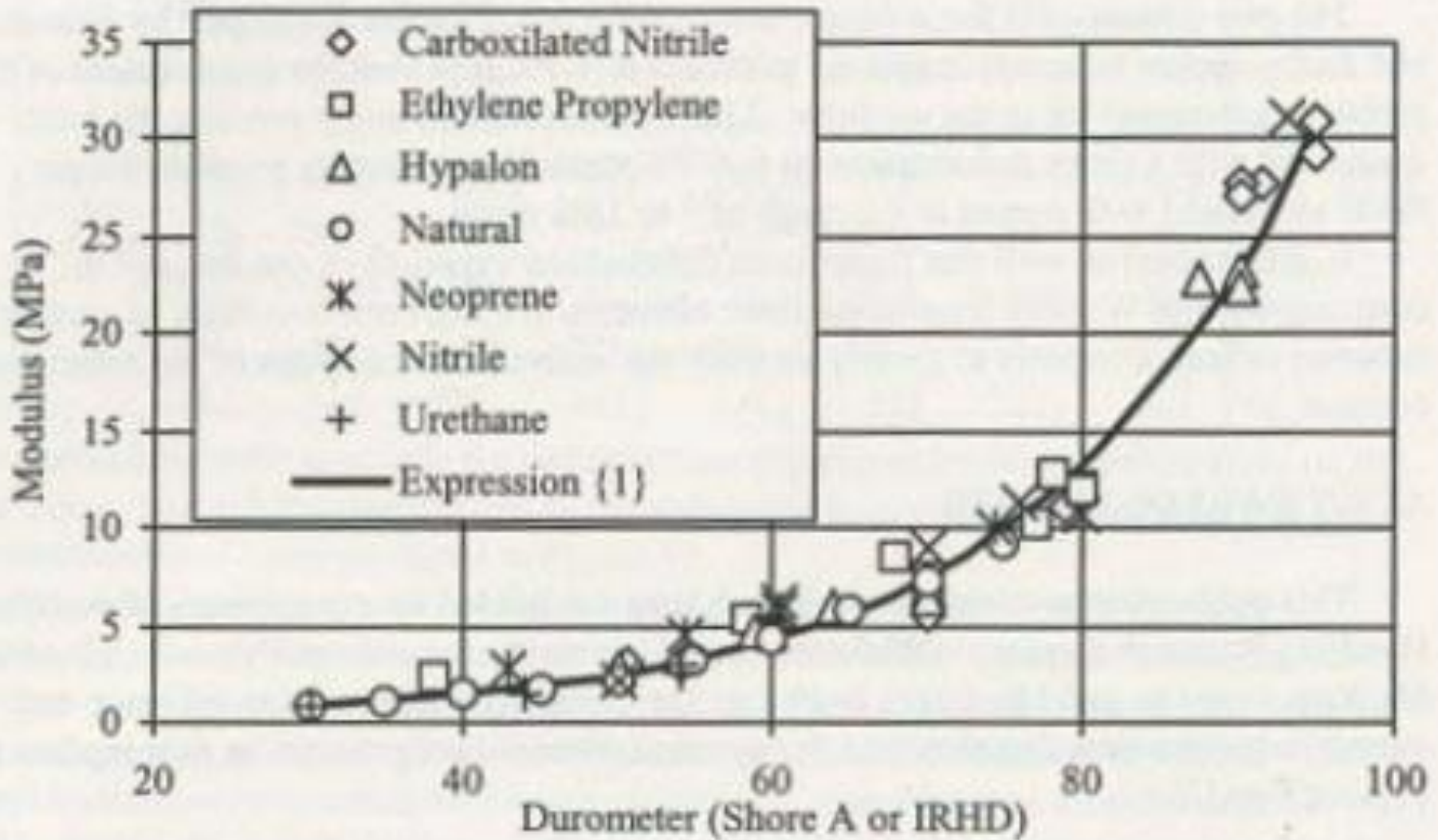
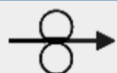


Figure 1 – The Relation between Young's Modulus and Durometer

☐ Metric Units
 ☒ US Units
 ☒ Text
 Rubber Roll Calculator
[Help](#)


Nip Load lbf	503	Shore	80
Rubber OD in	4	Rubber in	.25
Steel OD in	24	Steel ID in	23
Width in	30	Speed ft/min	40
Contact in	0.269	Indent in	0.005
Time msec	33.682	Max P kpsi	0.0798
		Crown mil	0.00

INPUTS**WEB**

poisson's	nu	0.46 --
Shore A	H(SA)	80 --
width	w(web)	30 in

ROLLERS

base roller radius steel	R2	12.0 in
rubber roller outer radius	R1	2.00 in
rubber roller core outer radius	RCO	1.75 in
rubber roller core inner radius	RCI	1.5 in
roller width	w(roller)	30 in

NIPPING

air pressure	P(nip)	40 psi
radius cylinders	r(cyl)	1 in
no. of cylinders	n	4 --
nip/cyl leverage	L(cyl)/L(nip)	1 --
speed	V	300 fpm

OUTPUTS**NIP FORCE / LOAD**

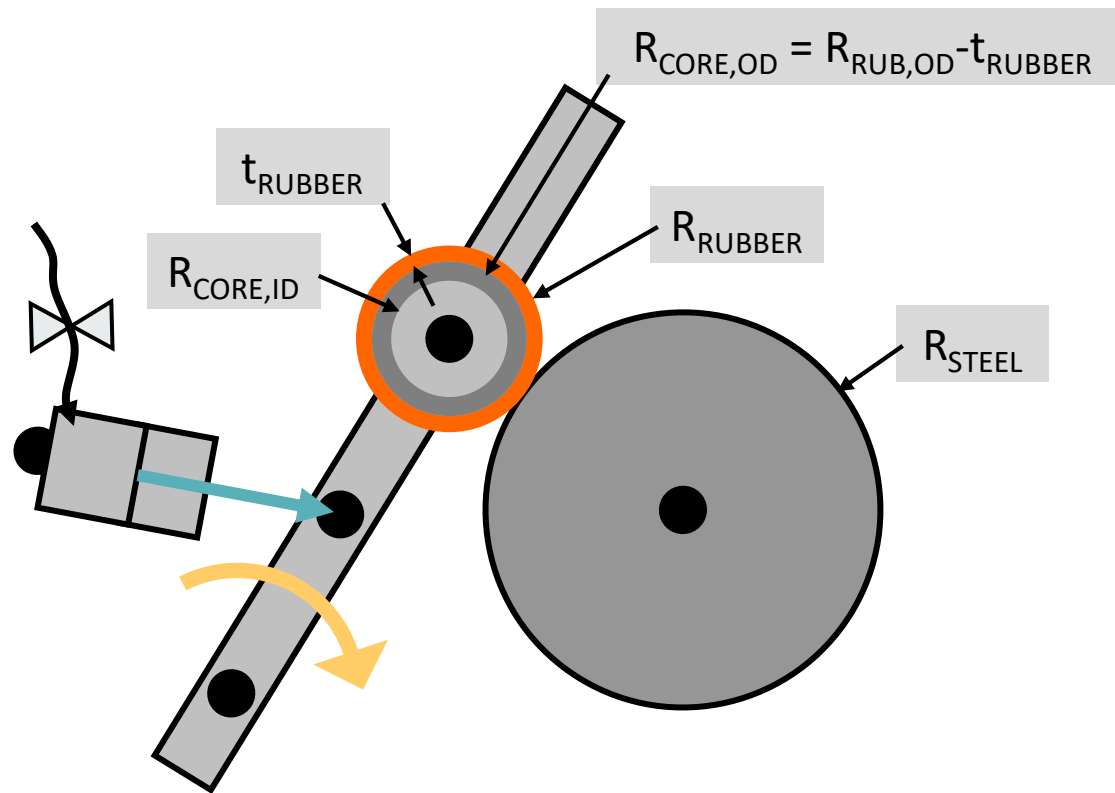
Nip force	F(nip)	503 lbs
Nip load	N(force/width)	16.8 lbs/in

INDENTATION / DEFLECTION

thickness	t(r)	0.25 in
Rubber modulus	Eo	1910 psi
radius equiv	req	1.7 in
indentation	delta	0.0053 in
footprint	b	0.27 in
%indent	$\Delta r/t(\text{rubber})$	2.1%
deflection	y	0.0017 in
deflect/indent	$=y/\Delta r$	33%

NIP PRESSURE / TIME

Pressure	Pavg	62 psi
Nip Time	t(nip)	0.004 s
overspeed	$\varepsilon(\text{theta})$	0.528% --



How is Rubber Hardness Measured?

Pusey and Jones

P&J is named for a ship building company from Delaware, USA.



Courtesy of Bareiss
Prüfgerätebau GmbH



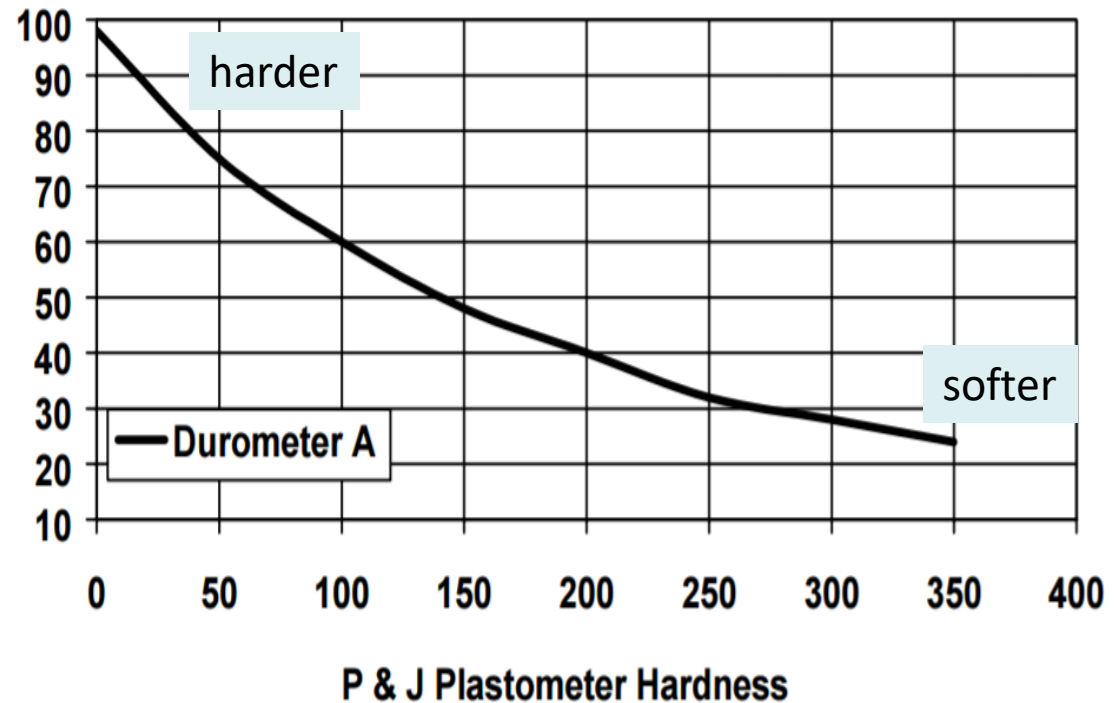
Antique P&J Hardness Tester
(found on Ebay)

Pusey : pronounced pew-sy

Comparing Hardness Scales

For Shore and IRHD, higher numbers are harder.

For P&J, lower values are harder.



Durometer A	100	95	90	85	80	75	70	65	60	55	50	45	40	35	30
BS Hardness	0	10	19	25	34	42	45	65	75	90	100	120	142	175	210
IRHD	100	95	90	85	80	74	68	64	62	54	49	44	39	35	28
Shore & Mason	100	98	96	95	94	93	92	90	85	80	78	72	65	52	40
P & J	0	10	25	32	43	58	70	86	100	118	130	132	155	220	
ASTM Tinius Olsen	0	5	10	18	22	30	39	44	55	68	80	100	112	145	172
Adams Densimeter	0	5	10	16	21	27	34	40	47	55	64	75	85	100	
Zet (Frank - Zwick)	0	7	12	17	25	35	45	53	61	72	85	100	115	140	180
Cussons	100	97	94	90	86	81	77	73	69	65	62	56	45	32	20

Rubber Performance vs Speed, Temperature

$$E_c = \left(1.375636 \ln(\dot{\epsilon}) + 29.29927 \right) \cdot e^{[0.0536 \cdot (IRHD - 0.101651 \cdot (Temp - 20))]}$$

$$(20^{\circ}C \leq Temp \leq 100^{\circ}C ; 30IRHD \leq IRHD \leq 90IRHD ; 0.002 \text{ in/in/sec} \leq \dot{\epsilon} \leq 0.2 \text{ in/in/sec})$$

Rubber Strain Rate

$$\dot{\epsilon} = \frac{\frac{\delta_{\max}}{t}}{\frac{a}{V}}$$

Temp C	Temp F	V fpm	V mpm	a/V 1/s	e-dot in/in/s	E(rub) 70A
20	68	0.3	0.1	4.1667	0.0096	976
20	68	3	0.9	0.4167	0.096	1111
20	68	30	9.2	0.0417	0.96	1246
20	68	300	92	0.0042	9.6	1381
40	104	300	92	0.0042	9.6	1238
60	140	300	92	0.0042	9.6	1110
80	176	300	92	0.0042	9.6	996
100	212	300	92	0.0042	9.6	893

At higher speeds, rubber is 40% stiffer.

At higher temperature, rubber is 40% less stiff.

$\delta_{\max}$ = maximum deflection, mm or in

t = covering thickness, mm or in

a = half footpring (b/2), mm or in

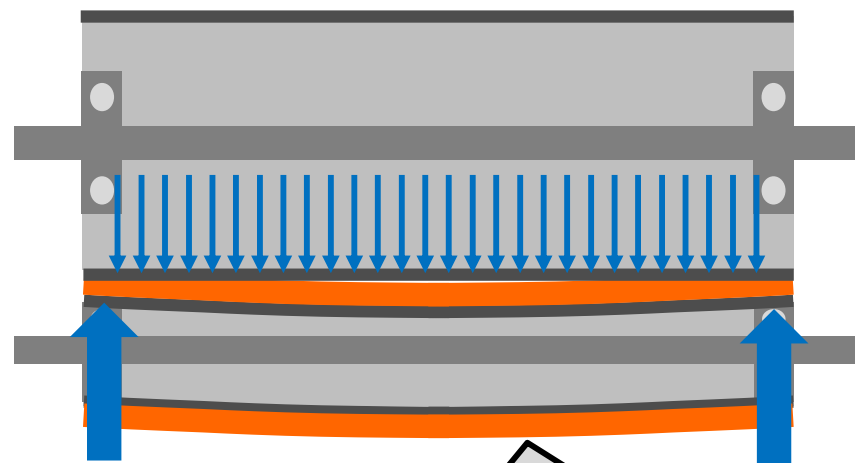
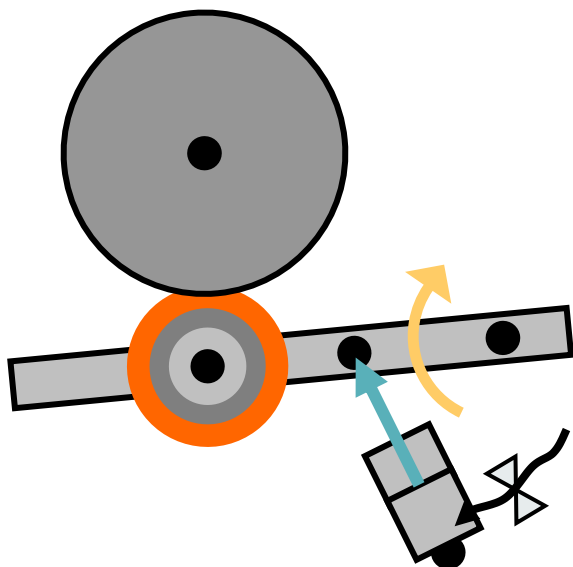
V = velocity, mm/s or in/s

A hot fast nip will behave like a slow room temperature nip.

Optimize a Nipped Process

- ▶ Nip force from Pressure and Geometry
- ▶ Effective width
- ▶ Nip footprint
- ▶ Nip average pressure
- ▶ Nip residence time
- ▶ Rubber covering indentation
- ▶ Roller deflection
- ▶ Compare indentation to deflection
- ▶ Compensate for deflection, if needed

Nip Deflection Compensation



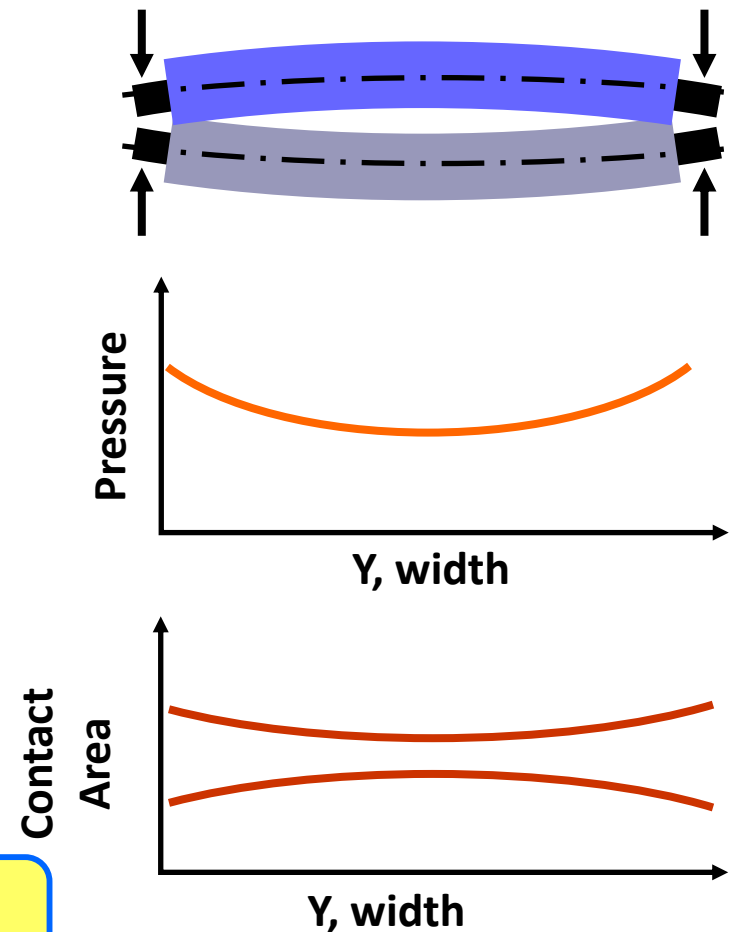
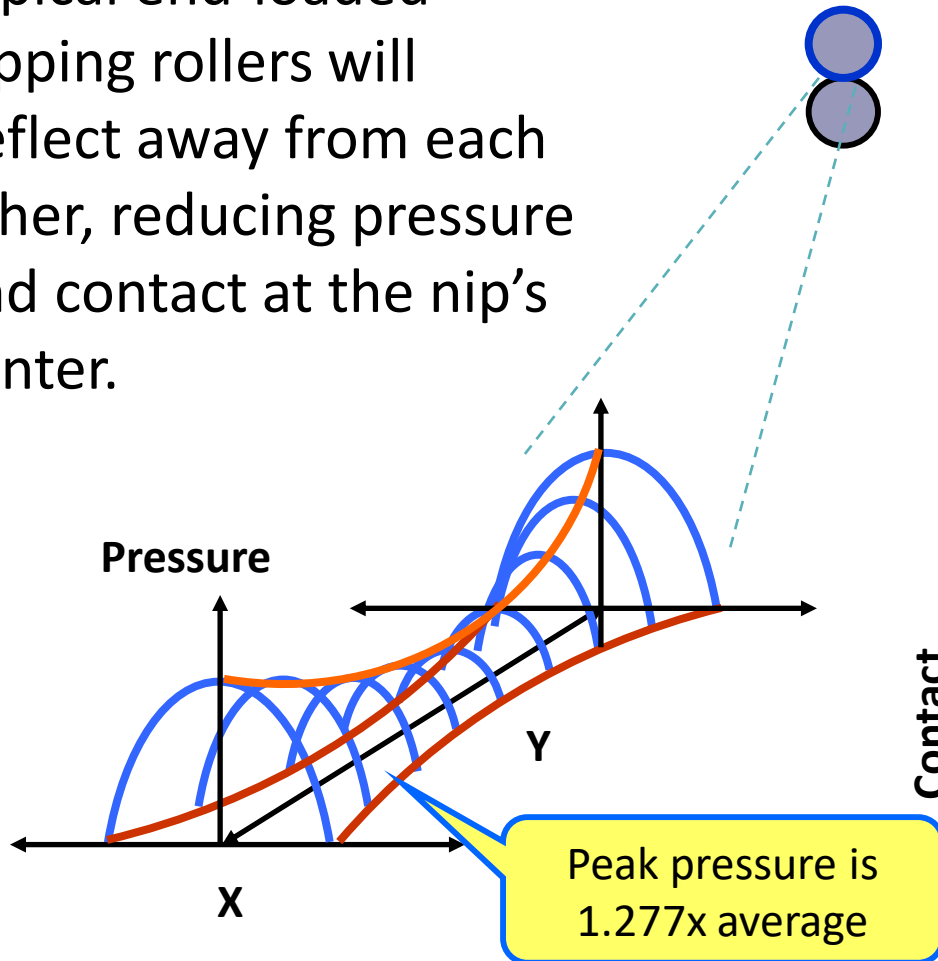
Roller bending from distributed load and end support

Bookshelf bending from distributed load and end support



Nip Pressure vs. Width

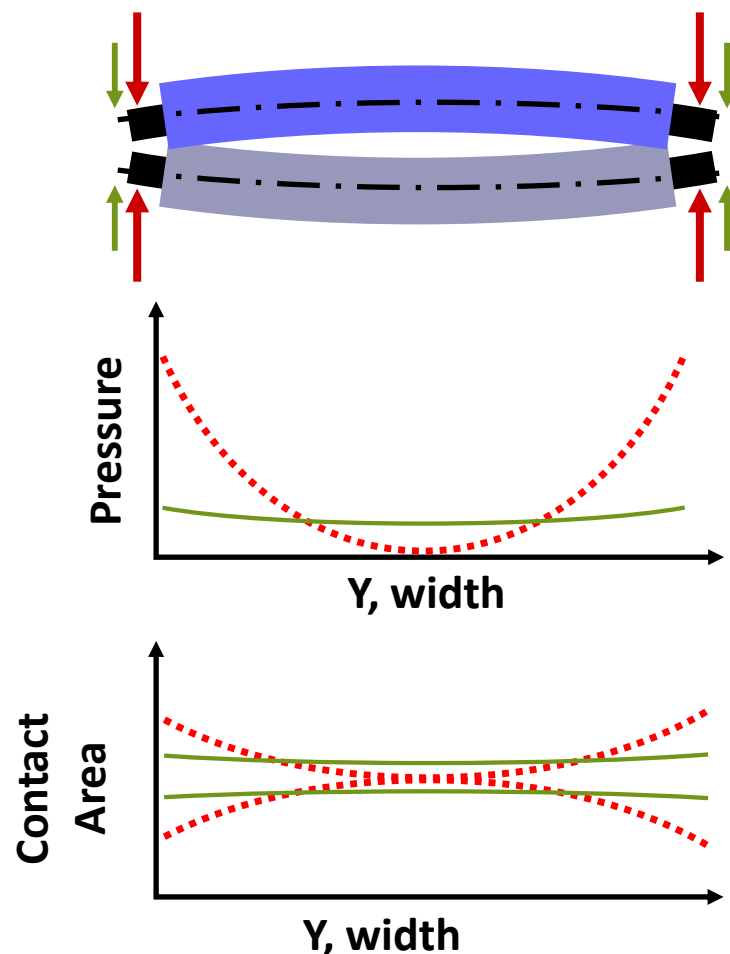
Typical end-loaded nipping rollers will deflect away from each other, reducing pressure and contact at the nip's center.



Nip Variations vs. Load

Nip deflection is directly proportional to load.

Excessive nip loads lead to increase variation and associated problems.

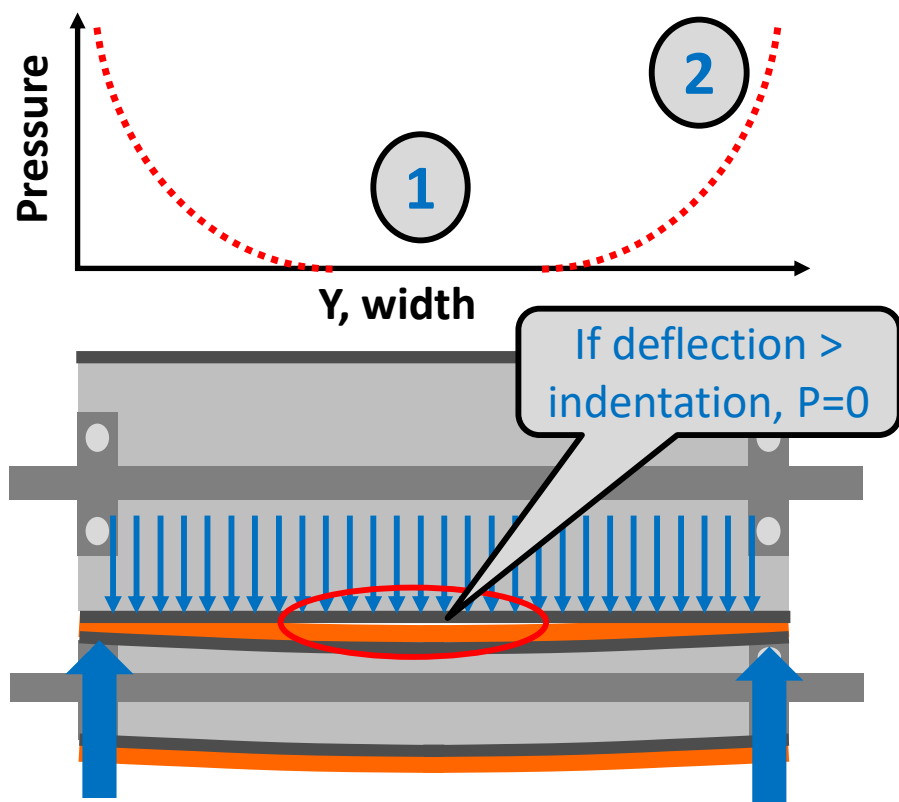


Nip Deflection Compensation – Thicker Rubber

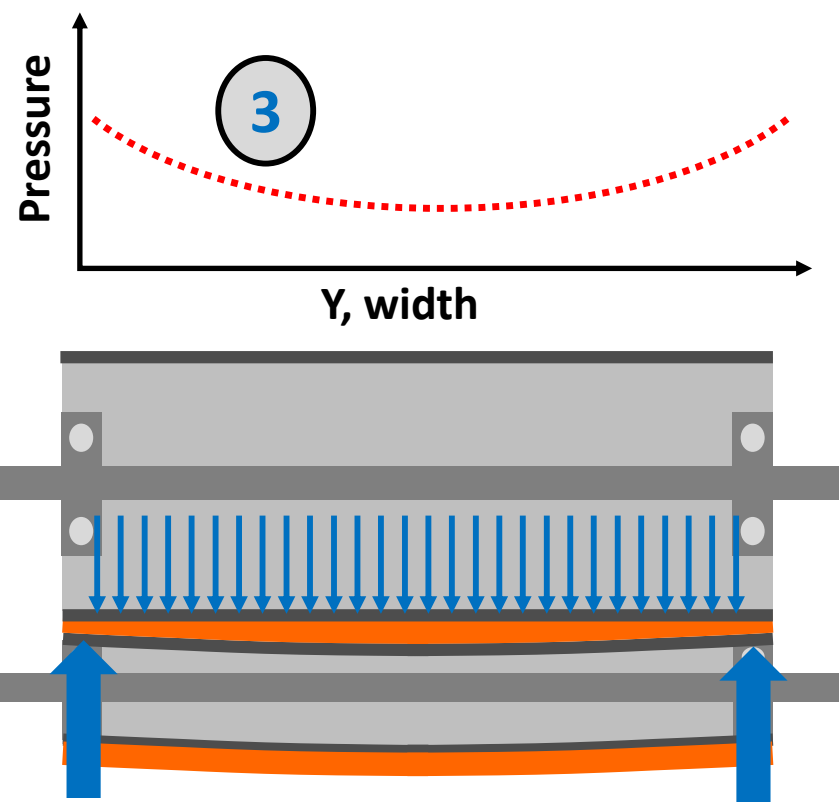
1: No rubber indentation, no pressure.

2: More indentation, more pressure.

3: Thicker rubber, less percent indentation, lower pressure, less pressure variations.

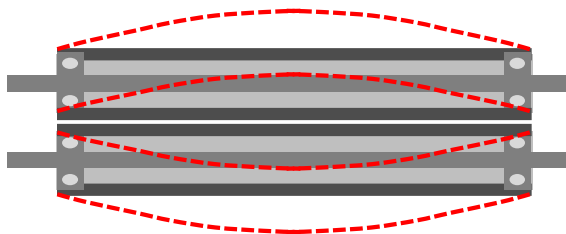


Thinner Rubber Covering



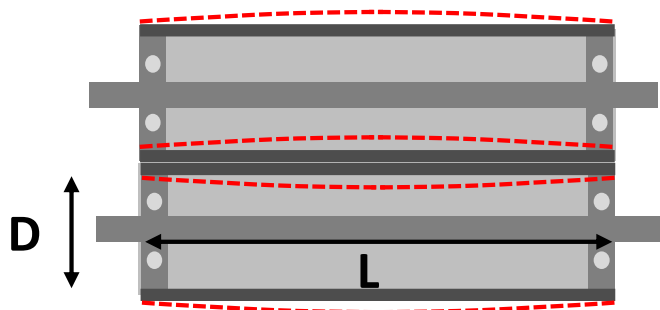
Thicker Rubber Covering

Nip Deflection Compensation

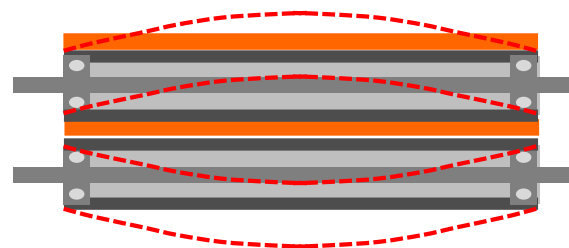


L/D too large:
Large deflection
under load.

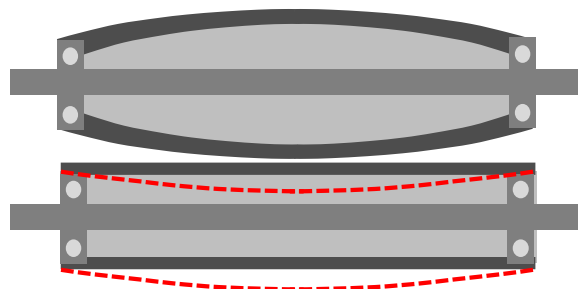
Increase Roller Diameter



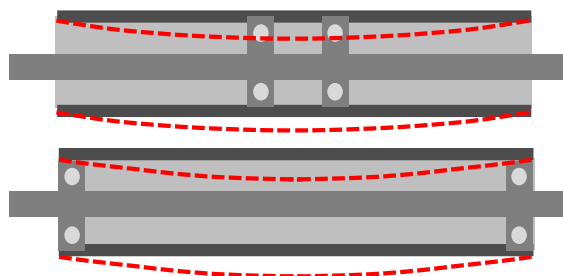
Add Compliant Covering



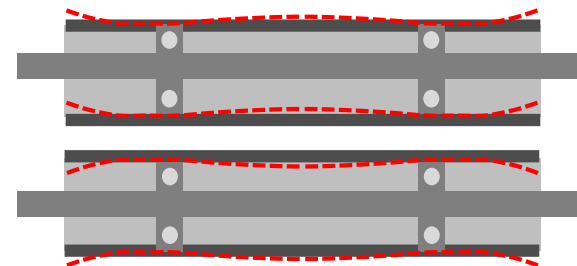
Crowning



Mirrored Deflection

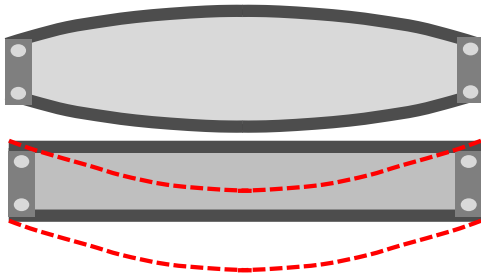
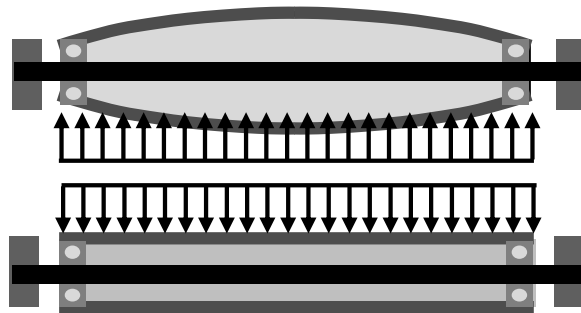


Airy Point Support

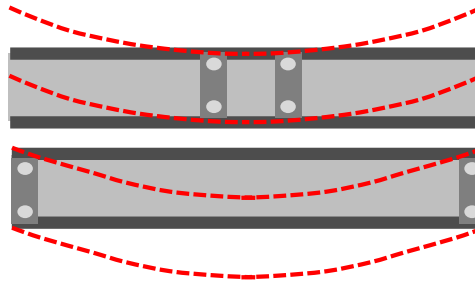
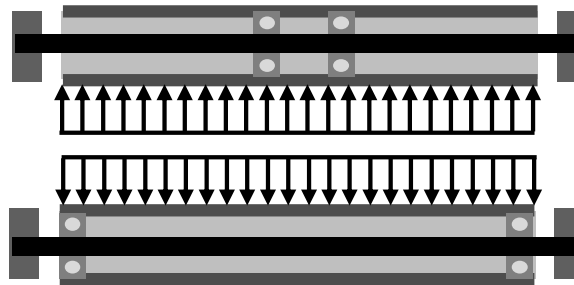


Minimizing Pressure Variation from Deflection

Crowned roller designed to match deflection.

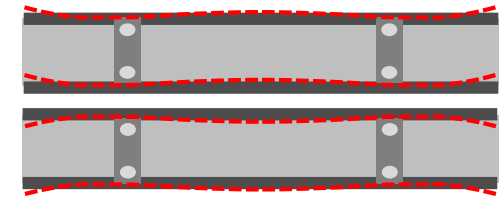
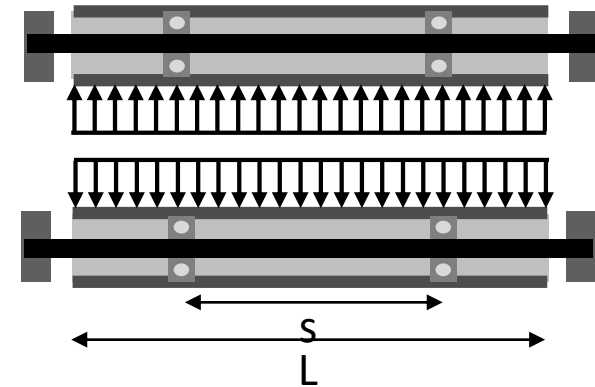


Centrally-supported roller nipped to end-supported roller.



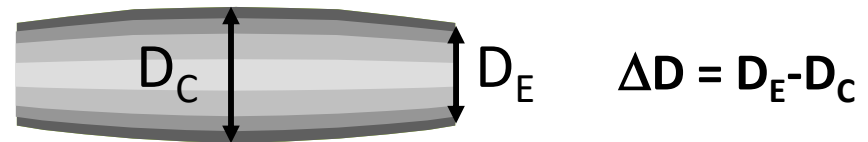
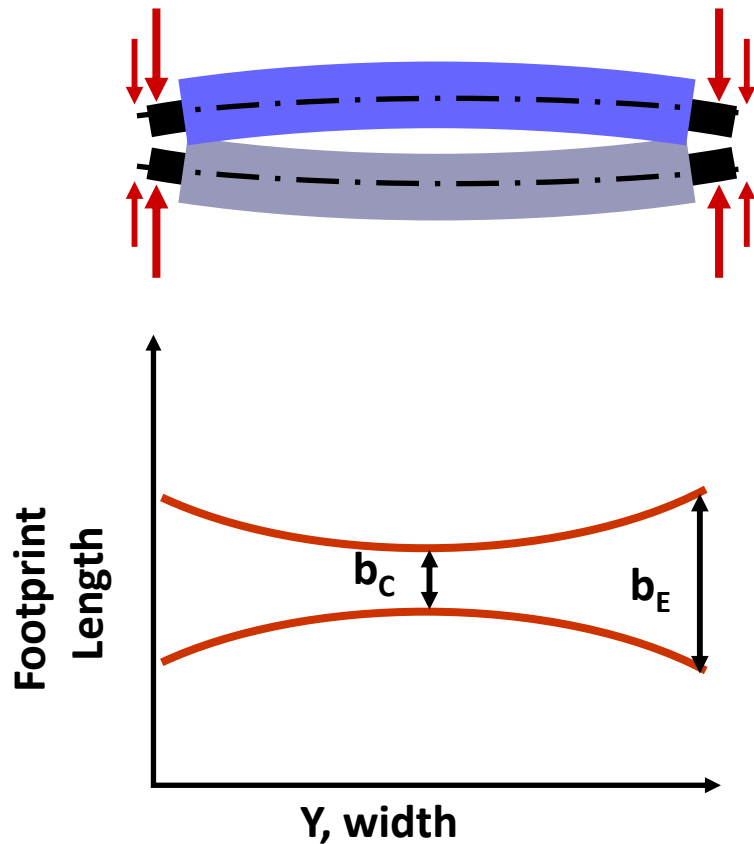
Nipped rollers supported at Airy Points.

$$s = \frac{L}{\sqrt{3}}$$



Calculating Crown from Footprint

Solution to uneven footprint is crowning one roller.



$$\Delta D = \frac{(b_E^2 - b_C^2)(D_1 + D_2)}{2D_1D_2}$$

ΔD = Diameter change needed (crown)

b_E = Footprint length at edge

b_C = Footprint length at center

D_1 = Diameter of roller 1

D_2 = Diameter of roller 2

Recommendation #9

Default should be cylindrical roller.

Crowned rollers are effective,
but should be reserved for extreme high
pressure processes.

Why? Crowned rollers compensate for deflection under a narrow load range, otherwise they create pressure variations. Larger diameter should be the first design option to fight deflection.



Recommendation #8

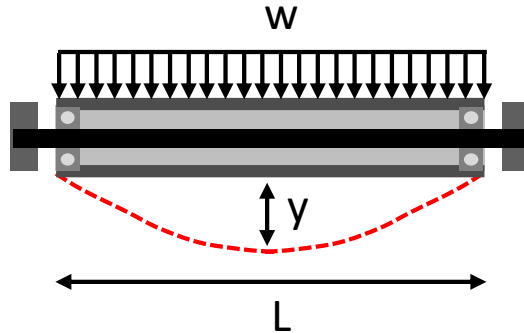
Keep deflection under process conditions to be less than 20 percent of rubber roller indentation (2-6 mils).

Why? Compliant rollers are designed to forgive many sins of variation – deflection is only one sin.



Shell Deflection and Support Position

**Simple Support Beam
w/ Uniform Distributed Load**

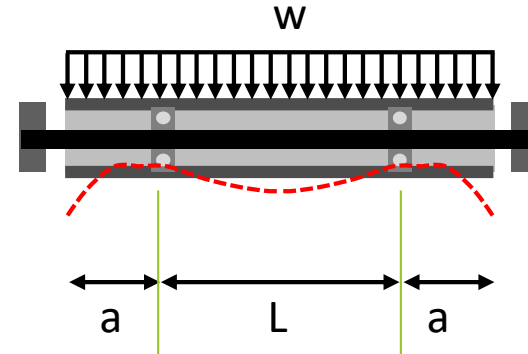


Deflection at the Center:

$$y_c = \frac{5WL^3}{384EI} = \frac{5wL^4}{384EI}$$

$$W = wL$$

**Simple Support Beam
w/ Uniform Distributed Load
and Symmetrical Overhanging Ends**



$$y_c = \frac{WL}{384EI} (5L^2 - 24a^2)$$

(These equation are the same if $a = 0$)

y_c = deflection at beam center, E = elastic modulus, I = bending moment of inertia,
 L = distance between simple supports, a = overhang length, w = load per length,
 W = total load (equal to $w \times l$), Total roller length = $(L+2a)$

Anti-Deflection Roller Design (1953)

Sept. 8, 1953

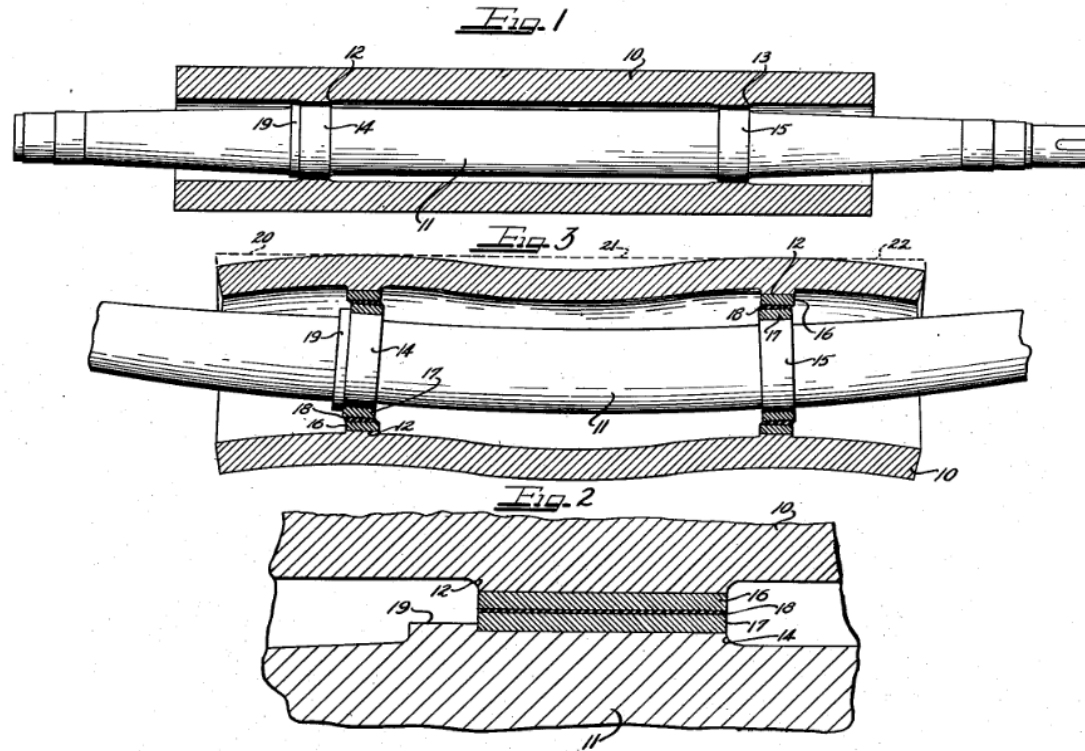
L. HORNBOSTEL

2,651,103

ANTIDEFLECTION ROLL ASSEMBLY

Filed April 9, 1948

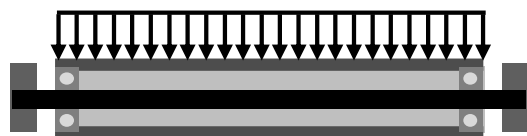
3 Sheets-Sheet 1



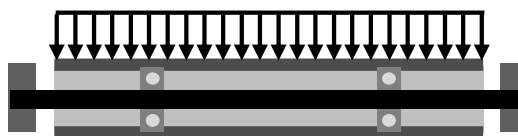
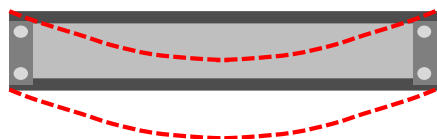
Inventor
LLOYD HORNBOSTEL

By The Firm of Charlesworths Attys.

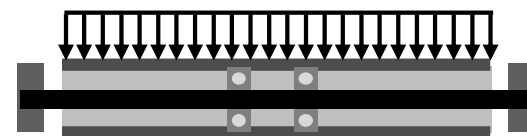
Shell Deflection and Bearing Position



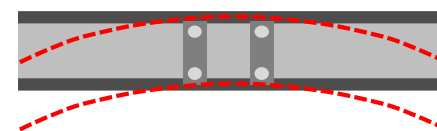
60-in.



34-in.

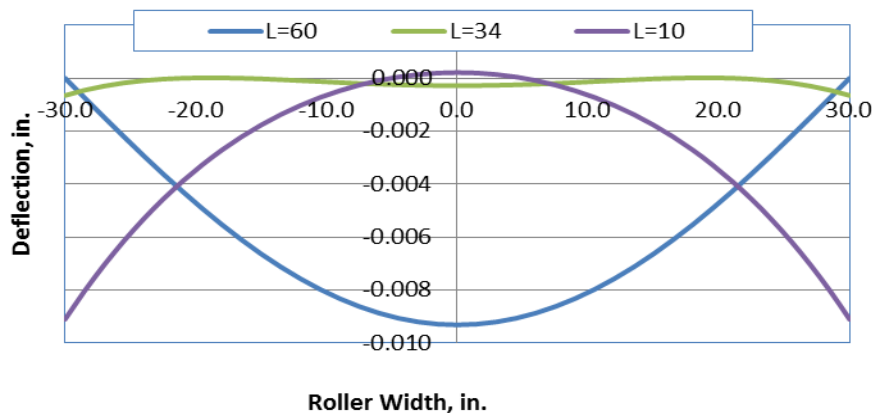


10-in.



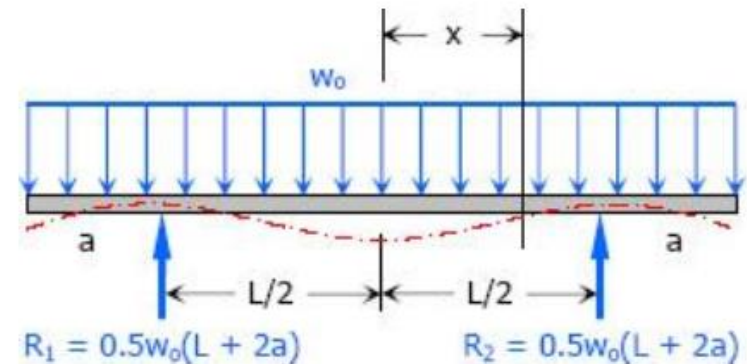
Red dashed lines show direction and magnitude of deflection from straightness.

Roller Deflection vs. Support Location



Anti-Deflection Roller Design

Uniform distributed load	W	10 lbs/in
Overall length	LT	60 in
Length between supports	LL	36 in
Length between support and beam end	AA	12 in
Young's modulus of elasticity	E	1.00E+07 psi
Bending inertia	I	18.70 in ⁴
Cylinder ID	DI	5.5 in
Cylinder OD	DO	6 in

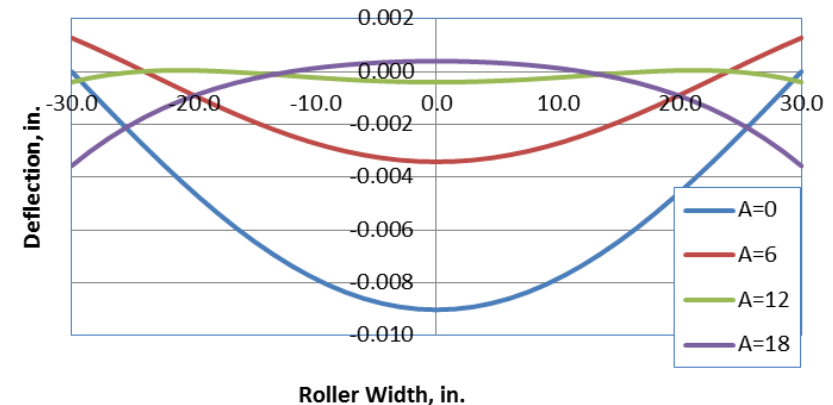


$$I_x = \pi (do^4 - di^4) / 64$$



- Deflection under uniform load is reduced 13X by moving the supports from the ends to Bessel points (22% from end).
- Doubling diameter reduces deflection by more than 8X.
- Changing from aluminum to steel reduces deflection by 3X.

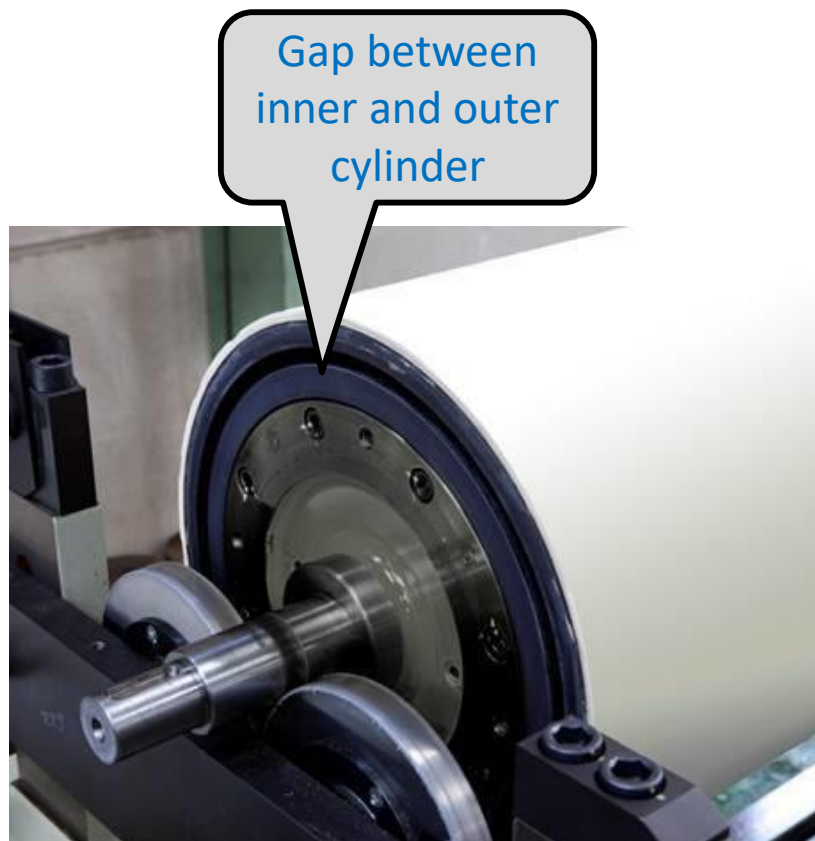
Roller Deflection vs. Support Location



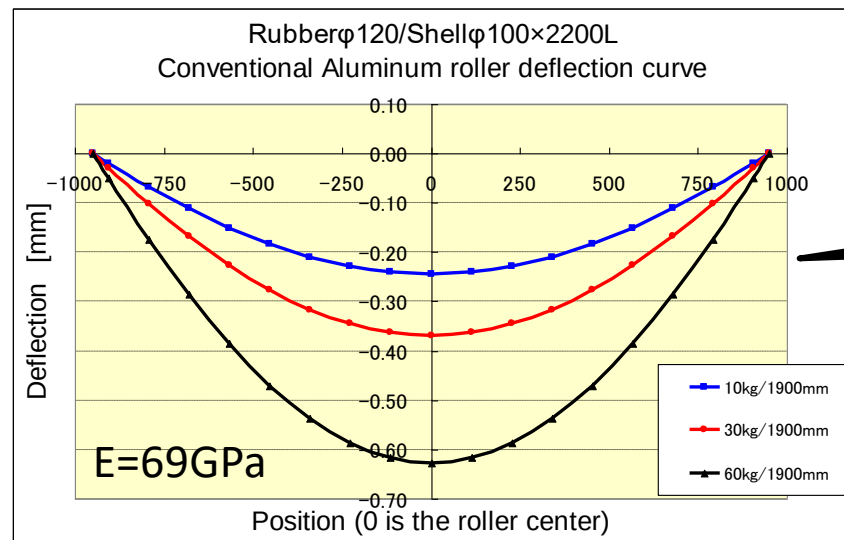
$$EIy(x) = \frac{1}{8}w(L + 2a)Lx^2 - \frac{1}{24}wx^4 - \frac{1}{16}w(L + 2a)^2x^2 + \frac{1}{384}wL^2(24a^2 - 5L^2)$$

Example: Double-Pipe Roller

Courtesy of Mitsubishi Chem.
Carboleader® Roller
φ342X9000mm CF Double Pipe Roller



Comparing Deflections by Roller Design



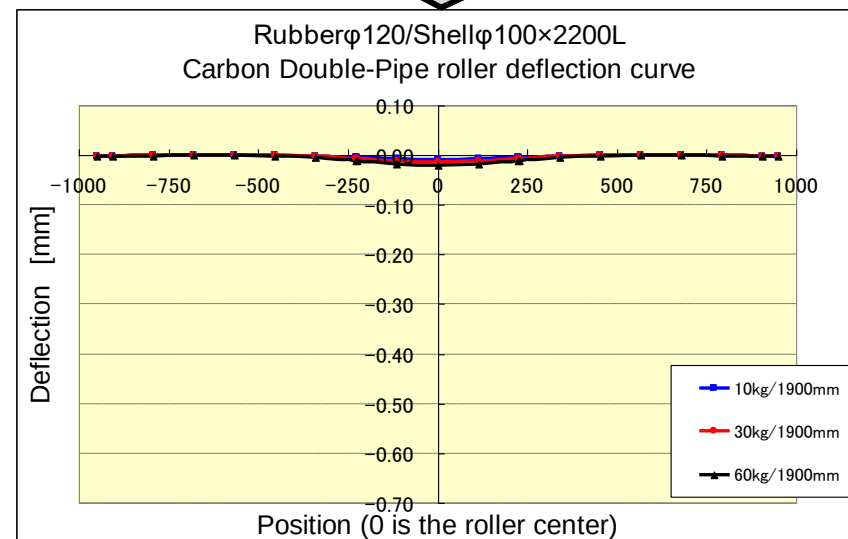
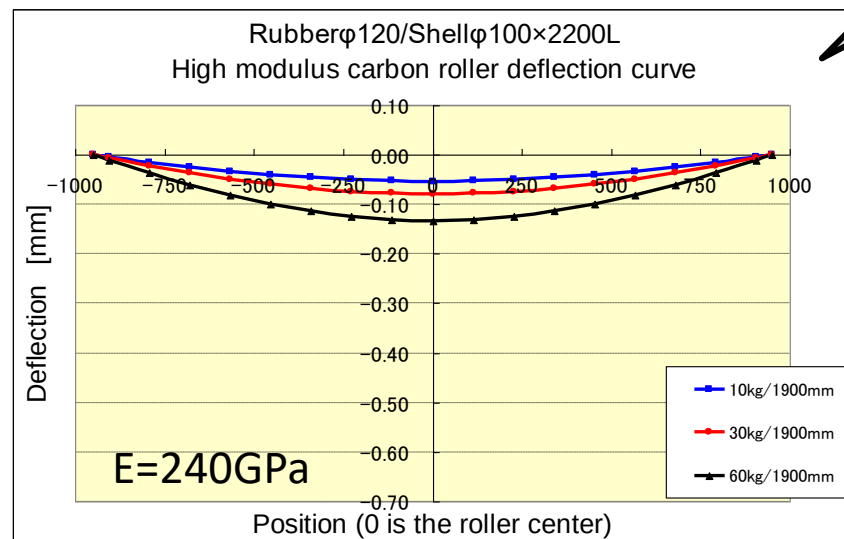
0.3, 0.9, 1.8 lbs/in over 75-in width

50, 150, and 300 N/m over 1.9m width

Aluminum Cylinder (single pipe)

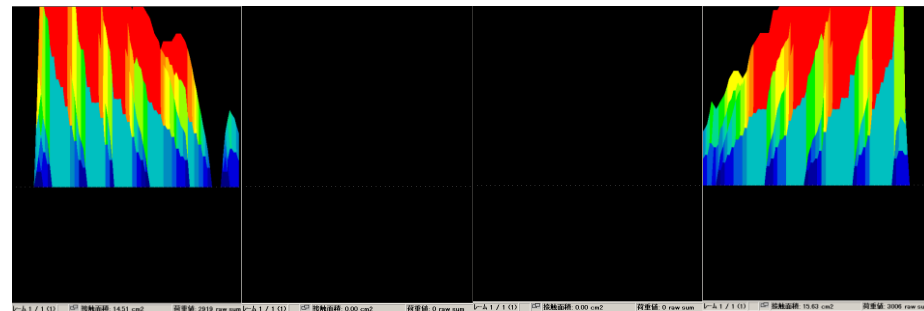
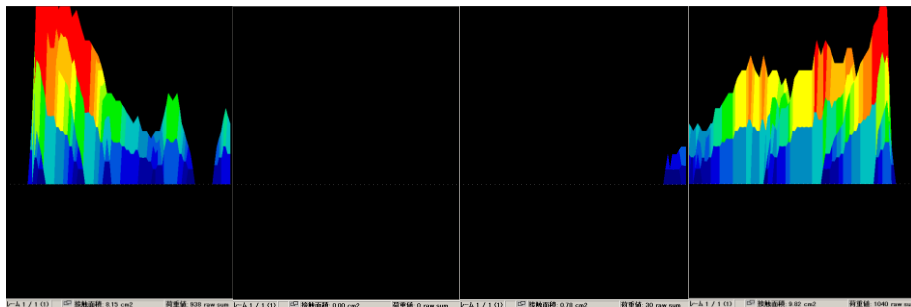
Carbon Fiber Cylinder (single pipe)

Carboleader® (double pipe)

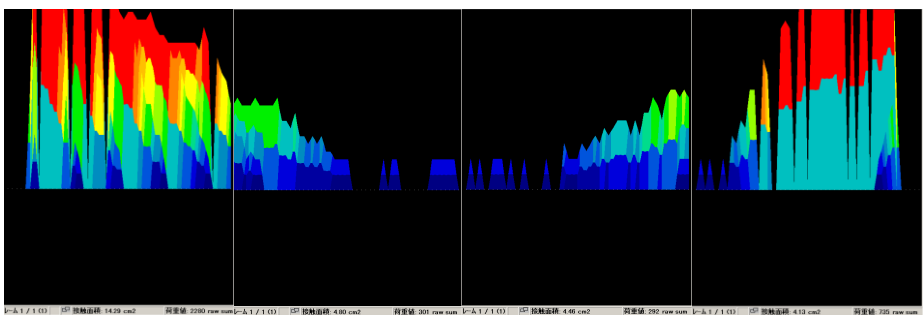
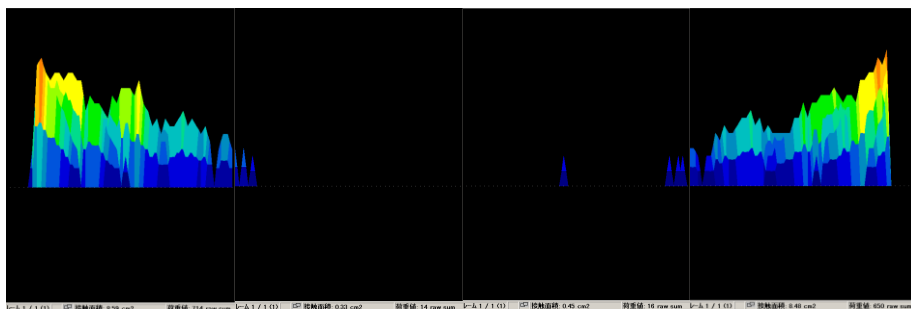


Pressure Mapping Low vs High Loads

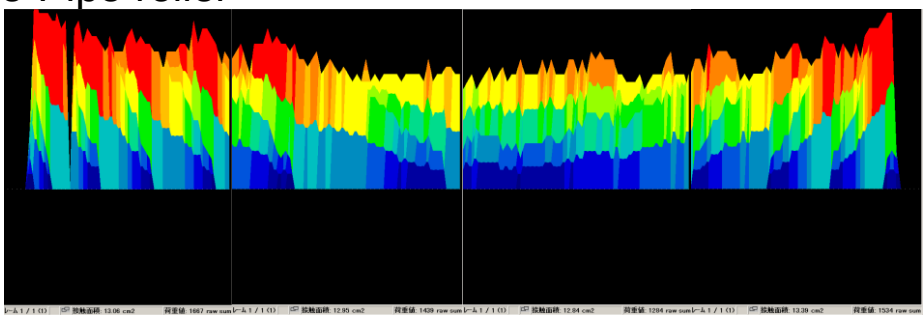
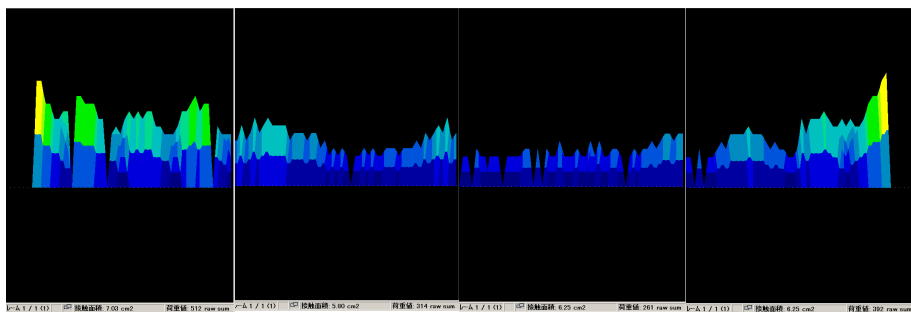
Aluminum roller (E=69GPa)



Carbon roller (E=240GPa)



Carbon Double-Pipe roller



Indentation vs Deflection, Pressure x Time?

INPUTS

WEB

poisson's	nu	0.46 --
Shore A	H(SA)	80 --
width	w(web)	30 in

ROLLERS

base roller radius steel	R2	12.0 in
rubber roller outer radius	R1	2.00 in
rubber roller core outer radius	RCO	1.75 in
rubber roller core inner radius	RCI	1.5 in
roller width	w(roller)	30 in

NIPPING

air pressure	P(nip)	40 psi
radius cylinders	r(cyl)	1 in
no. of cylinders	n	4 --
nip/cyl leverage	L(cyl)/L(nip)	1 --
speed	V	300 fpm

OUTPUTS

NIP FORCE / LOAD

Nip force	F(nip)	503 lbs
Nip load	N(force/width)	16.8 lbs/in

INDENTATION / DEFLECTION

thickness	t(r)	0.25 in
Rubber modulus	Eo	1910 psi
radius equiv	req	1.7 in
indentation	delta	0.0053 in
footprint	b	0.27 in
%indent	$\Delta r/t(\text{rubber})$	2.1%
deflection	y	0.0017 in
deflect/indent	$=y/\Delta r$	33%

NIP PRESSURE / TIME

Pressure	Pavg	62 psi
Nip Time	t(nip)	0.004 s
overspeed	$\varepsilon(\text{theta})$	0.528% --

Deflection must be less than indentation. Prefer deflect/indent ratio < 20%

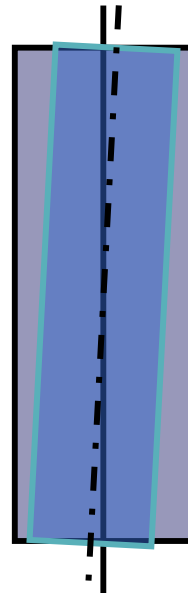
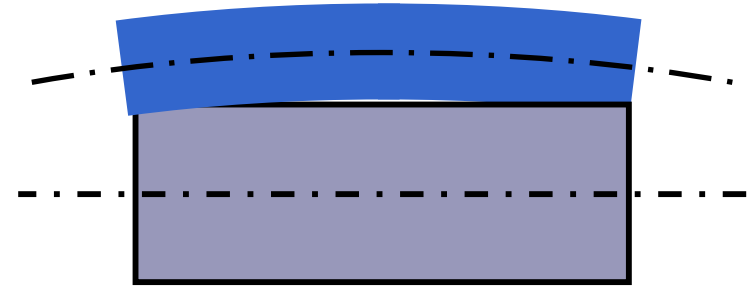
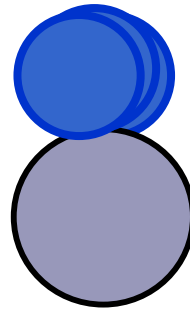
Pressure and time requirements are dependent on process needs.
Is time important? Sometimes high pressure-short time = low pressure-long time

Nip Deflection Compensation

3. Skewed Roller Axes

Imagine a banana
wrapped around a
cylinder...

Normally, I advise
against intentionally
skewing any roller, but...



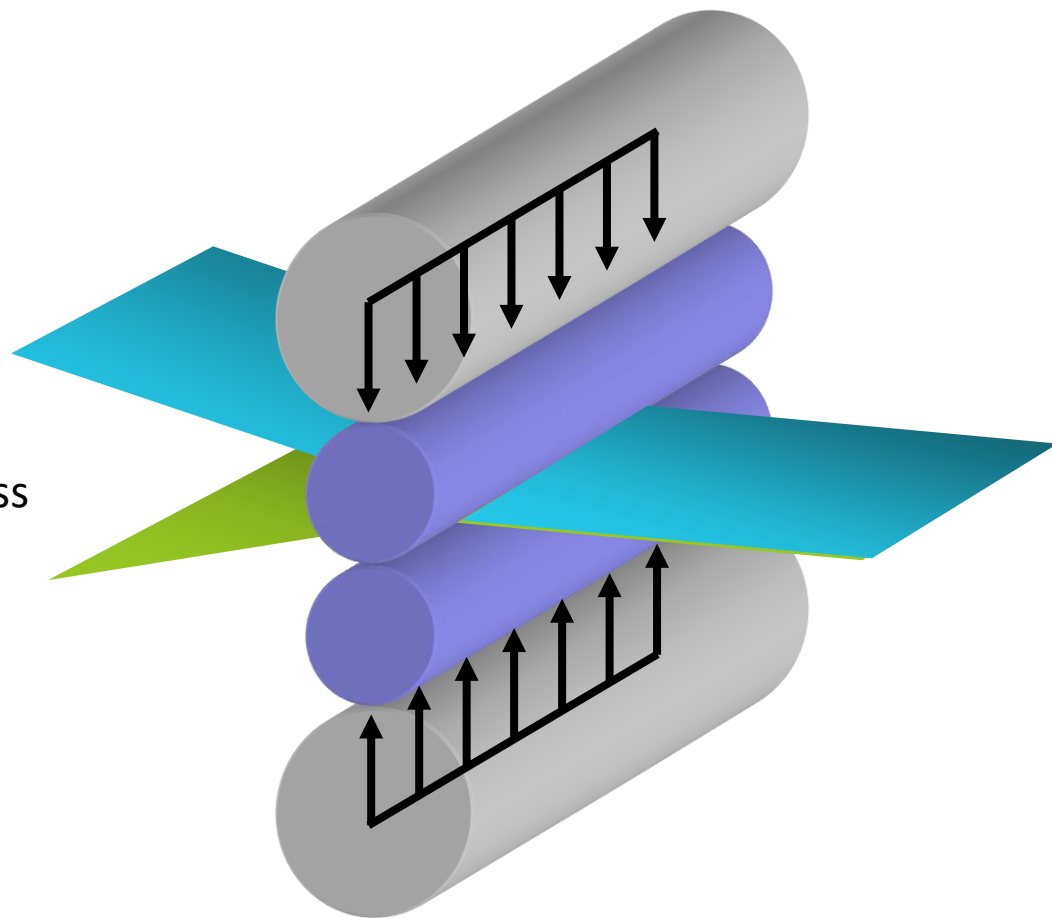
Nip Deflection Compensation

6. Roller Stacks

End-loaded roller nips are more prone to deflection.

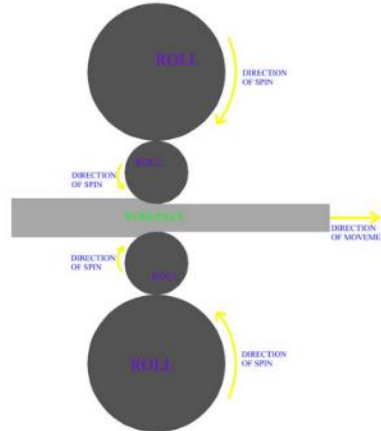
Roller stacks distribute load across the width, greatly reducing deflection.

Nip stacks are commonly used in high load nips.

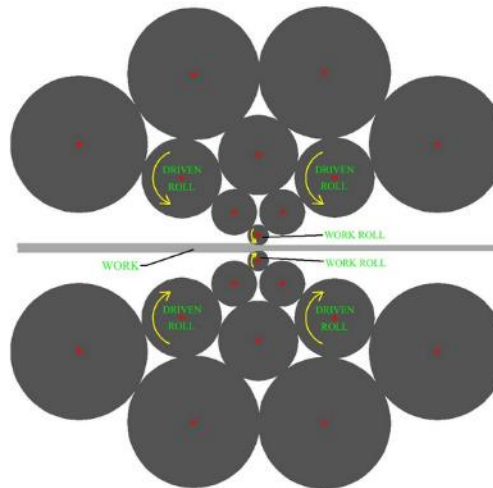


Ultra High Load Nipped Processes

Four High Mill



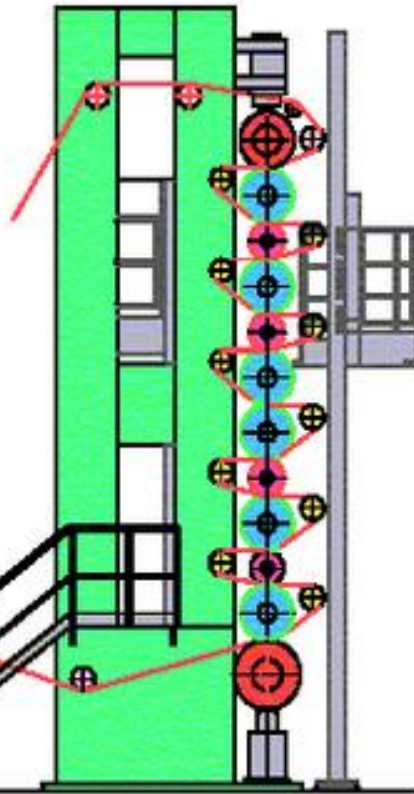
Sendzimir Mill



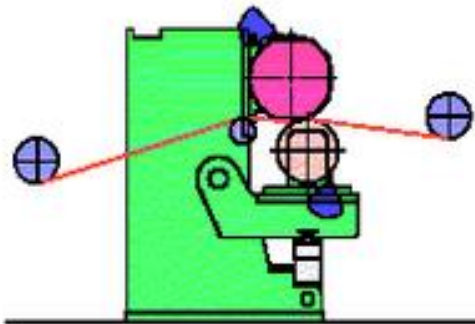
http://thelibraryofmanufacturing.com/metal_rolling.html

Calendering and Super Calendering

Super Calender Machine

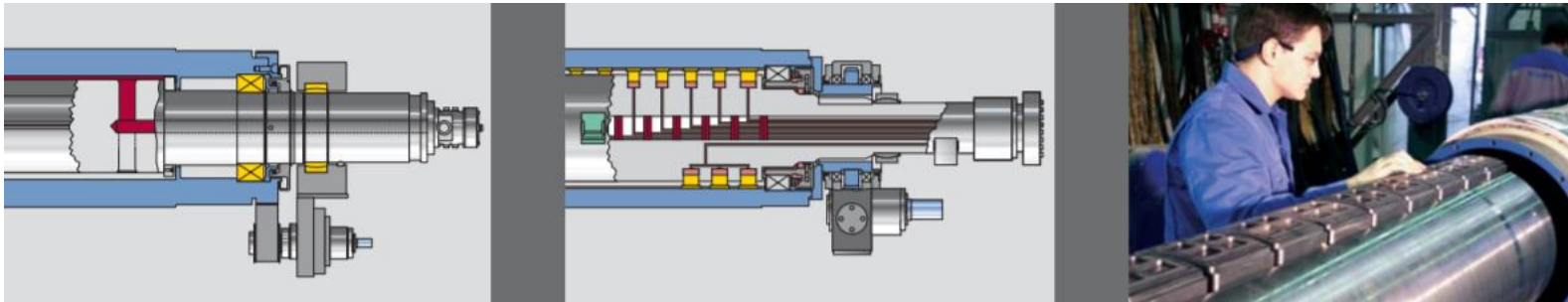


Calender Machine

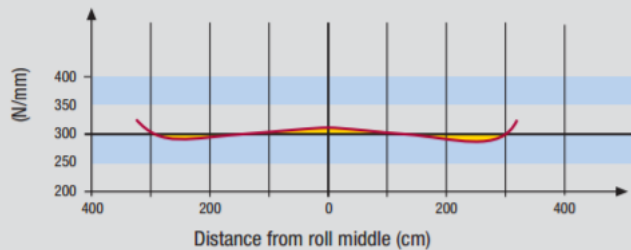


Courtesy of Valmet

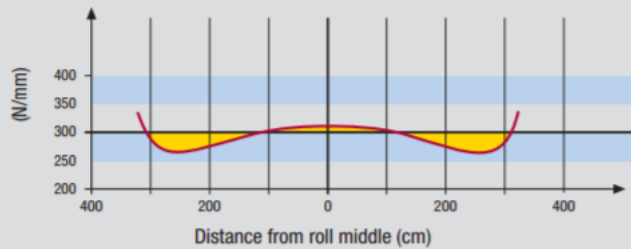
Swimming Rollers – Voith



Line load with Econip™ X roll



Line load with standard Econip™ roll



Technical data	Ecosoft™	NipcoFlex™
Speed	Up to 1,500 m/min	Up to 1,500 m/min
Number of nips	1 – 4	1
Shoe length	–	Up to 250 mm
Line load	Up to 500 N/mm	Up to 1,200 N/mm
Roll surface temperature	220 °C	250 °C
Heating	inductive	inductive and additional Calcoil
Additional extras	Steam moistener, water spray device and infrared preheating	
Paper roll width	640 mm	640 mm
Roll diameter	1,200 mm	1,200 mm
Core diameter	76 or 150 mm	76 or 150 mm

Swimming Rollers - Valmet

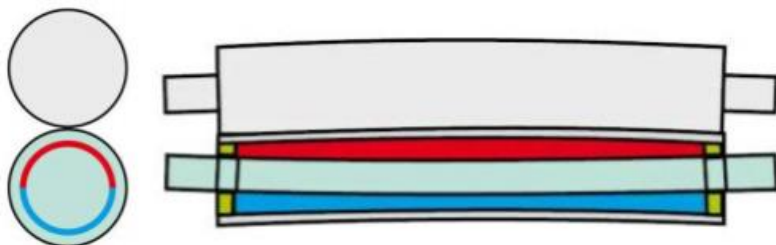


Fig. 14 Swimming roll (Sym roll)

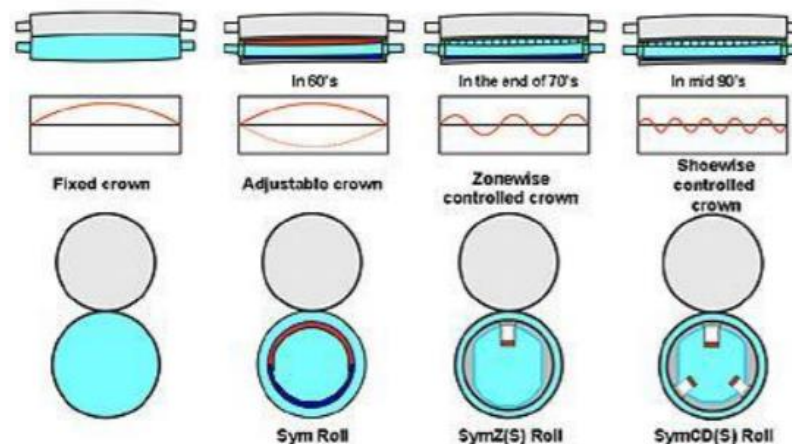


Fig. 19 Schematic overview of DC rolls with different functions

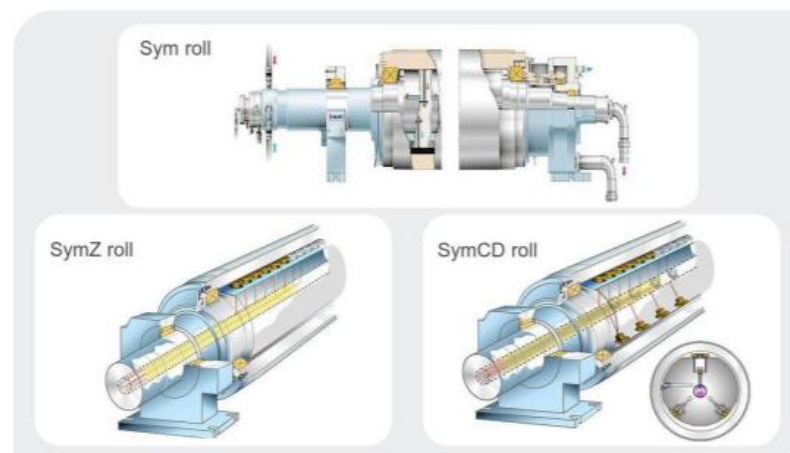


Fig. 20 Structural overview of DC rolls with different profiling functions

Valmet's recent calender technologies

High bulk calendering, OptiCalender Compact and effective machine calender rebuilds -

Hidehiko Yamazaki

Valmet K.K.

Jari Ilomäki and Ville Tiistola

Valmet Technologies, Inc.

Additional Nipped Roller Considerations

- ▶ Nip Measurements (Footprint, Pressure Profile)
- ▶ Rubber Compounds / Properties
- ▶ Driving Nipped Rollers and Tension Control
- ▶ Web Strain, Feed Rate, and Draw Control
- ▶ Lubrication (gas or liquid)
- ▶ Traction
- ▶ Web Path Geometry
- ▶ Wrinkling, Baggy Webs, Lateral Motion
- ▶ Lamination and Curl

Example of Rubber Compounds

From Menges Roller:

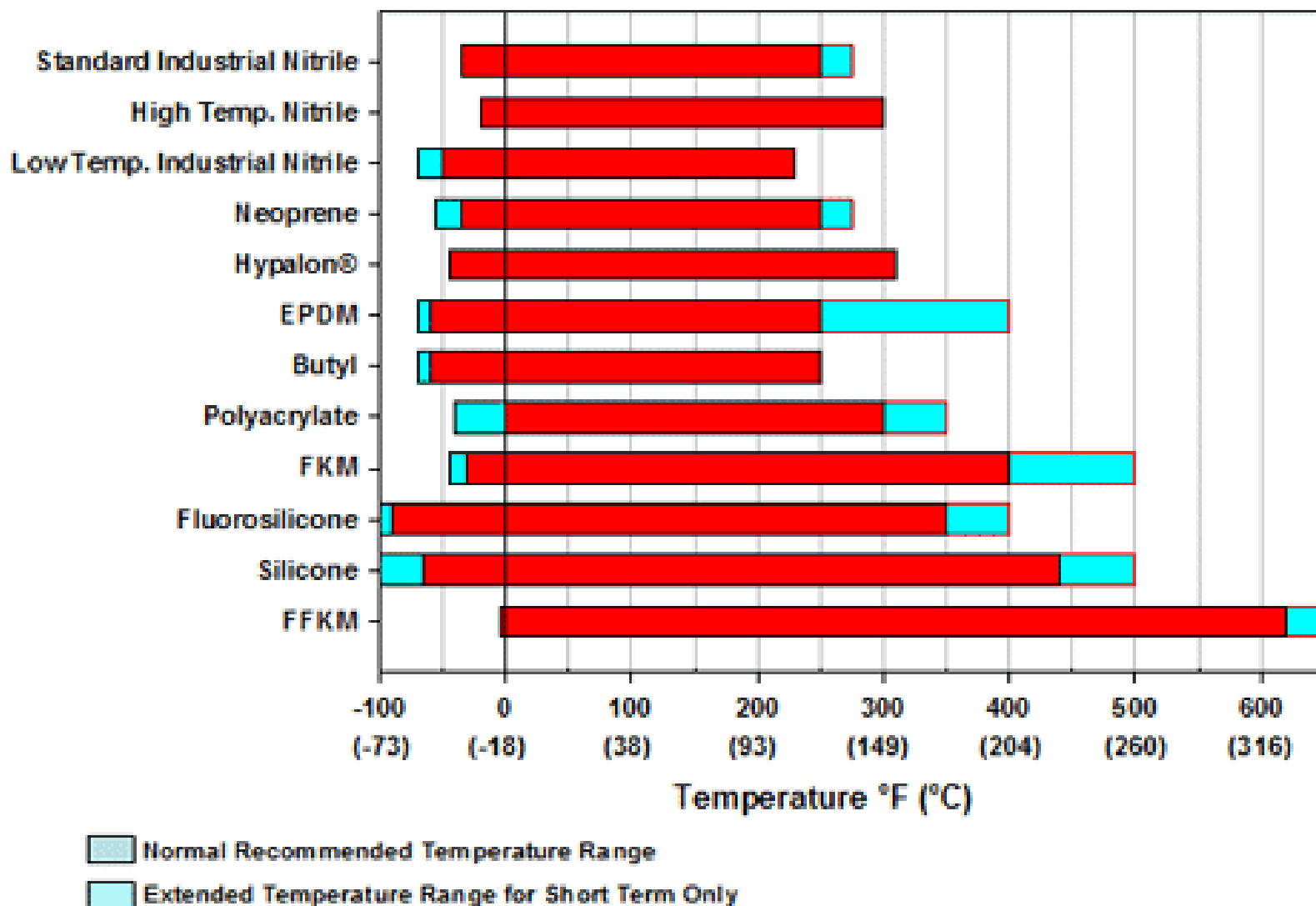
- Thermal Plastic Urethanes (TPUs)
- Hypalon & Hypalon Blends (for Corona Treater purposes)
- Nitrile (BNR/BUNA)
- Neoprene
- EPDM
- EPDM-Silicone Blends
- Silicone (many High-Release & Heat-Resistant options)
- Viton
- Hypalon
- Teflon
- Natural and Synthetic Rubbers
- Anti-Static Blends
- FDA-Approved Blends (for Food & Pharmaceutical uses)

Rubber compound selection is complicated decision. The most common answer is “What has your used before and did it work?”

Important rubber compound properties:

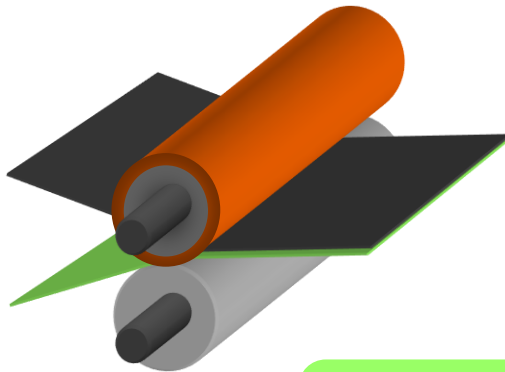
- Temperature range
- Solvent or water resistance
- Ozone resistance
- Wear resistance
- Cut resistance
- Electrical resistance
- Surface roughness

Rubber Compounds and Temperature



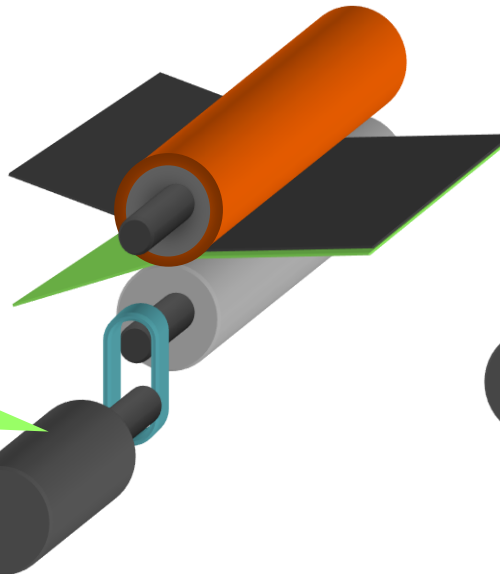
Courtesy of www.jjshort.com

Laminating Drive Options

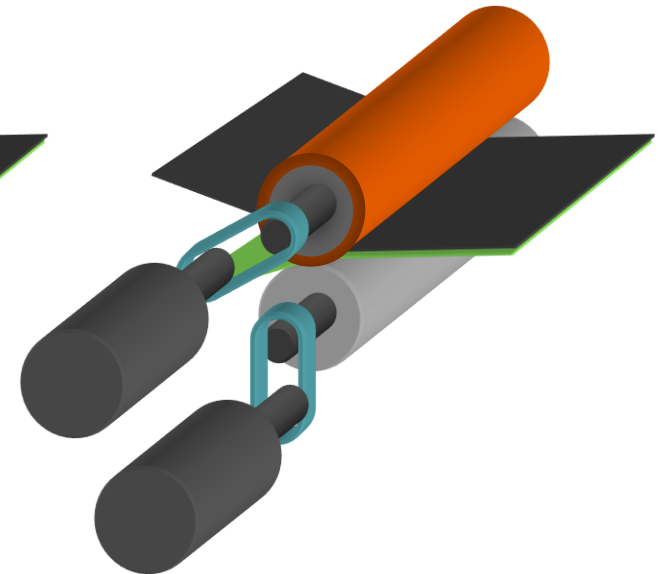


Most
Common

Neither Roller
Driven

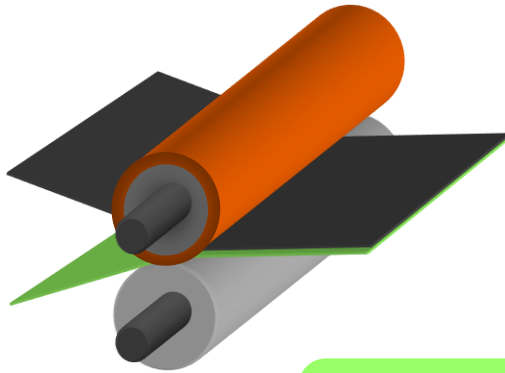


Driven Steel Roller
Idling
Rubber Roller



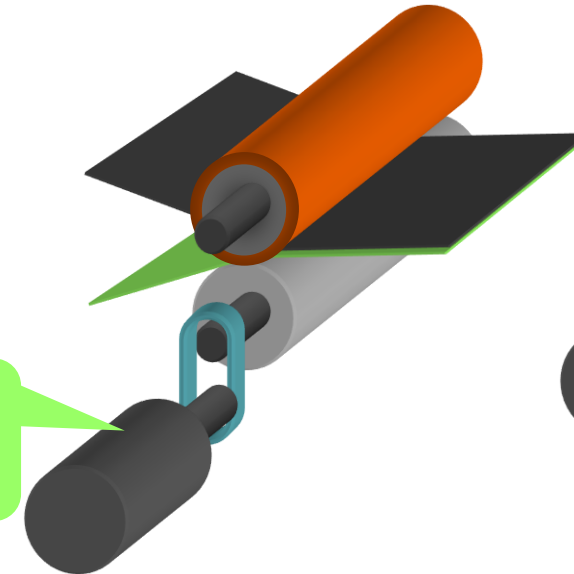
Direct-Driven Steel
Roller
Torque-Driven
Rubber Roller

Laminating Drive Options

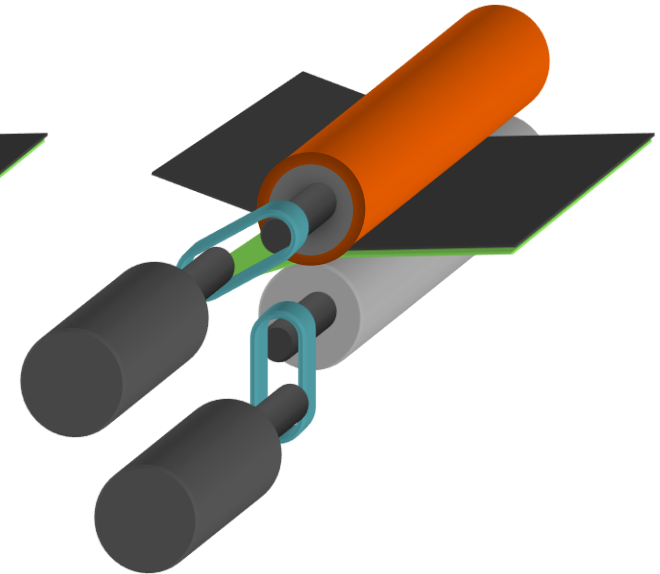


Most
Common

For low tension laminating, often leads to problems from drag and inertia.

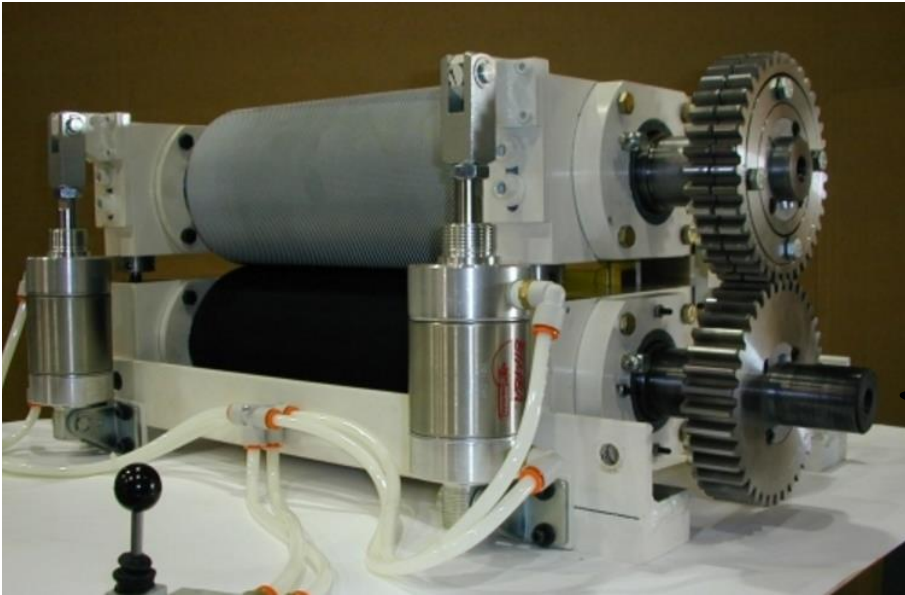


Good for most products, but may have too much shear thicker laminates



One in tension mode, one in torque, most advanced laminating.

Driving Both Nipping Rollers



Example of driving both rollers in a nipped system.

industrial-engraving.com

Nipping and Lamination Key Points

Compliance and Deflection:

- Two hard rollers may be used when the product is compliant.
- At least one compliant roller is preferred for pressure controlling processes.
- Rubber rollers should have 10-30 mils indentation under anticipated process pressures.
- $L/D < 5$ or Keep deflection under process conditions to be less than 20 percent of rubber roller indentation (2-6 mils).
- Default should be cylindrical roller. Crowned rollers are effective, but should be reserved for extreme high pressure processes.
- Crowned rollers can compensate for deflection at a set load value, but not over a wide load range.

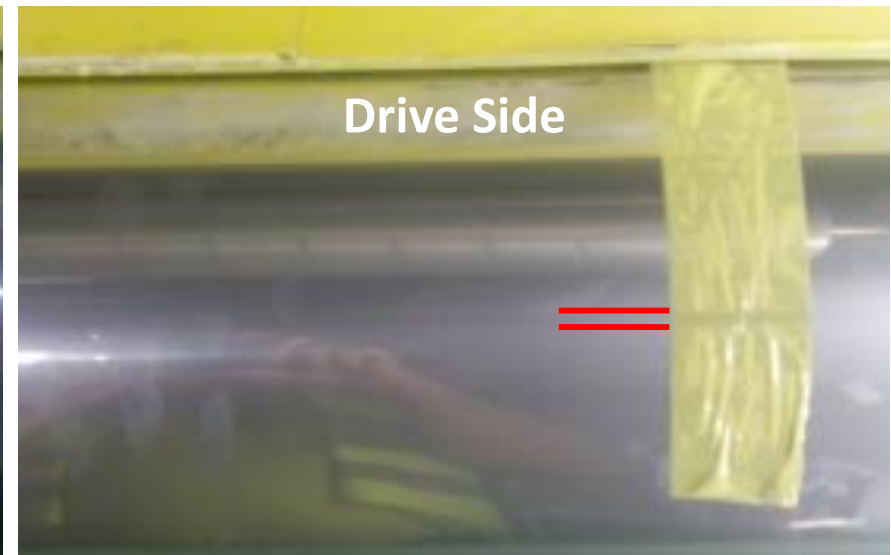
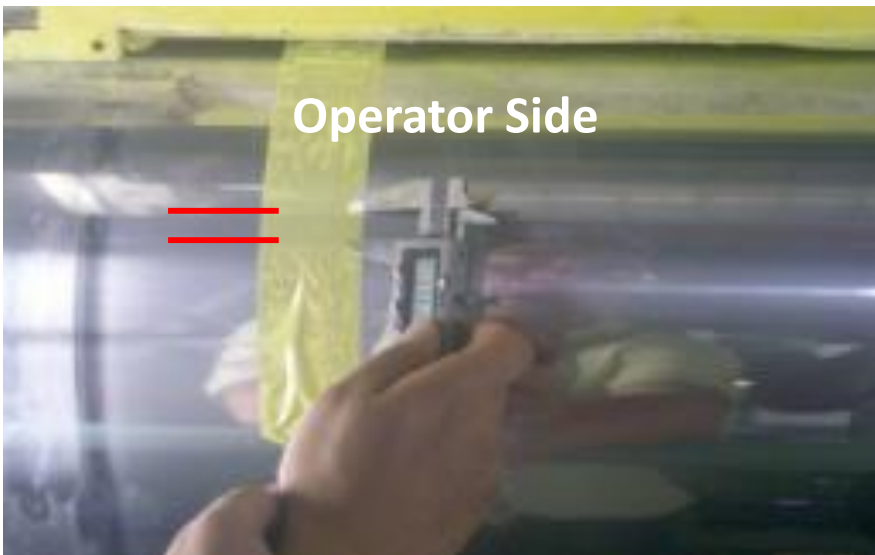
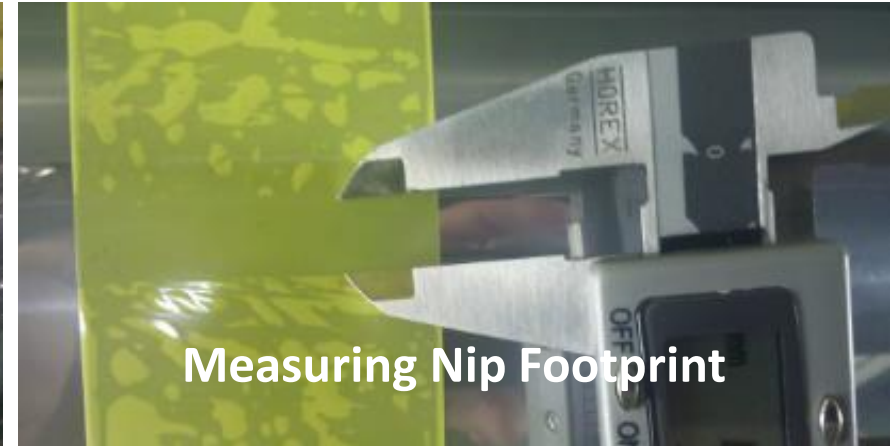
Recommendation #14

Use nip footprint measurements to verify crossweb nip uniformity.

Why? Footprint is a easy measurement and a good indicator of crossweb pressure.



Tape Bubble Footprint Measurement



Tape Bubble Footprint Measurement



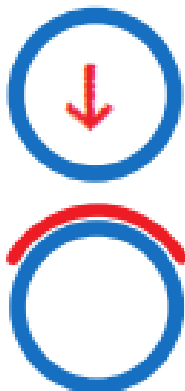
Colored adhesive polypropylene splicing tape is a good candidate for this measurement.



Interpreting Nip Footprints

Static Nip Impression

FIGURE 5



Dynamic Nip Impression

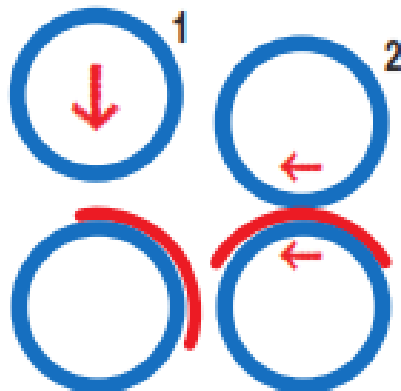


FIGURE 6

Good



Crown Too Low for Given Load



Crown Too High for Given Load



Unbalanced Loading or Misaligned



Banding – Roller Wear, Grinding



Good



Crown Too Low for Given Load



Crown Too High for Given Load



Unbalanced Loading or Misaligned



Banding – Roller Wear, Grinding



Menges Roller Company

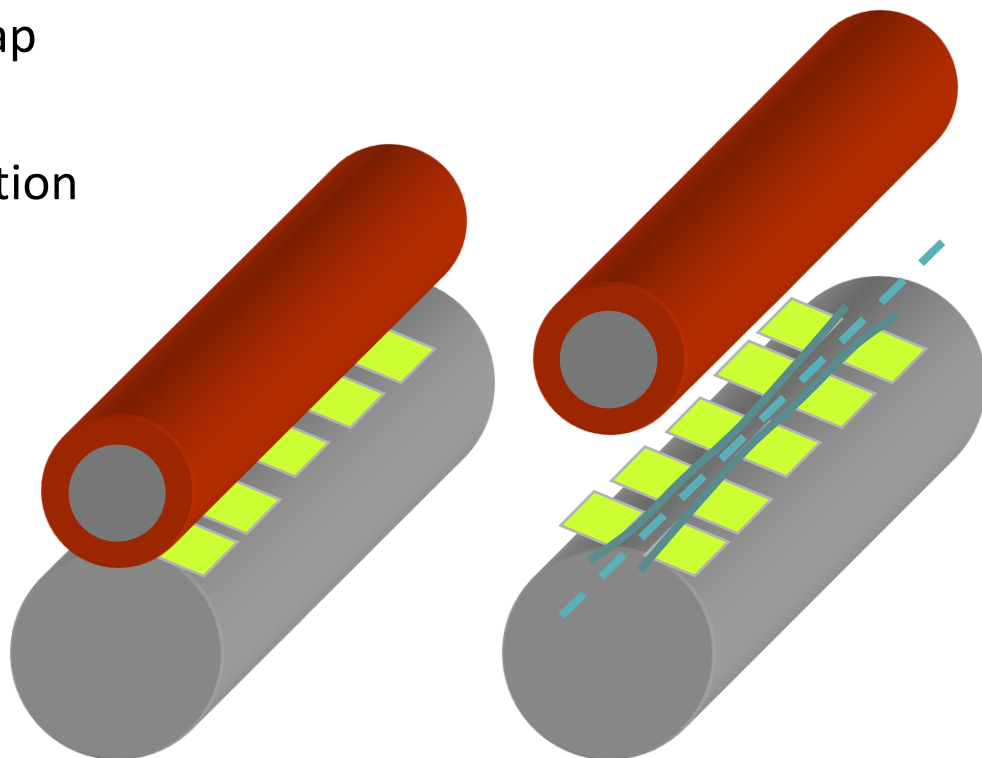
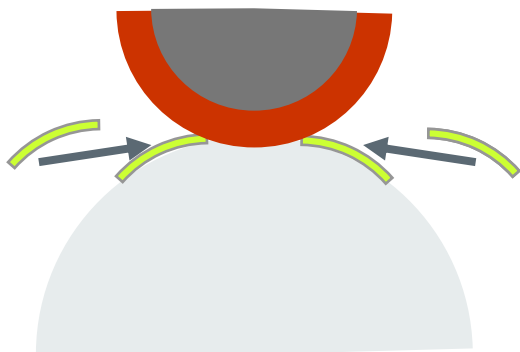
**260 Industrial Dr
Wauconda, IL 60084-1077**

(847) 487-8877

Nip Footprint Measurement 1

Measure nip contact area using a series of sticky notes.

1. Close the nipping roller and apply full load (with nip not rotating).
2. Place a series of Post It™ notes on one roller, pushing them as far into the nip point as possible and stick them in place.
3. Open the nip and measure the gap between opposing notes.
4. Look for MD contact length variation as a sign of non-uniform nip pressure.

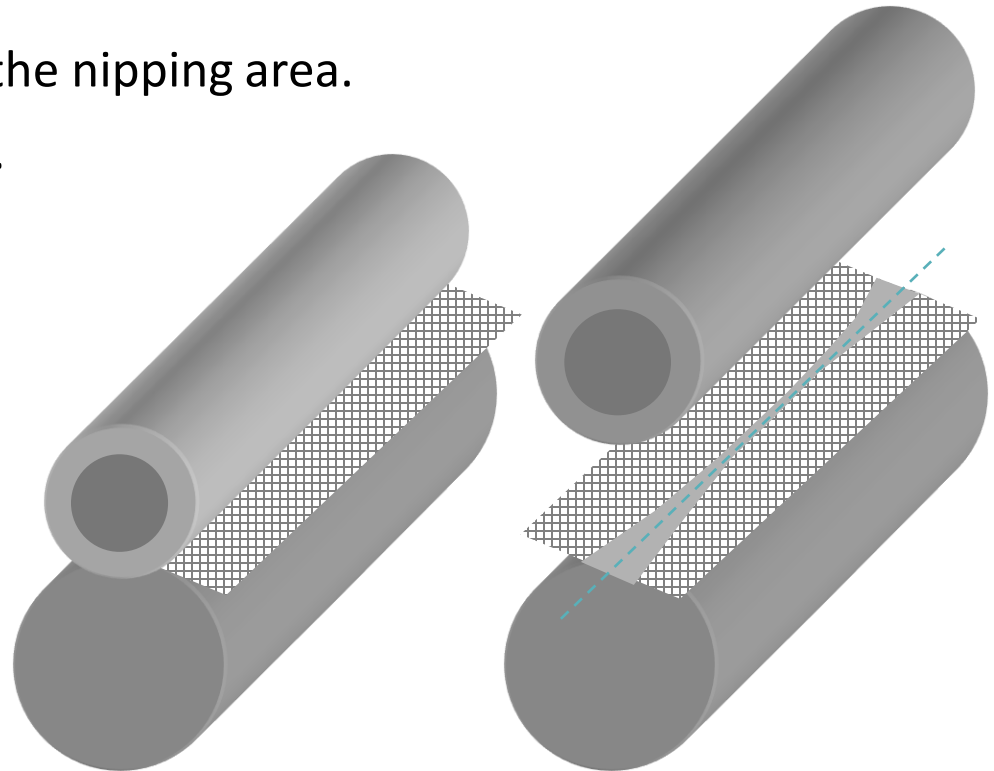


Nip Footprint Measurement 2

Measure nip contact area using knurled foil.

(Best for high pressure steel-steel nips.)

1. Open the nipping rollers
2. Place a strip of knurled foil across the nipping area.
3. Close the nip with the desire force.
4. Open the nip and inspect the area where the knurl was flattened.
5. Look for MD contact length variation as a sign of non-uniform nip pressure.



Contact:

Stowe Woodward - Neenah, WI

Amy Babich 920-729-7000 ext. 24

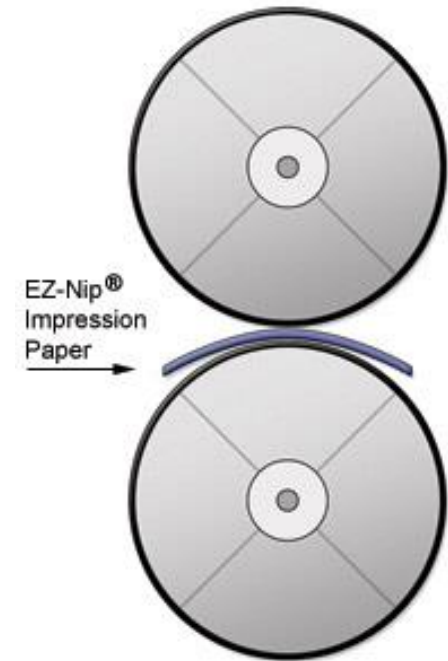
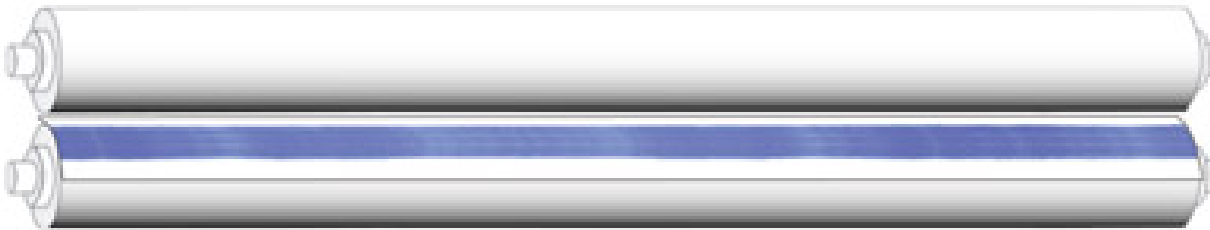
Nip Footprint Measurement 3

EZ-Nip[®]

EZ-Nip - \$300 / 200 ft.

A single sheet pressure sensing film.

Thickness	15 mils (0.076 mm)
Humidity Range	10-90% RH
Pressure Minimum	300 PSI (20.8 kg/cm ²)



www.sensorprod.com/nip.php

Nip Footprint Measurement 4



P-Nip - \$2000

Sensor Products Inc.
300 Madison Avenue
Madison, NJ 07940 USA

Phone: 1.973.884.1755

Fax: 1.973.884.1699

Toll Free (U.S. Only): 1.800.755.2201

Email: sales@sensorprod.com

www.sensorprod.com/nip.php

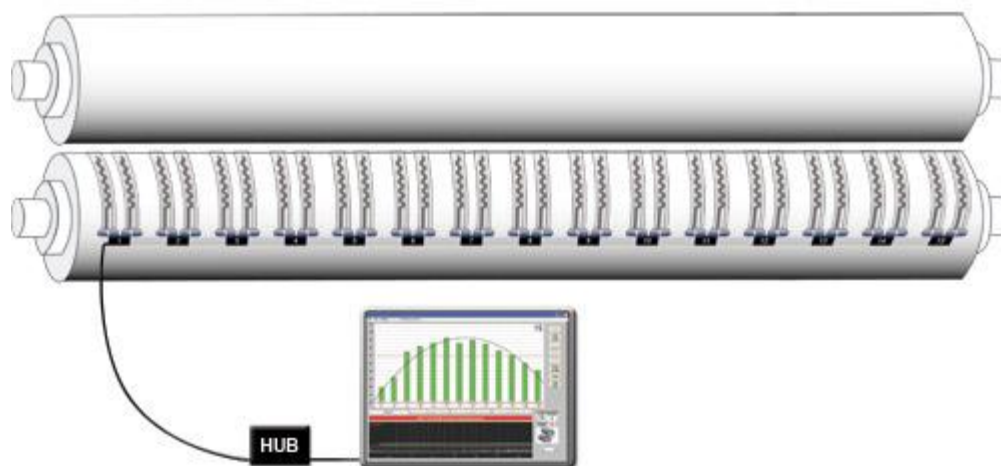
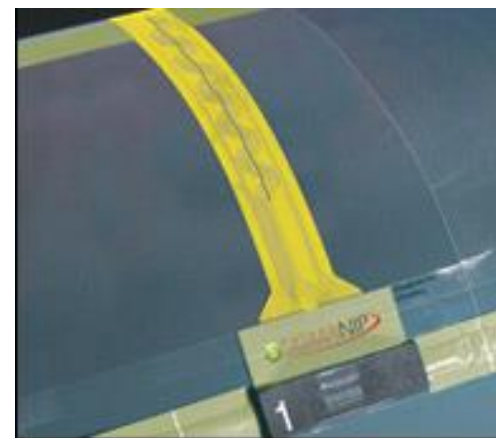


Nip Footprint Measurement 5



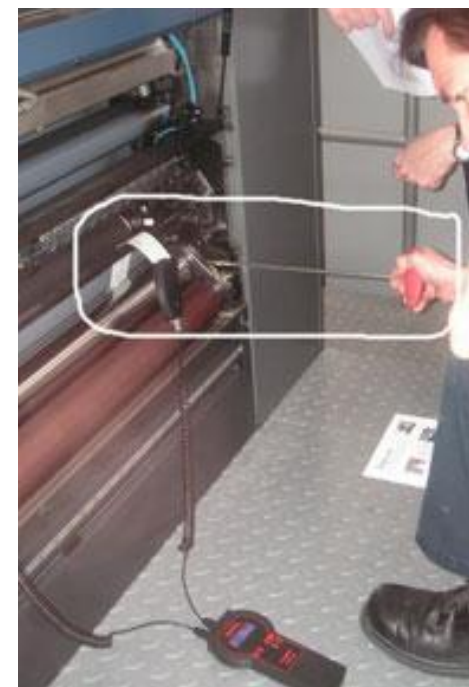
Sigma-Nip - Depending on the length of the chain and how many sensors, cost is from \$8k to \$16K.

For 60" around \$10K



www.sensorprod.com/nip.php

Nip Footprint Measurement 6



DIGI~Nip Roller Measuring &
Setting Gauge

\$2,750

Mfr: www.diginip.fr

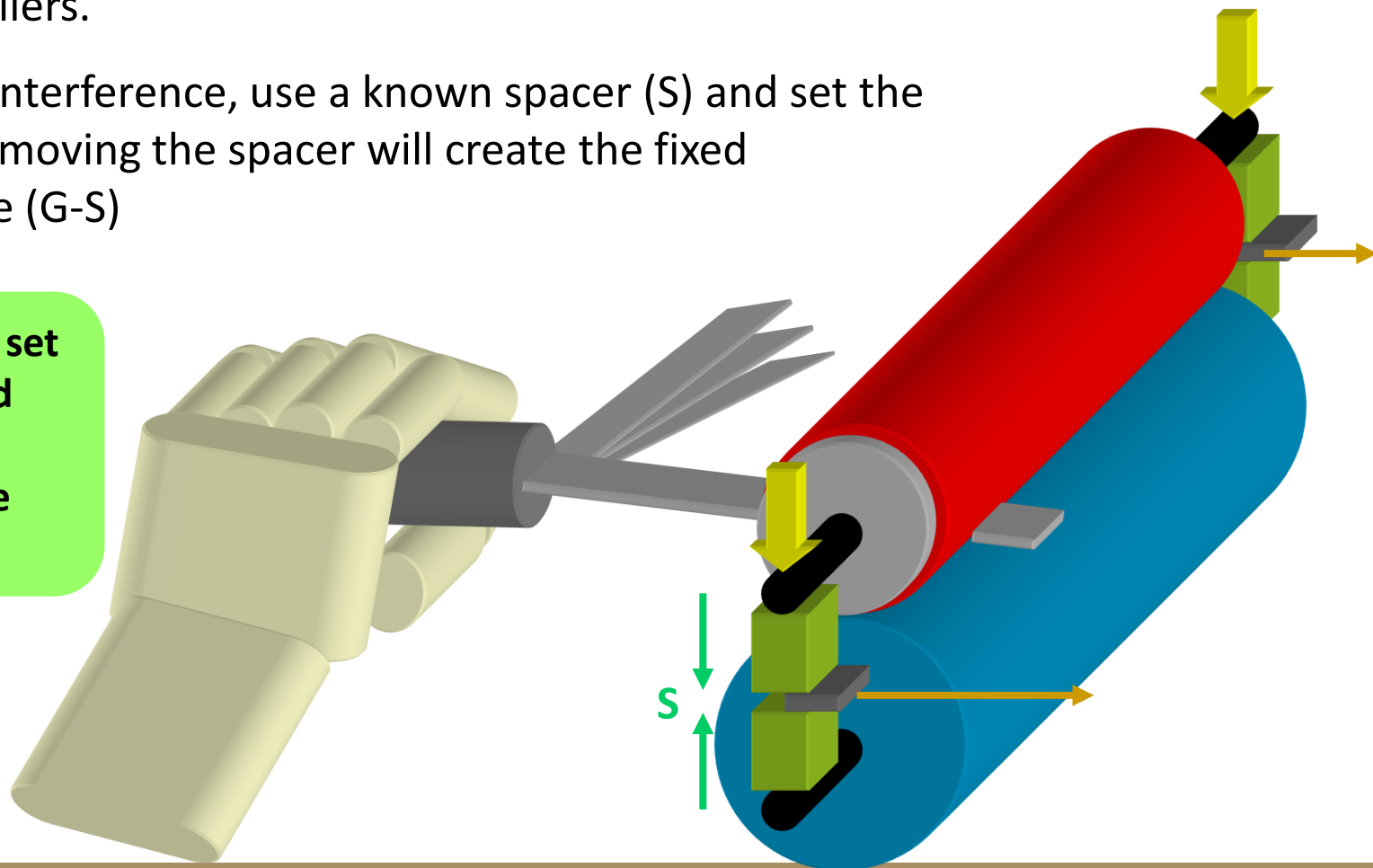
US Distr.: www.sunshinepaper.com



Measure Nip Gap

- For a fixed gaps, use a feeler gauge to measure the space between rollers.
- For a fixed interference, use a known spacer (S) and set the gap (G). Removing the spacer will create the fixed interference (G-S)

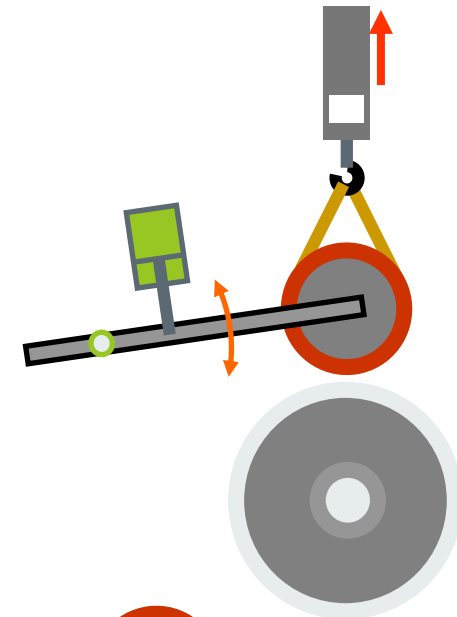
Some nips are set using a fixed opening or interference dimension.



Measure Nip Load

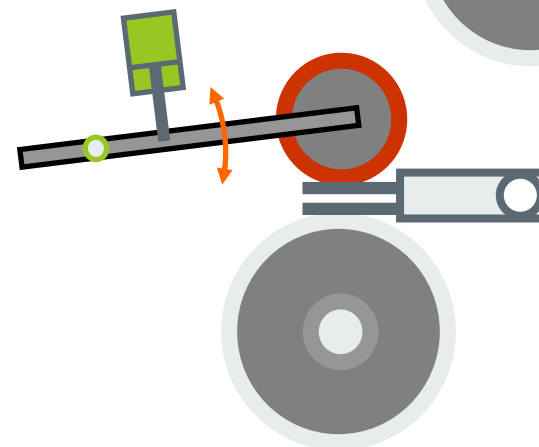
Method 1 (lower nip loads)

1. Wrap a rope around the nipping roller.
2. Attach the rope to a handheld force gauge.
3. Measure the force at various input pressures.
4. Divide the sum of the forces by web width to convert to pounds per inch load.



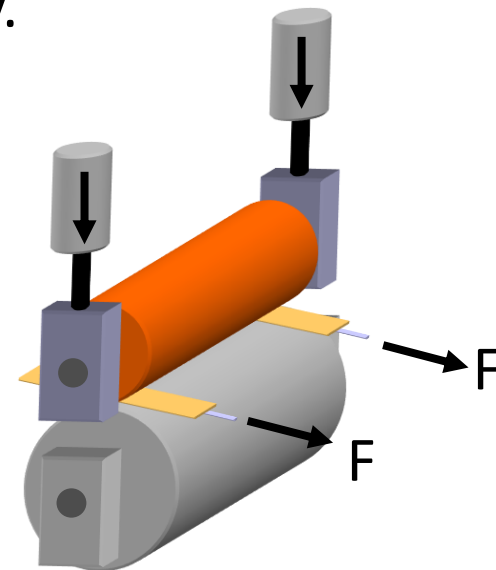
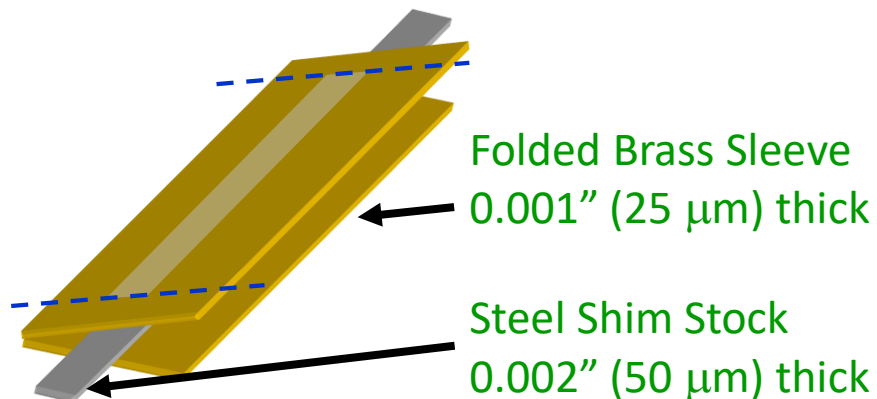
Method 2 (higher nip loads)

1. Insert two compression force gauges on either side of a nip.
2. Close the nip and load at various levels.
3. Divide the sum of the forces by web width to convert to pounds per inch load.



Nip Load Measurement

Insert a steel-brass 'hot dog' into the loaded nip and measure the force to slide the steel strip. Calculate the nip load from the measured frictional force, COF, and geometry.



$$N = \frac{F}{2\mu w_s}$$

μ = Brass-Steel Friction Coefficient (use 0.25)

N = Nip load per width (lbs/in or N/m)

F = Frictional force to slide steel vs. brass (lbs or N)

w_s = Width of steel strip

Nip Load Measurement 2



STRANDBERG ENGINEERING LABORATORIES, INC.

1302 N. O. HENRY BLVD. (U.S. 29 N.) • GREENSBORO, N.C. 27405 • U.S.A.

TEL: (336) 274-3775 • FAX: (336) 272-4521 • EMAIL: sensors@strandberg.com • <http://www.strandberg.com>

Sensor Dimensions 11.0 "(279mm) long,
0.6" (15mm) wide,
0.016" (0.4mm) thick

Range 0-22.0 lbs (0-10.0 kg)

Resolution 0.1 lb or kg

Repeatability $\pm 2.5\%$ of measured value

22 lbs over 6.6 in² =
145 psi

Nip Check 8800 = \$885

Spectrum 4 (seven sensor options) = \$1850



NIP PRESSURE

Nip Footprint Measurement 7



Roll Nip Indicator = €1300

Dynamic Sensor = €195

Static Sensor = €160

Stefan Wegdell

NIP CONTROL AB

Gamla Skolvägen 36

133 35 Saltsjöbaden, Sweden

Phone: +46 (0)8 556 164 77

Fax: +46 (0)8 717 5526

Mobile: +46 (0)708 220659

stefan.wegdell@nipcontrol.com

The sensors are good for more than 500 measurements.

Nip Width: 2 - 20 mm

Roller Diameter: 30 - 200 mm

Nip Temperature: 20 - 50° C

Roller Surfaces:

Metal / Rubber, Hard Plastic / Rubber, Rubber / Rubber

Rubber Hardness: 20 - 60° shore A

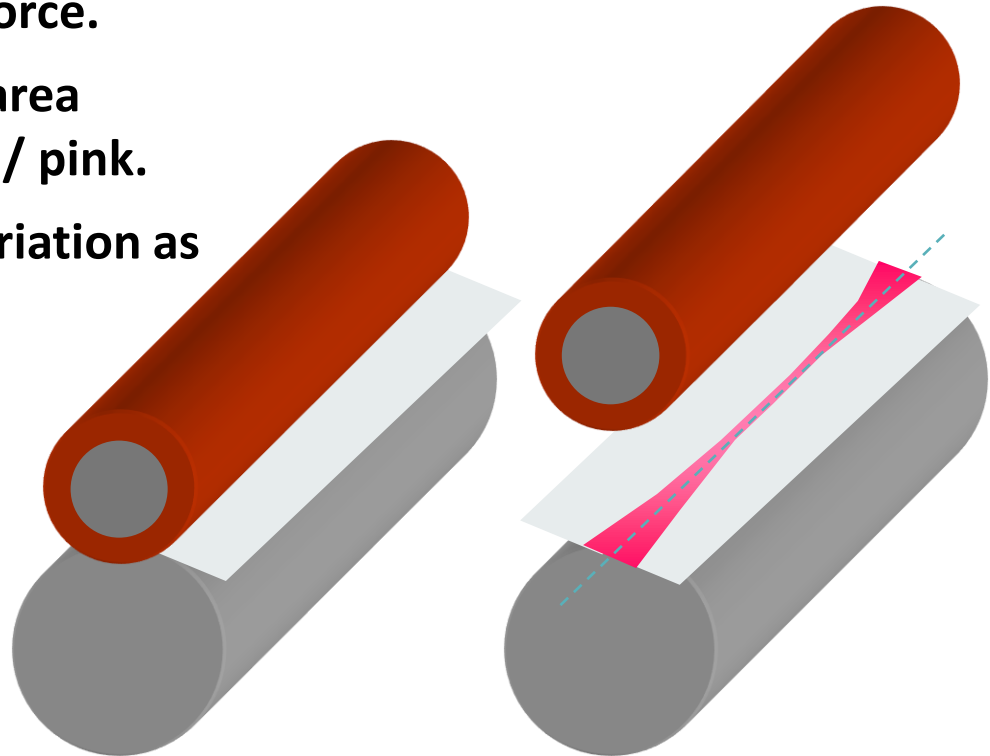
www.nipcontrol.com

Colorimetric Film Measurement 1

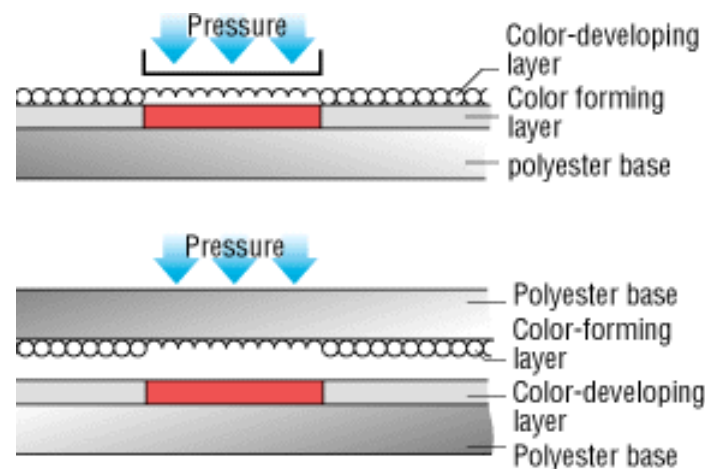
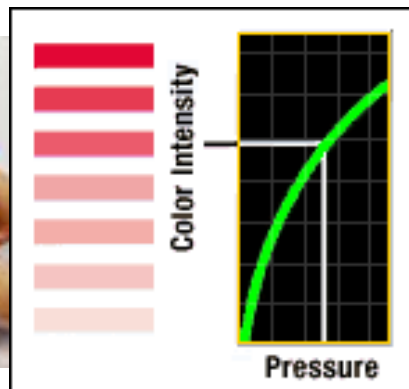
Measure nip contact area using colorimetric pressure sensing film.

1. Open the nipping rollers
2. Place a strip of colorimetric film across the nipping area.
3. Close the nip with the desire force.
4. Open the nip and inspect the area where the film has turned red / pink.
5. Look for MD contact length variation as a sign of non-uniform nip pressure.
6. If valuable, measure with color densitometer and convert to pressure.

***Can also be used by feeding through a running nip.**



Colorimetric Pressure Film



Film Type	Pressure Range	Dimensions	Cost	Layer(s), Thickness
Pressurex Micro*	2 - 20 psi	11" x 17" per sheet	\$580/box	Double, 8 mils
Ultra Low	28 - 85 psi *	13.1' x 10.6"	\$580/box**	Double, 8 mils
Super Low	70 - 350 psi	16.4' x 10.6"	"	Double, 8 mils
Low	350 - 1,400 psi	39' x 10.6"	"	Double, 8 mils
Medium	1,400 - 7,100 psi	39' x 10.6"	"	Single, 4 mils
High	7,100 - 18,500 psi	39' x 10.6"	"	Single, 4 mils

*Most likely range for internal roll pressure sensors

**Price equates to \$0.07-0.33/in²

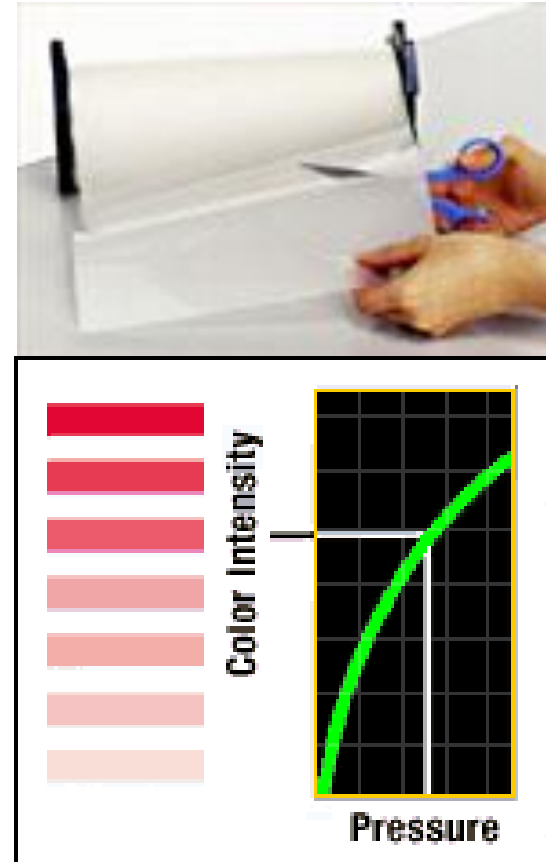
Colorimetric Pressure Film

Available from:

Tekscan, Inc.
307 West First Street.
South Boston, MA. 02127-1309, USA
tel: 800.248.3669 / 617.464.4500

<http://www.tekscan.com/pressure-indicating-film.html>

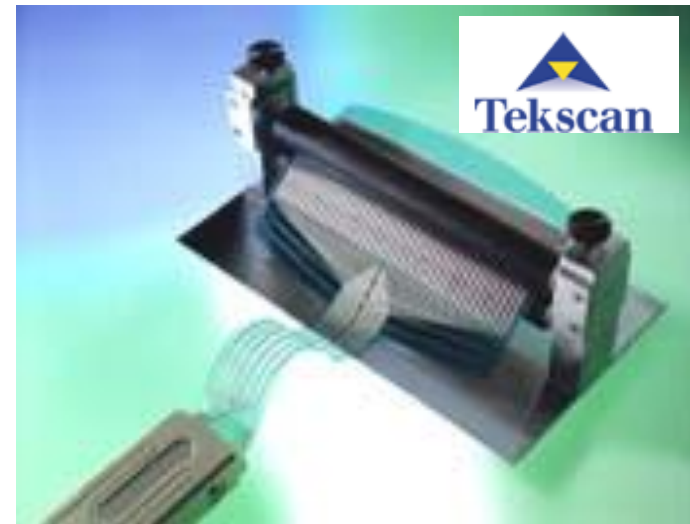
Sensor Products Inc.
188 Rt. 10 Suite 307
East Hanover, NJ 07936-2108
800-755-2201
www.sensorprod.com/xsensor.html



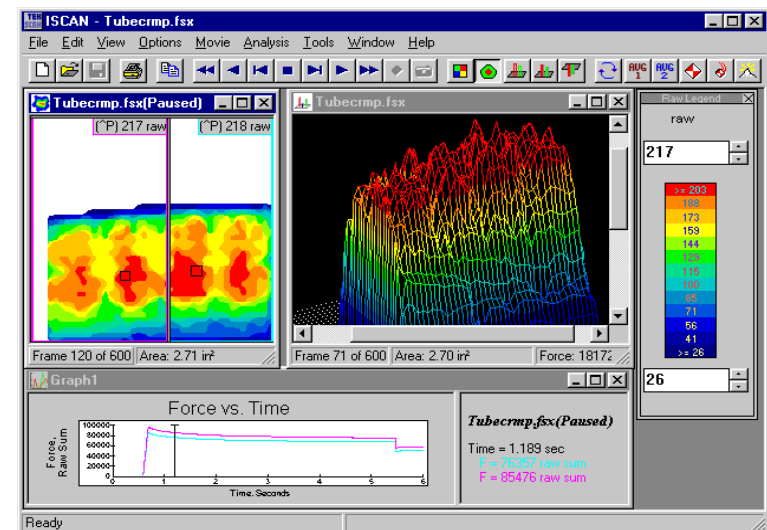
Nip Pressure Measurement 1

Tekscan's sensors detects pressure changes as a electrical resistance drop though a conductive ink.

The array sensor is 0.005" thick and read every 0.1" over a 1" by 16" area.



Tekscan®
307 West First Street
South Boston, MA 02127-1309
800-248-3669
www.tekscan.com
System Cost ~\$25000

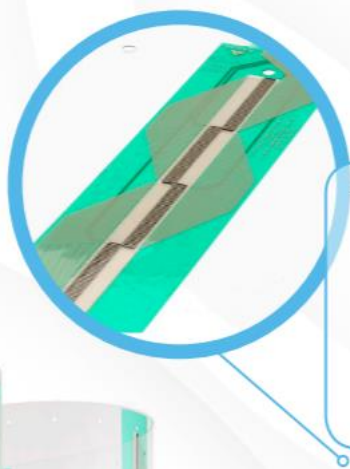
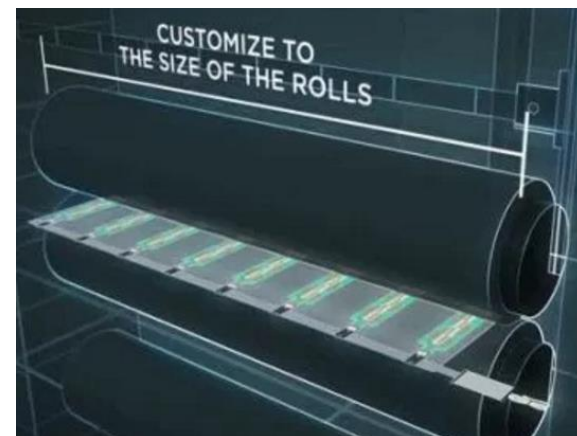
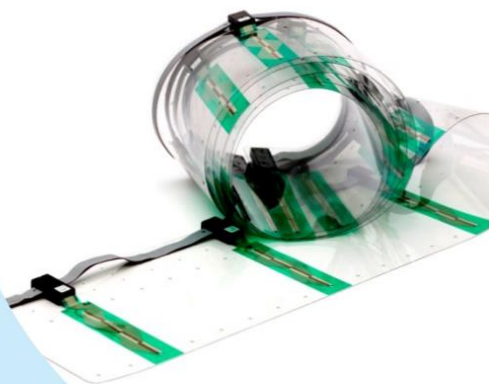


Tekscan NPAT



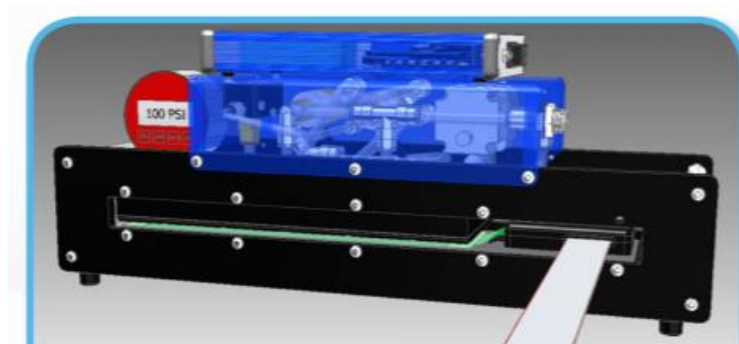
Nip Pressure Alignment Tool™

Nip Width & Relative Pressure Distribution



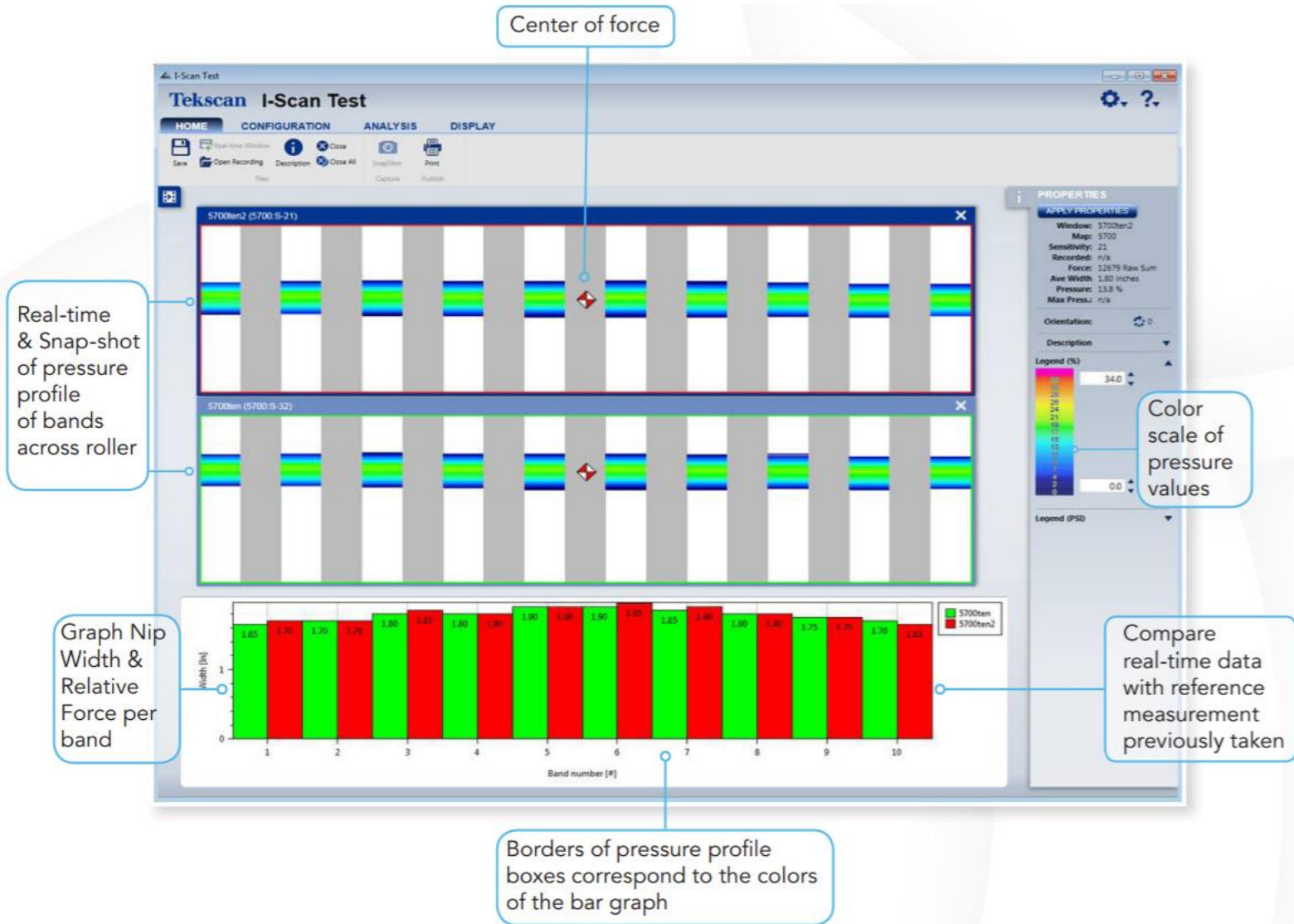
Sensor Band:

- 2 sensor model resolutions to support different nip sizes
- Standard band measures pressures from 25- 4,500 psi (172- 31,026 kPa). Also available in custom pressure ranges measuring up to 7,500 psi (51,711 kPa)

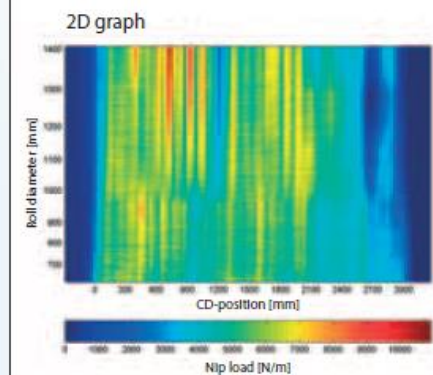
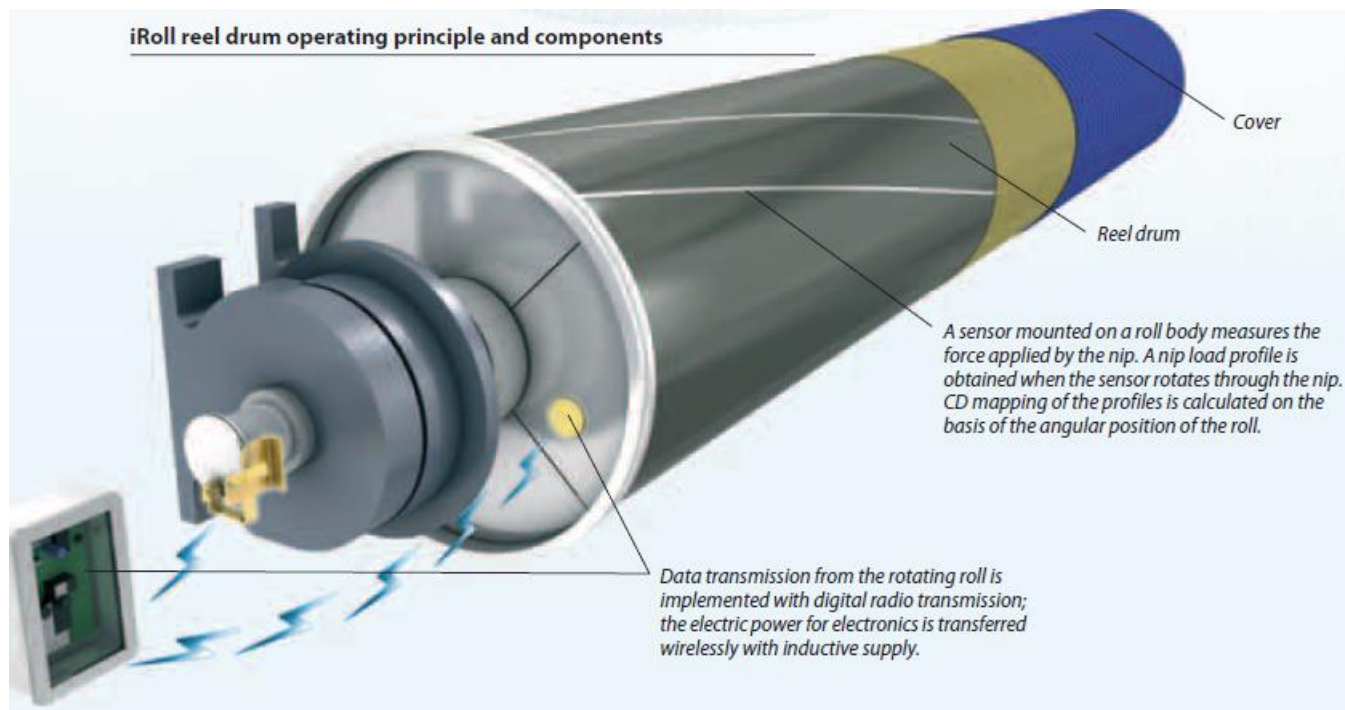


Sensor equilibrator / calibrator

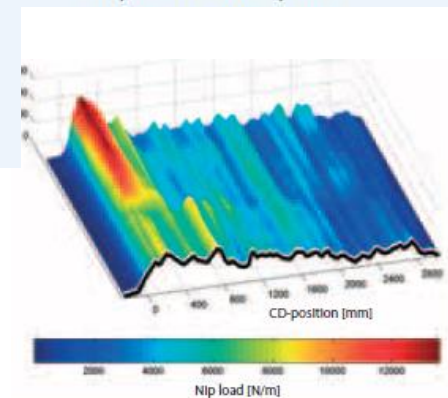
Tekscan NPAT



Nip Pressure Measurement 3



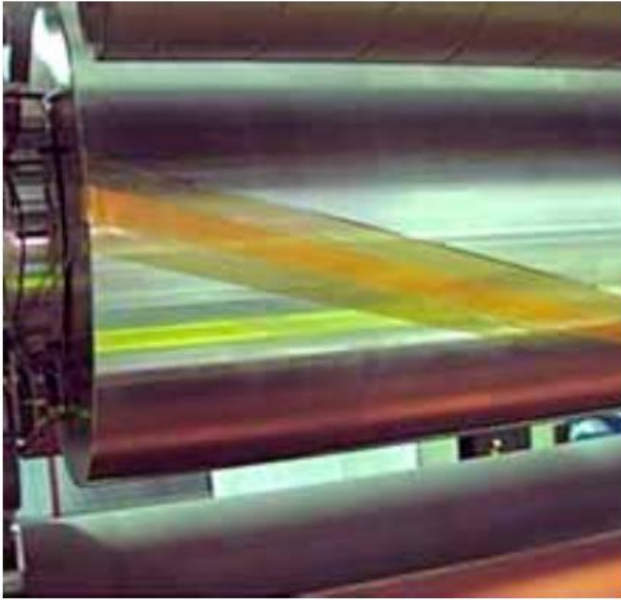
A reeling nip load profile shows bumps and valleys coming from paper profiles, roll alignment, nip load control problems, etc. With iRoll it is also possible to monitor nip straightness and prevent skewed roll profiles.



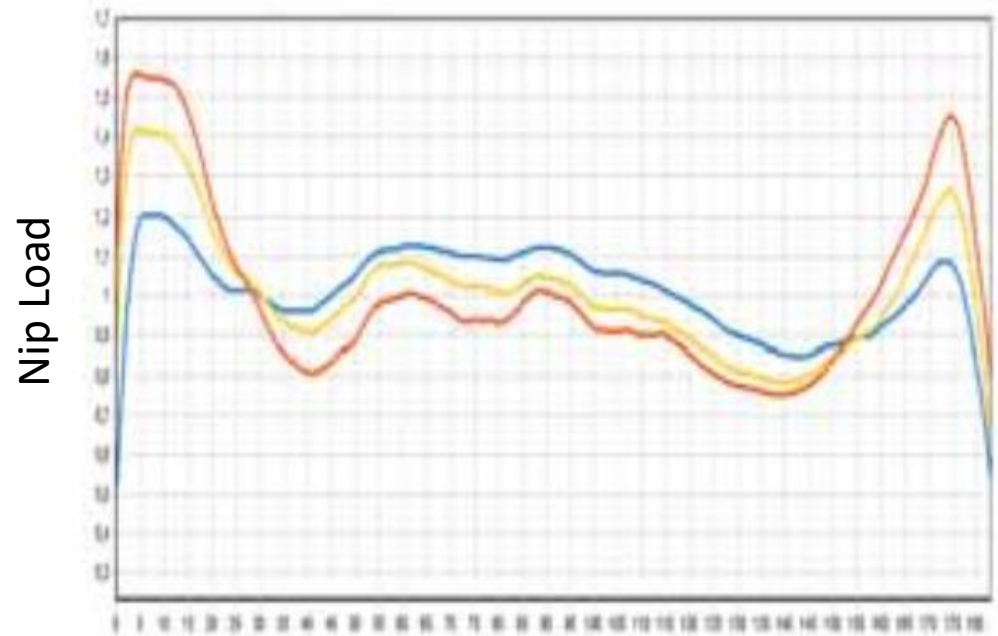
A profile map indicates a bump on the other edge originating from a fault in an earlier process. The profile map above is measured online directly from the process without delay.

Stowe Woodward
1 Technology Park Drive
Westborough, MA 01581
508-616-9458
www.stowewoodard.com

Temporary Nip Profile Measurement



iRoll Portable Lite sensor on a laminating machine roll.

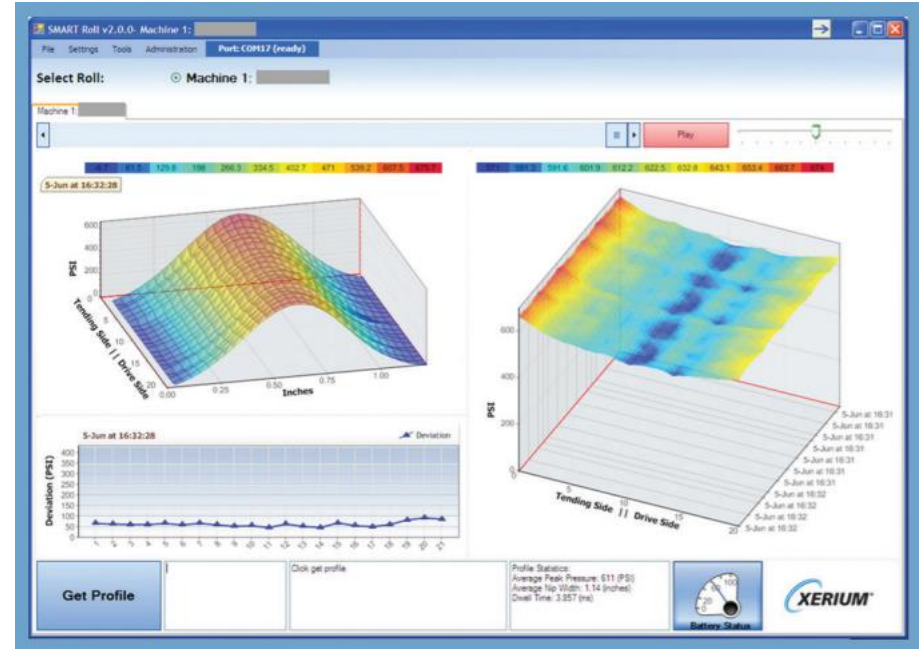
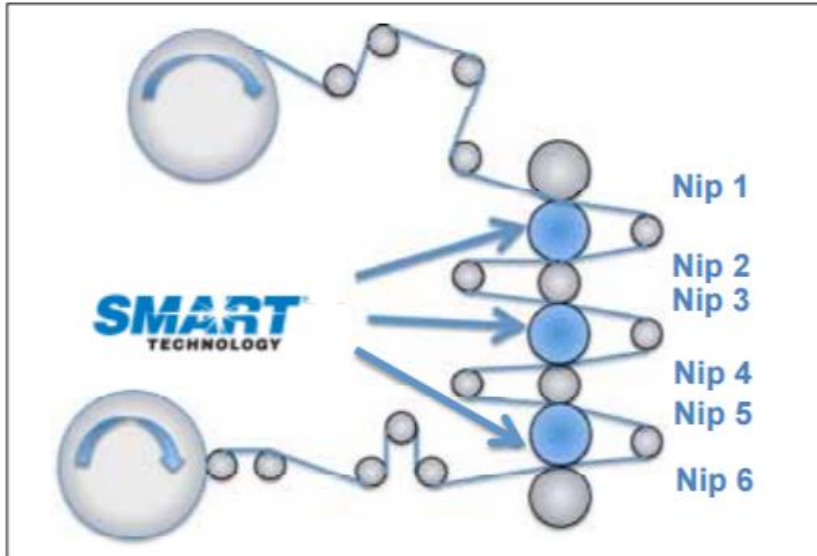


Nip Width

Features

- iRoll Portable Lite sensors are mounted with high strength taping to the roll surface
- Measurement electronics are mounted on the roll end or journal
- Measurements are done in dynamic conditions whilst the rolls are rotating, and information is given on how the friction, nip load control devices, rod loading devices, etc. are working
- Several nips can be analyzed at the same time
- Measurement and analysis can be done within a short shutdown (6 to 8h)
 - Installation of measurement equipment takes abt. 2 to 4 hours, measurements and tests abt. 2 to 3 hours, removing of the equipment takes abt. 1 hour
- Measurement data is analyzed online and illustrated in color maps and graphs

Pressure Sensing Roller Cover - Xerium



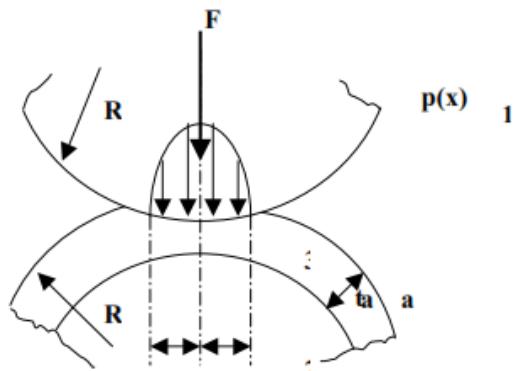


Figure 2-1: Rubber covered roller in contact with a rigid roller

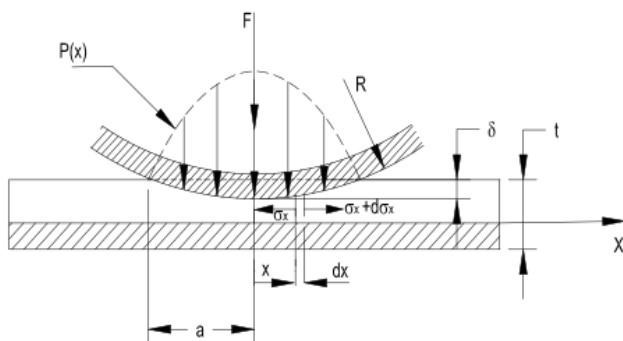


Figure 2-3: Definition of variables for Johnson's model

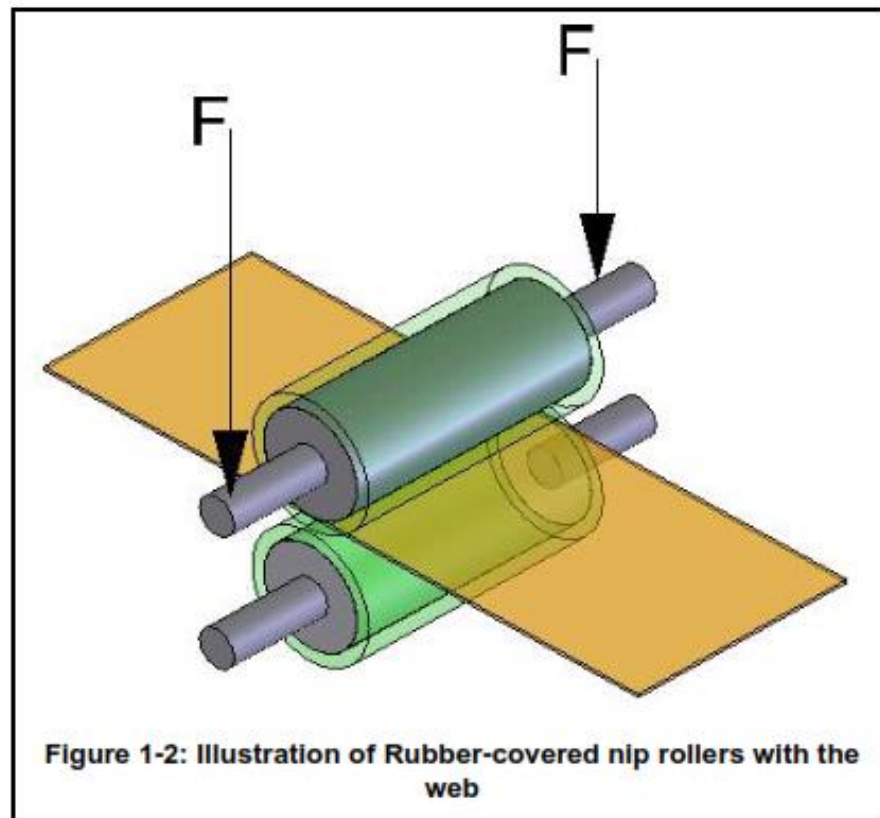
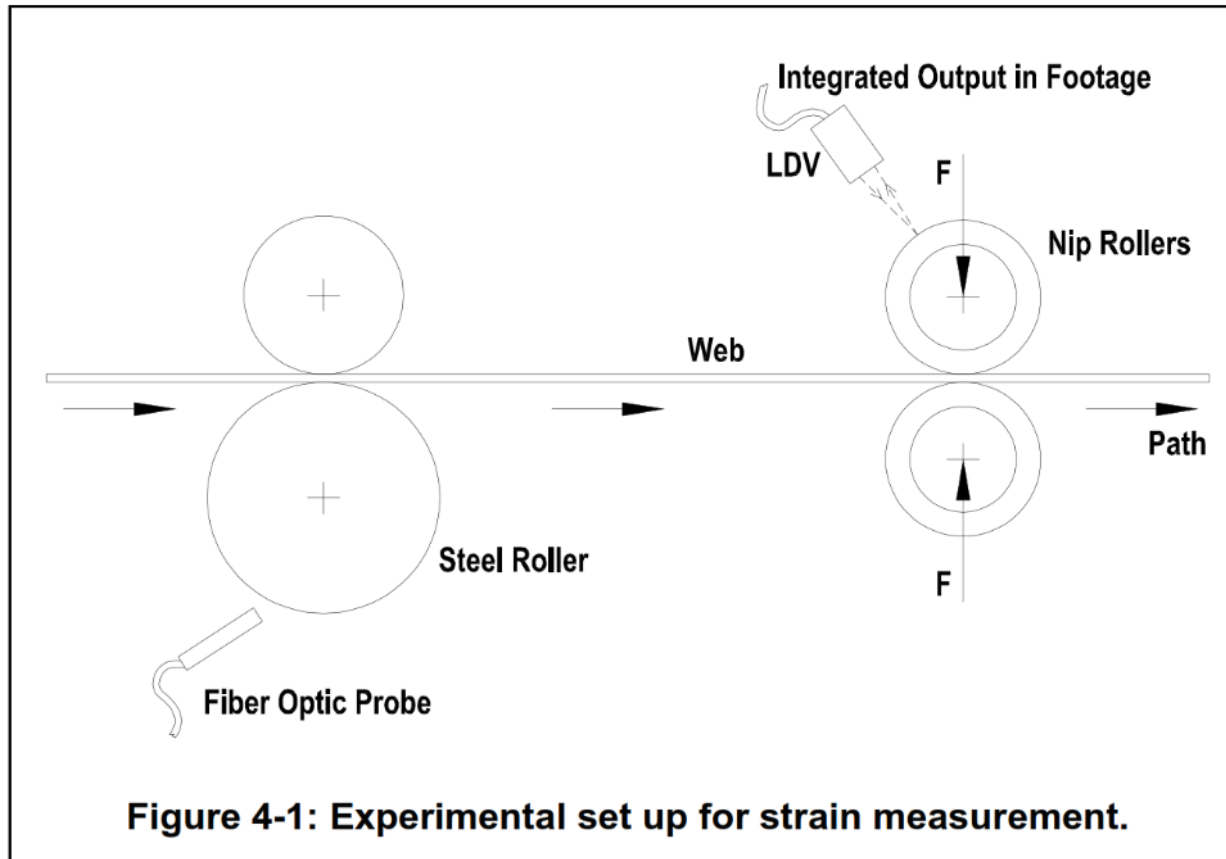


Figure 1-2: Illustration of Rubber-covered nip rollers with the web

Measuring Speed Increase of Rubber Nips



LATERAL STEERING OF THE WEB IN
A DIFFERENTIALLY LOADED NIP OF
RUBBER COVERED ROLLERS

By
KAVITHA BALAIYAN
OSU 2005

$$\varepsilon_{\theta} \approx \frac{\delta}{4t}$$

Good, J.K., Proceedings of the WHRC Semi Annual Technical Review and Industrial Advisory Board meeting, Oct. 1999. (Not published)

Web Feed Rate Differential vs Nip Load

Case	Roller A			Roller B		
	OD (in.)	CT (in.)	HARDNESS (Shore A)	OD (in.)	CT (in.)	HARDNESS (Shore A)
1	3.5	0.75	40	2.9	---	RIGID
2	3.5	0.75	40	3.5	0.75	40
3	2.5	0.25	80	2.9	---	RIGID
4	2.5	0.25	80	2.5	0.25	80
5	5.0	0.50	60	2.9	---	RIGID
6	5.0	0.50	60	5.0	0.50	60
7	7.5	0.75	80	2.9	---	RIGID
8	7.5	0.75	80	7.5	0.75	80
9	6.5	0.25	60	2.9	---	RIGID
10	6.5	0.25	60	6.5	0.25	60
OD ---- Outer Diameter of the Roller						
CT ---- Rubber Cover Thickness						

Web Feed Rate Differential vs Nip Load

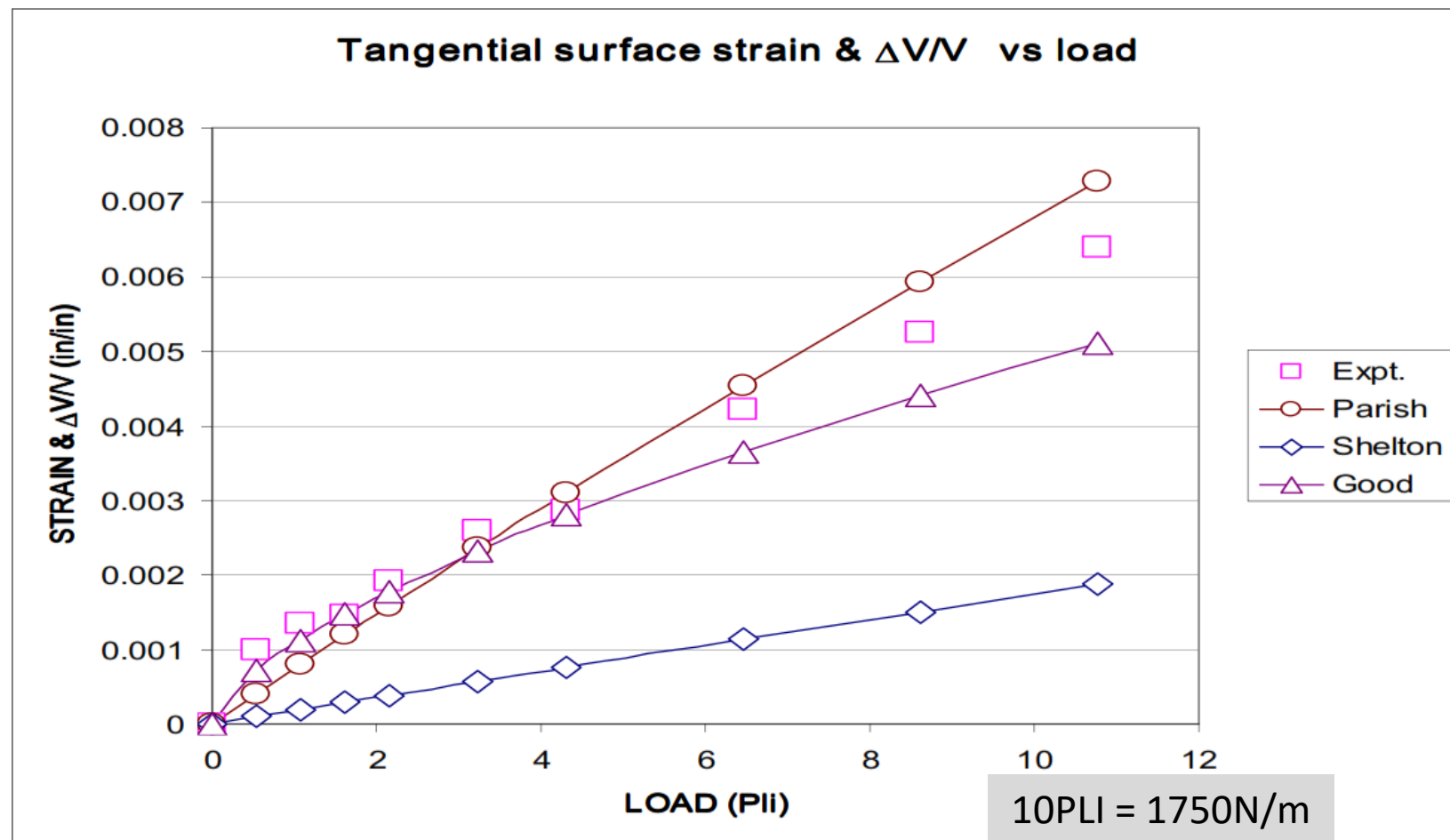


Figure B-3: Tangential Strain vs. Load / indentation for roller case 3.

Smaller Diameter Rubber Nip = More ΔV

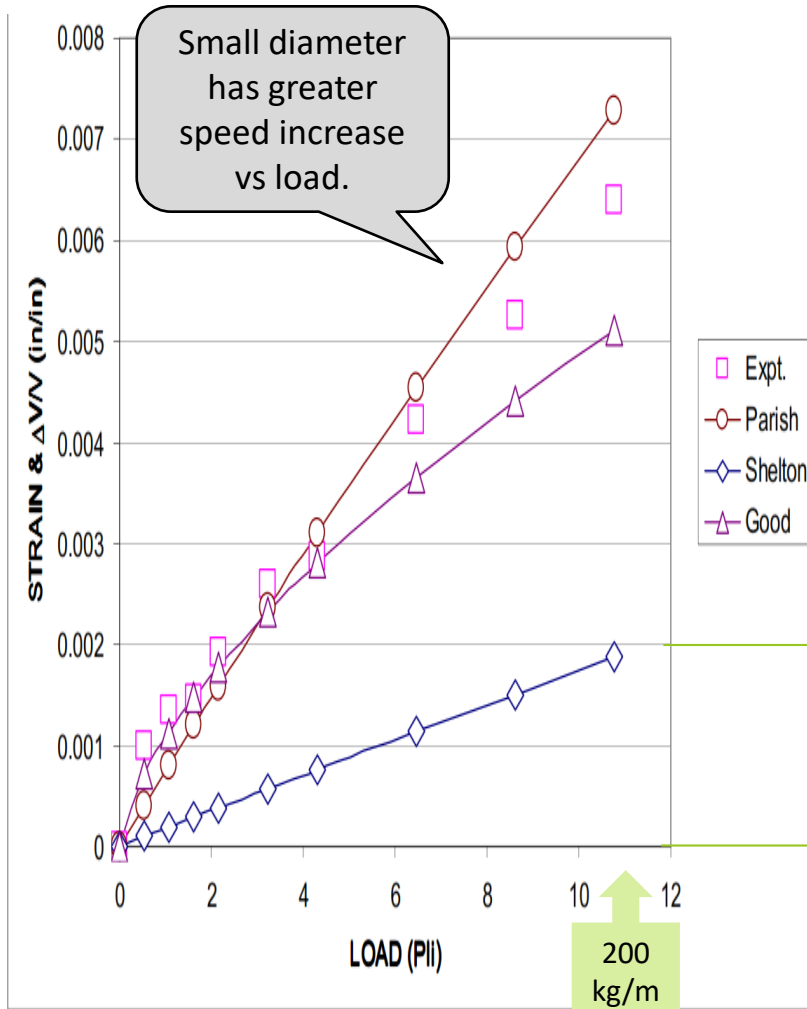


Figure B-3: Tangential Strain vs. Load / indentation for roller case 3.

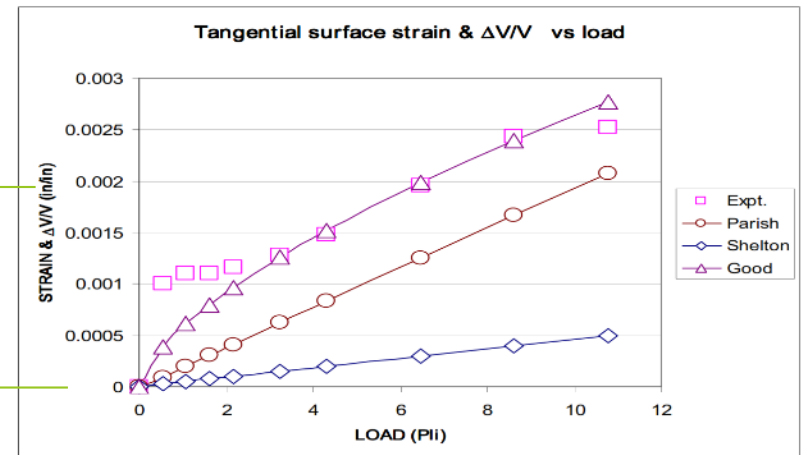
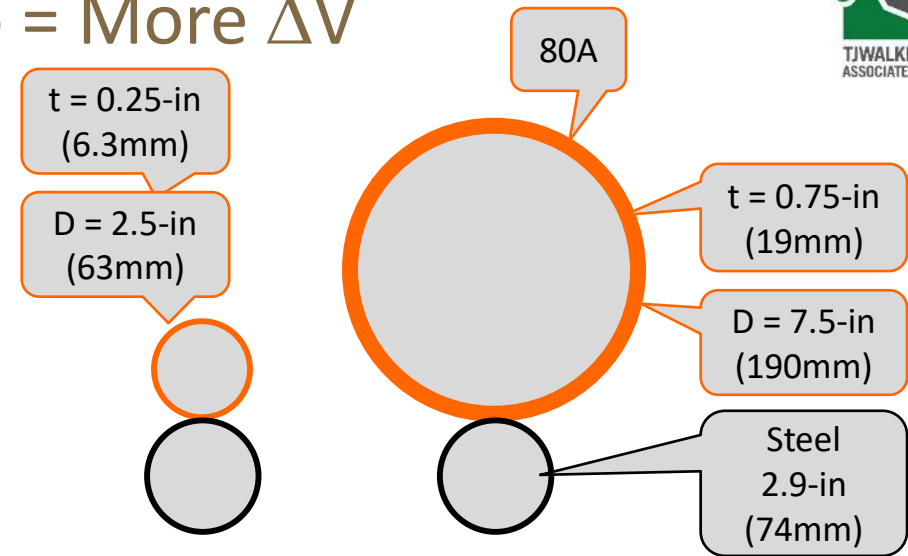


Figure B-6: Tangential Strain vs. Load / indentation for roller case 7.

100
kg/m

200
kg/m

Rubber-Rubber Slightly Less ΔV than Rubber-Steel

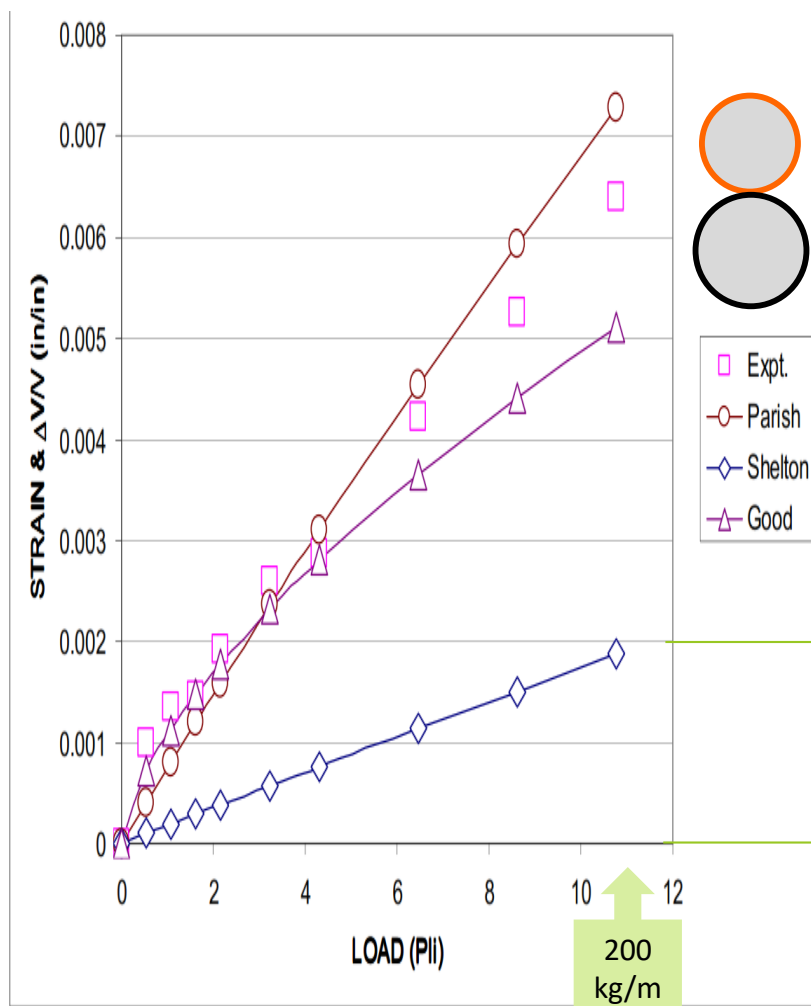
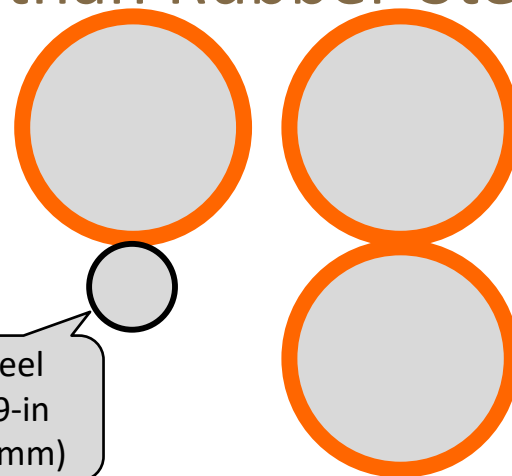
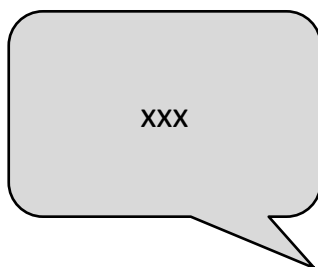


Figure B-3: Tangential Strain vs. Load / indentation for roller case 3.



Figure B-4: Tangential Strain vs. Load / indentation for roller case 4.

Rubber-Rubber Slightly Less ΔV than Rubber-Steel



80A
0.75-in
(19mm)

7.5-in
(190mm)

Steel
2.9-in
(74mm)

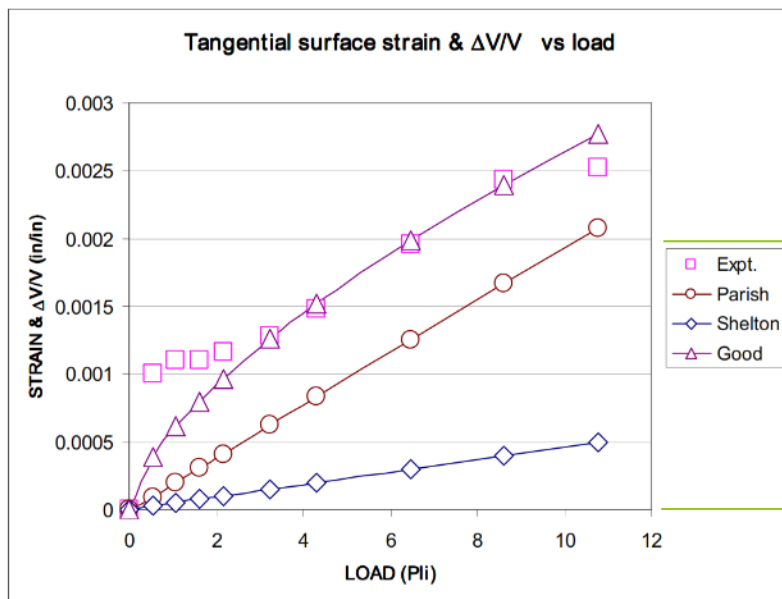


Figure B-6: Tangential Strain vs. Load / indentation for roller case 7.

200
kg/m

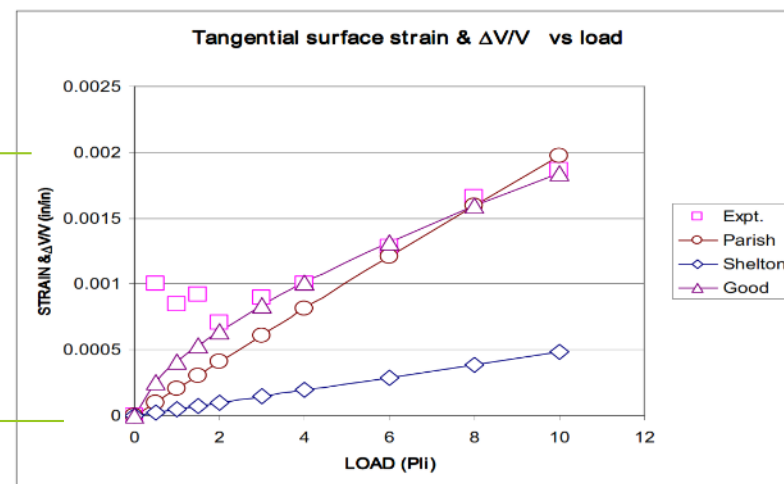
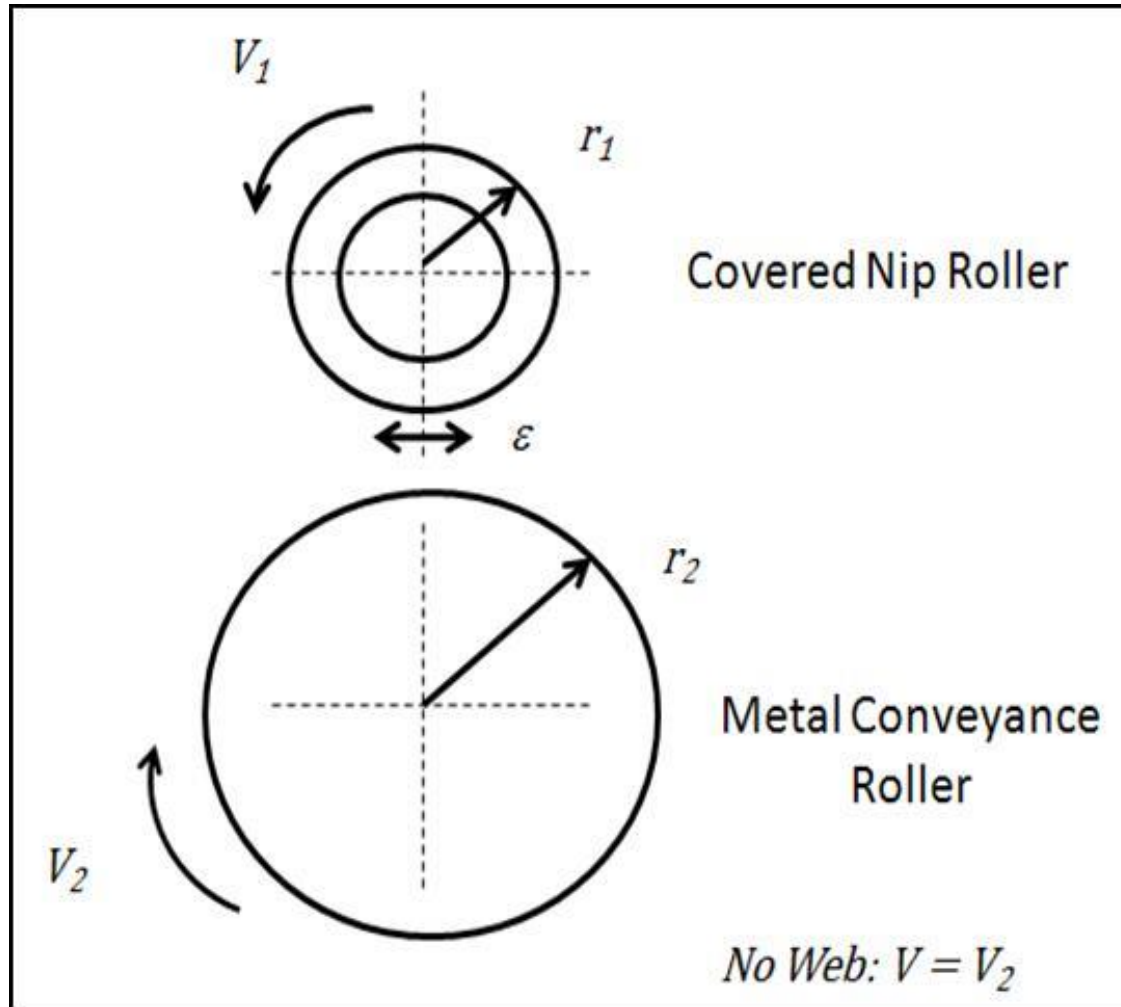


Figure B-7: Tangential Strain vs. Load / indentation for roller case 8.

100
kg/m

200
kg/m

Relative Speeds in Steel-Rubber Nips



Equations:

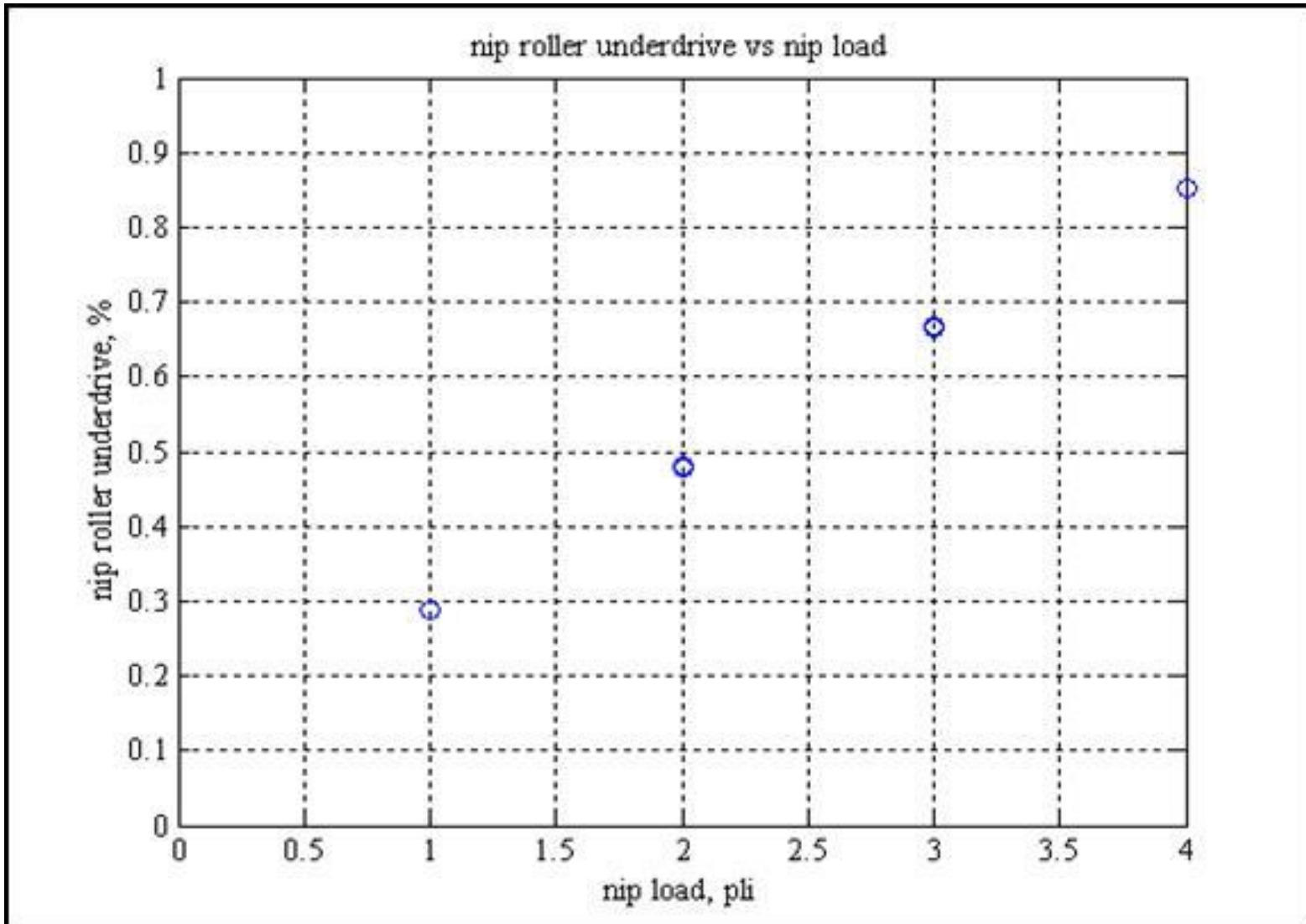
$$t_2 = \frac{2\pi r_2}{V}$$

$$t_1 = \frac{2\pi r_1(1+\varepsilon)}{V}$$

Solve for ε

$$(1 + \varepsilon) = \frac{r_2 t_1}{r_1 t_2}$$

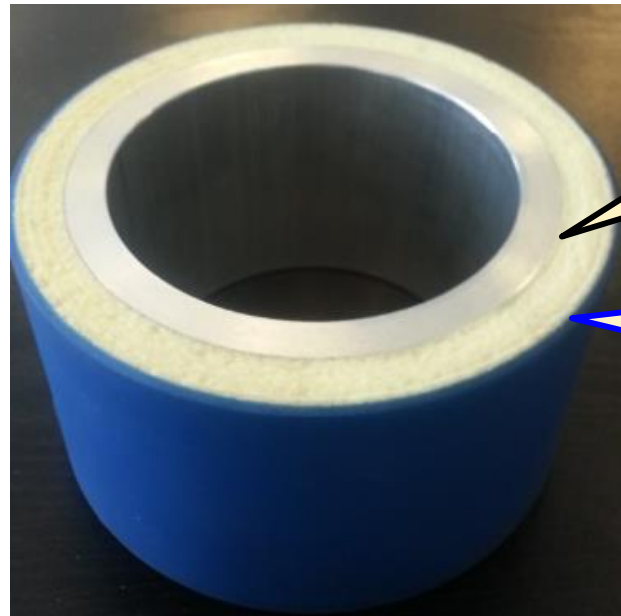
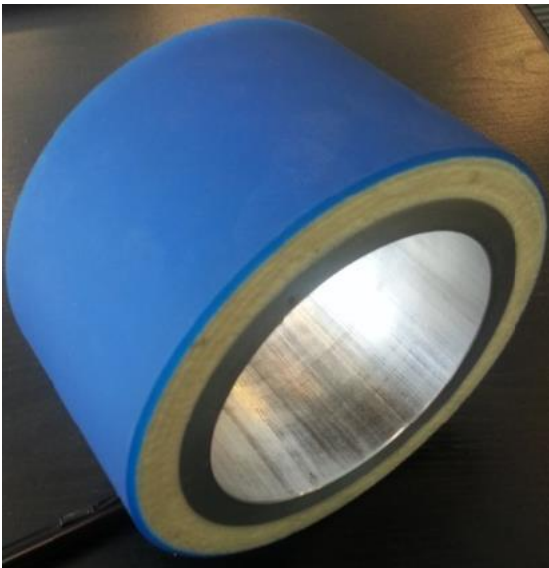
Relative Speeds in Steel-Rubber Nips



Challenges of Co-Axial Winding

Thickness Profile and Shared Nip Roller

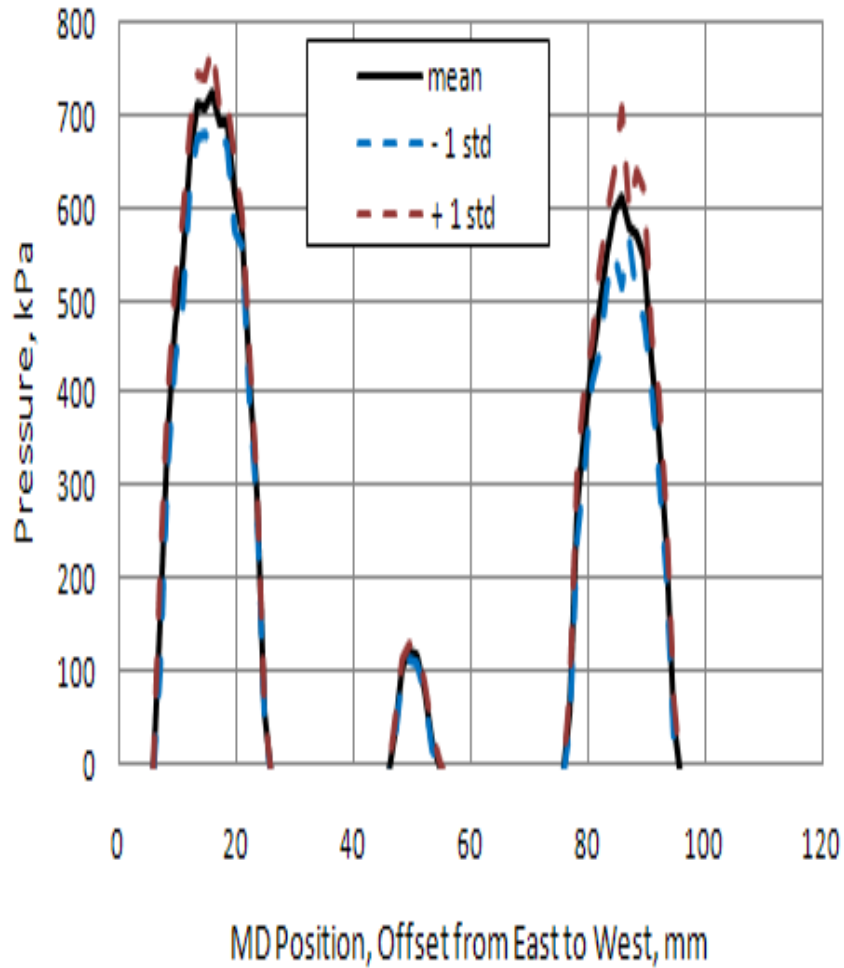
- ▶ The dual Durometer roller covering provide extended compliance around roll diameters with better abrasion resistance than single low Durometer rollers.



Inner thick covering of soft foam rubber.

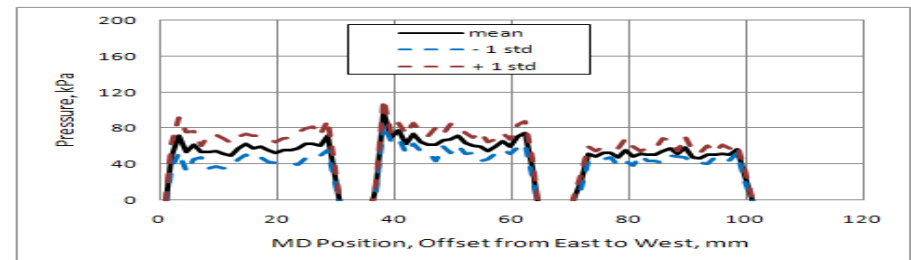
Outer thin covering of high durability rubber.

MD Nip Pressure – Single v. Dual

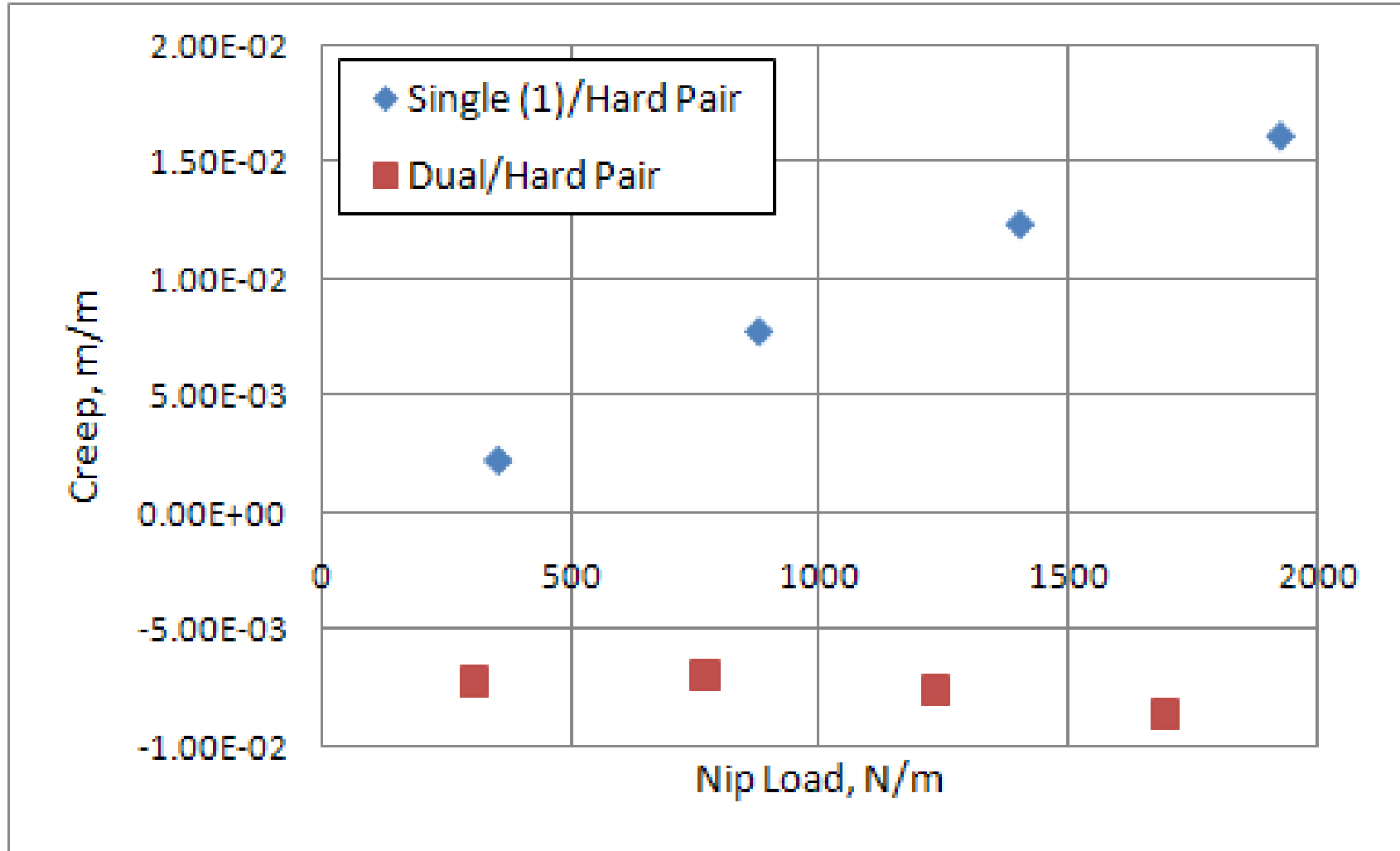


Single Durometer
Nipping Rollers

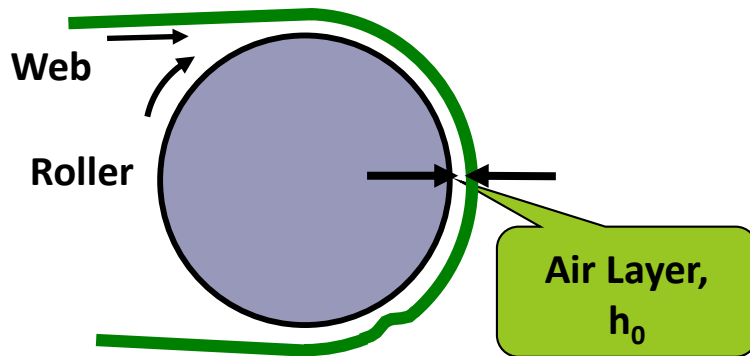
Dual Durometer
Nipping Rollers



Nipped Roller Creep

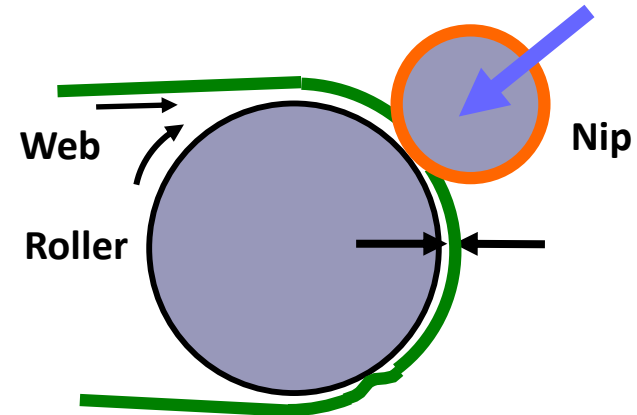


Wrapped vs Nipped Roller Lubrication



$$h_0 = 0.643r \left(\frac{6\eta(V_W + V_R)}{T} \right)^{2/3}$$

h_0 = Steady-state float height, in.
 r = Roller radius, in.
 V_W = Web speed, in/s
 V_R = Roller speed, in/s
 T = Tension, pli



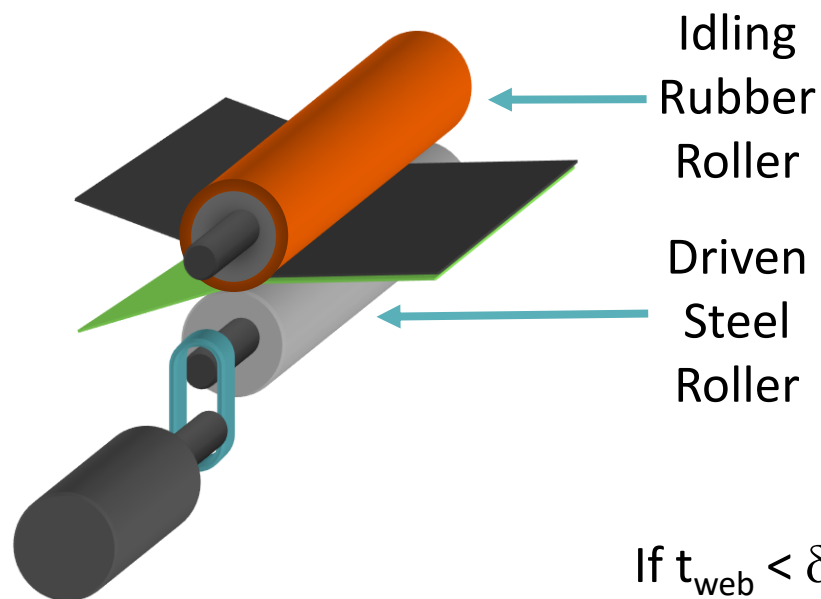
$$h_0 = 4\eta V \frac{r_{eq}}{N}$$

r_{eq} = Equivalent roller radius, in.
 η = Air Viscosity,
 V = Process speed, in/s
 N = Nip load, pli

Tire Traction – Dry vs Wet

- ▶ Jones and Childers report [coefficients of friction](#) of about 0.7 for dry roads and 0.4 for wet roads. The tread design represents an "all weather" compromise. If you were an Indianapolis race driver, you would use "slick" racing tires with no tread. On dry surfaces you might get as high as 0.9 as a coefficient of friction, but driving them on wet roads would be dangerous since the wet road coefficient might be as low as 0.1 .
- ▶ <http://hyperphysics.phy-astr.gsu.edu/>

Nipped Roller Traction



What friction coefficient controls nipped roller traction?
Web-steel? Web-rubber?

t_{web} = web thickness

δ_{rubber} = rubber indentation

If $t_{web} < \delta_{rubber}$

Function of both web-steel
and web-rubber traction



If $t_{web} > \delta_{rubber}$

Function of web-steel only.

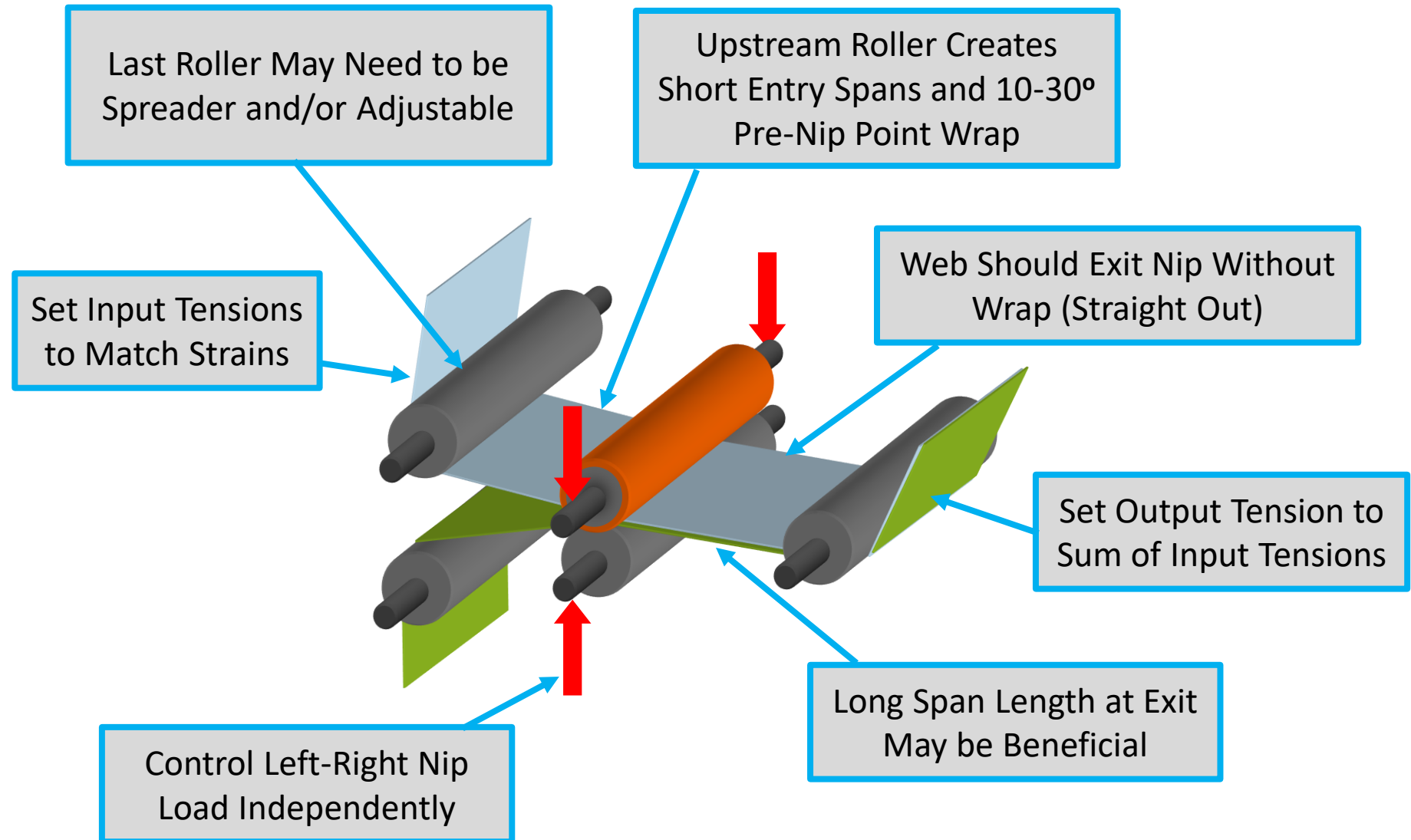


Nipping and Lamination Key Points

How To Nip:

- Pivoting nips are preferred to linear sliding nips.
- Load should be delivered via pneumatic or hydraulic pressure.
- Both rigid and articulating nipping systems can be successful.
- Use flow control to prevent nips from slamming shut or flying open.
- Nips should open when process is stopped for more than a few minutes.
- Nip point should be well-lit and easily viewable.
- Avoid wedging nips.

Nipping and Lamination Best Practices



Understanding Nipped Roller Systems

- ▶ Load vs Indentation
- ▶ Footprint
- ▶ Pressure
- ▶ Deflection
- ▶ Web Strain and Feed Rate
- ▶ Lubrication (gas or liquid)
- ▶ Traction
- ▶ Web Path Geometry
- ▶ Wrinkling, Baggy Webs, Lateral Motion

Ideal Laminator Key Points

Wrinkles:

- *Entry rollers for each web should create short spans and small pre-nip point wrap angles.*
- *Avoid pre-nip wrinkling with a spreader immediately upstream of the nip point.*
- *Avoid 90-degree wraps on nipping rollers to prevent deflection-induced tracking wrinkles.*
- *Minimize accumulation wrinkles by pulling out bagginess with tension.*
- *Avoid larger wrap angles on all rubber nip rollers.*
- *Use large wrap angles only when pre-heating is needed.*
- *For cambered webs, use a pre-nip skewing roller to avoid wrinkles.*

Recommendation #3

Minimize accumulation wrinkles
by pulling out bagginess with
tension.

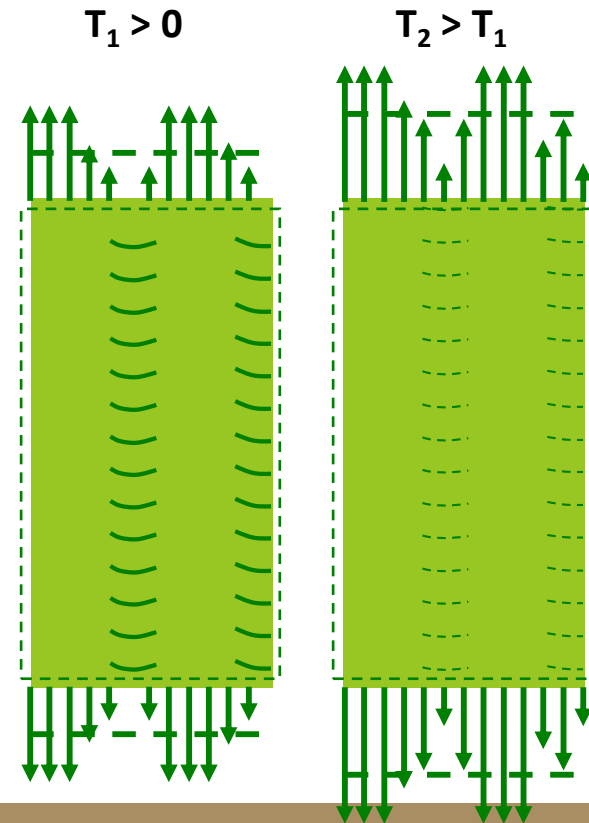
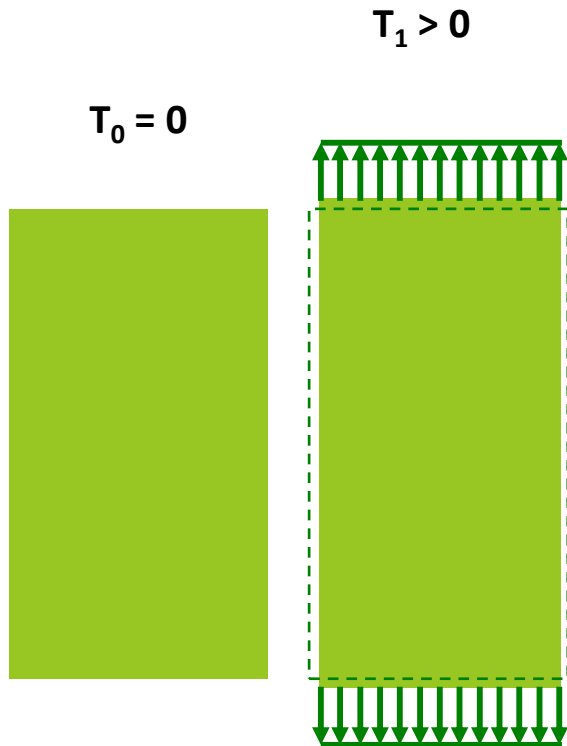
Why? Non-slipping nipped roller regulate speed and length of web. Any excess length (bagginess) will accumulate and wrinkle.



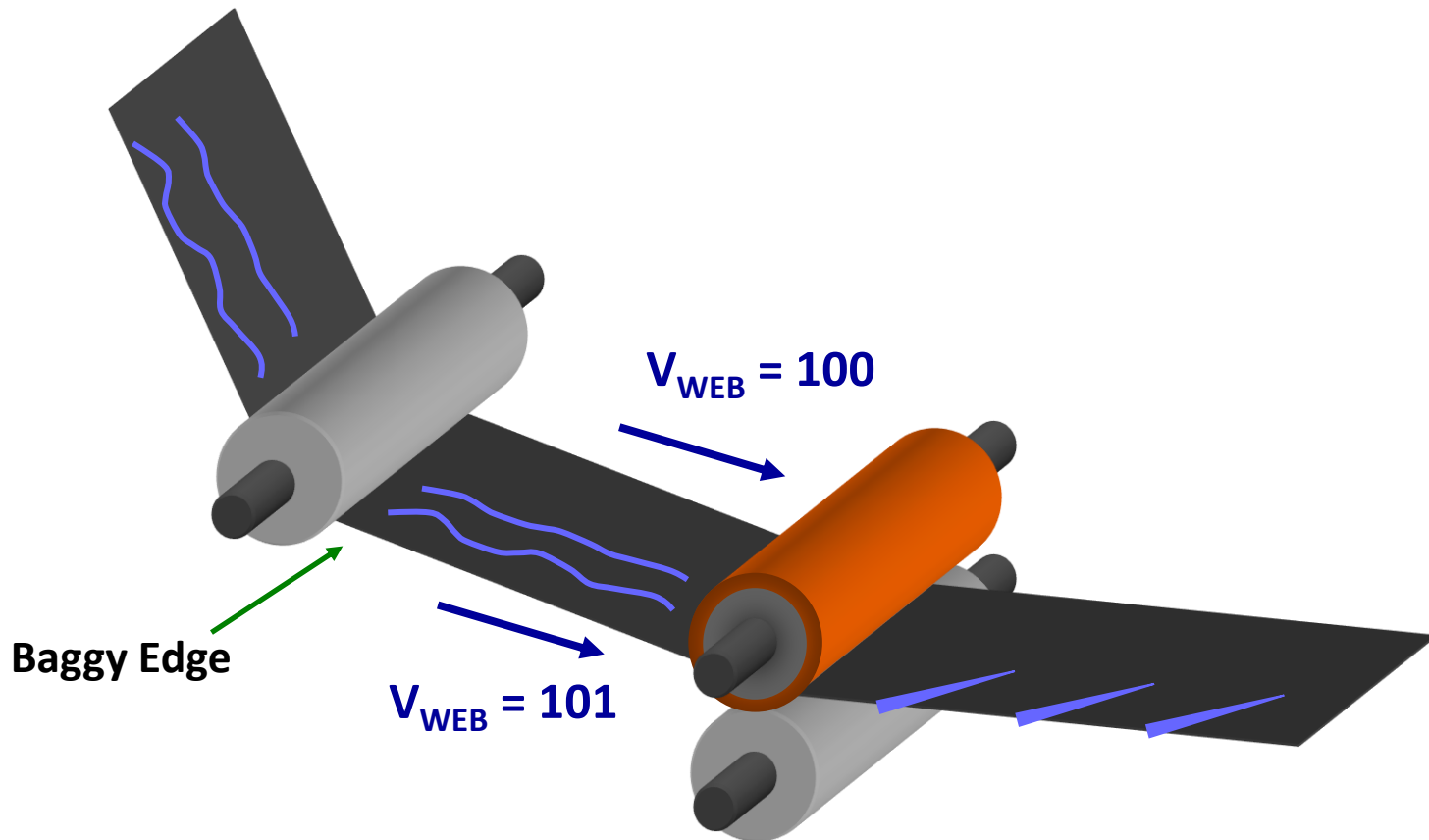
Tensioning Baggy Webs

The ideal web carries tension uniformly across the web width.

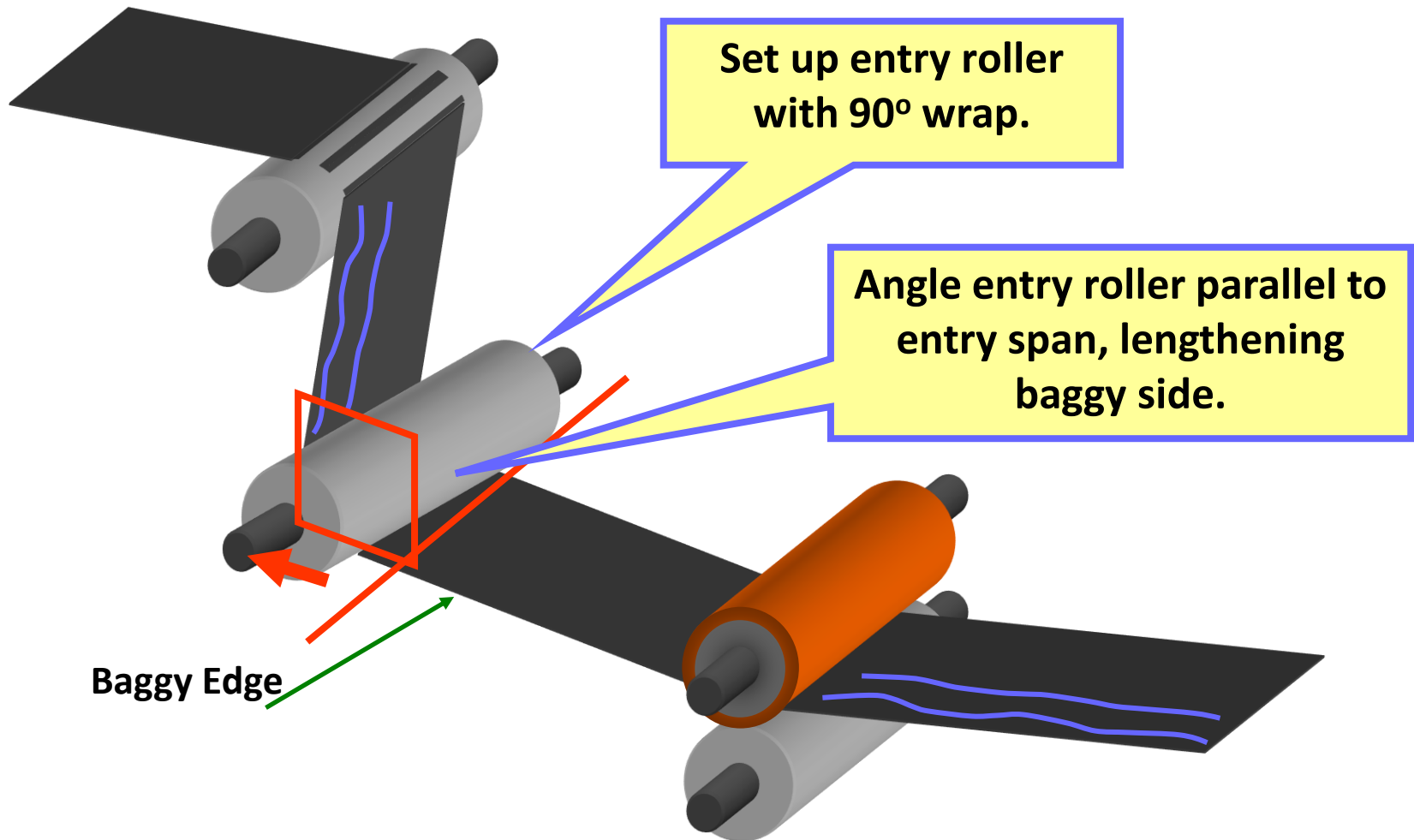
For an imperfect web, tension stretches the short lanes first. When short lanes are stretched to equal the long lanes, the web appears taut.



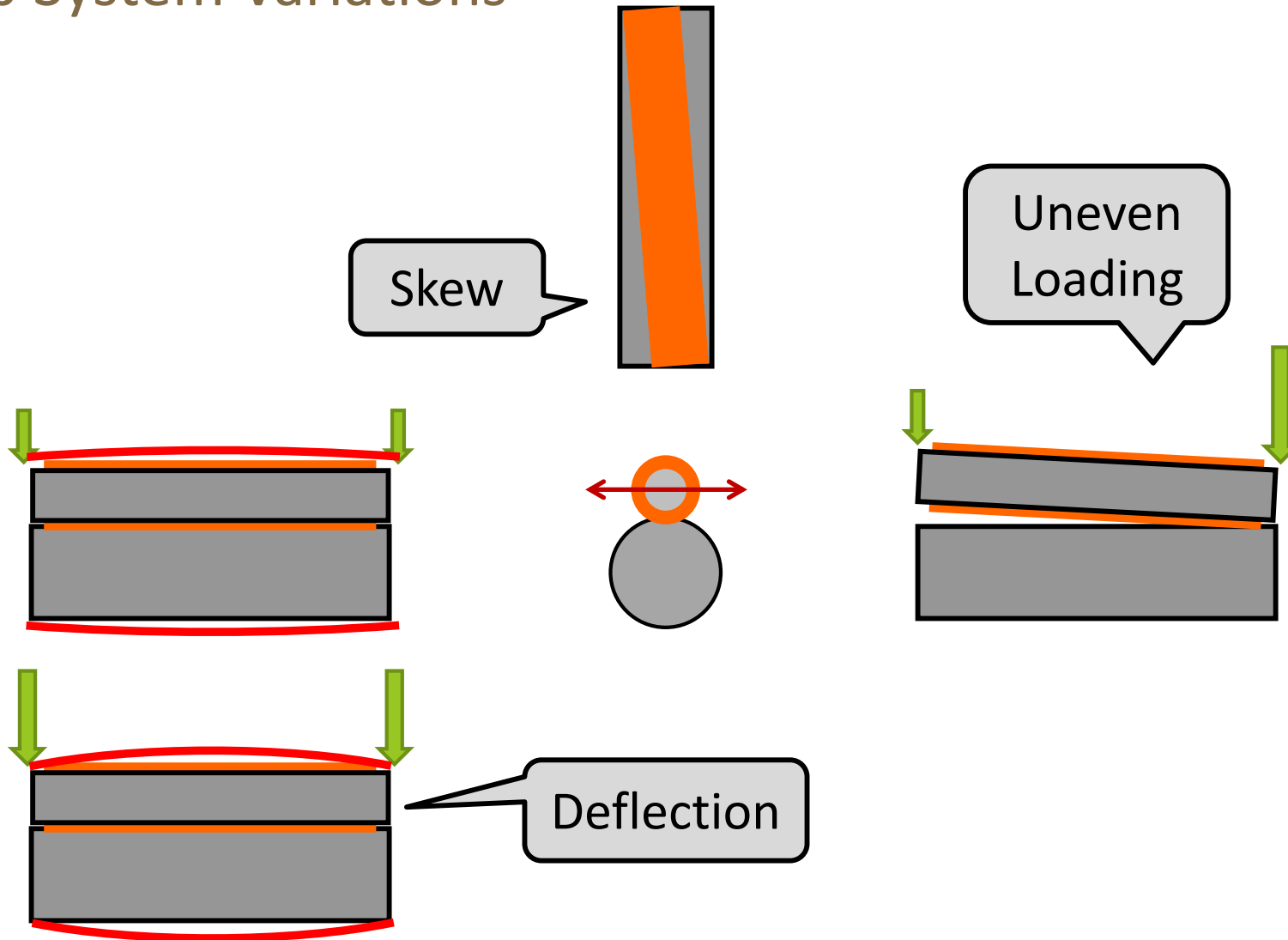
Worst Case: Baggy Web + Nips



Adjustable Roller: Best Practice



Nip System Variations



Entry Web Path Options: Steel-Rubber Nips

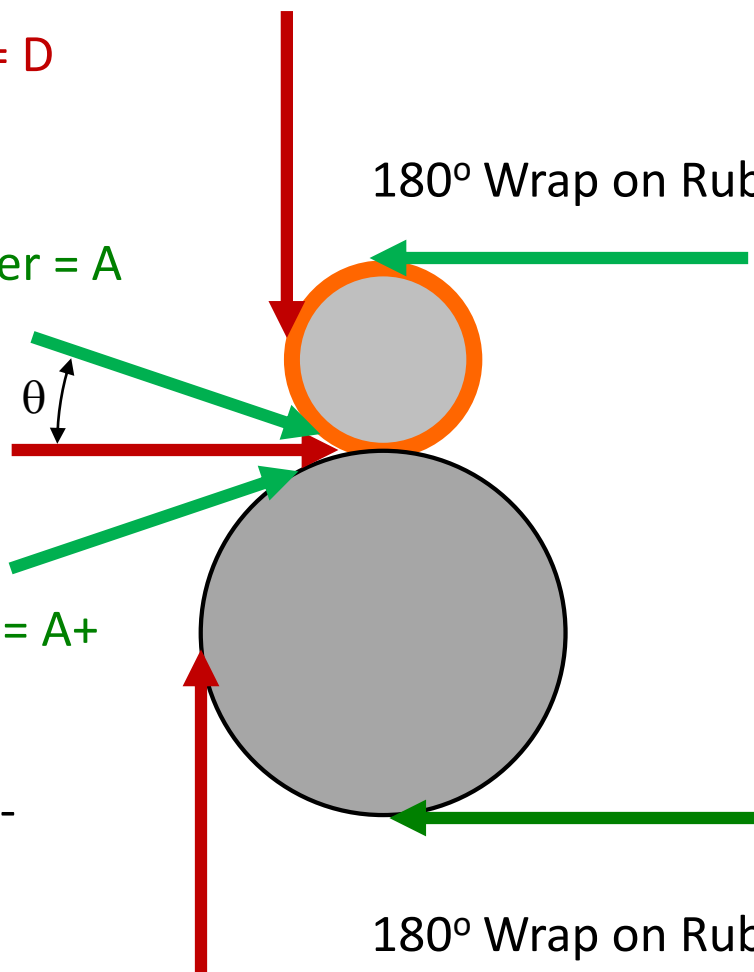
90° Wrap on Rubber Roller = D

10-30° Wrap on Rubber Roller = A

Tangential (0° Wrap) = D-

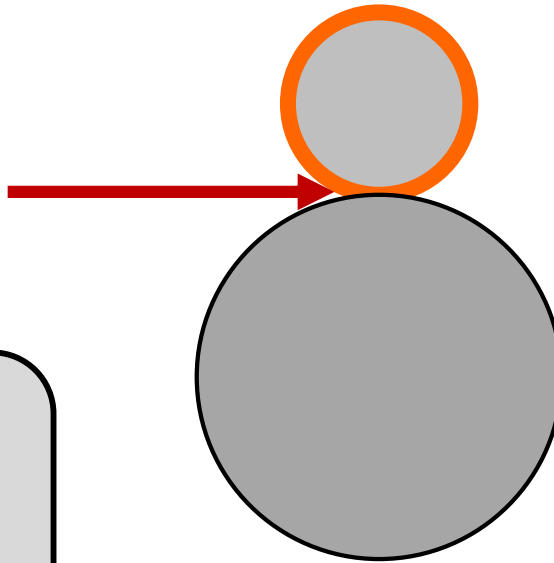
10-30° Wrap on Steel Roller = A+

90° Wrap on Steel Roller = C-



Entry Web Path Options: Steel-Rubber Nips

Tangential (0° Wrap) = D-

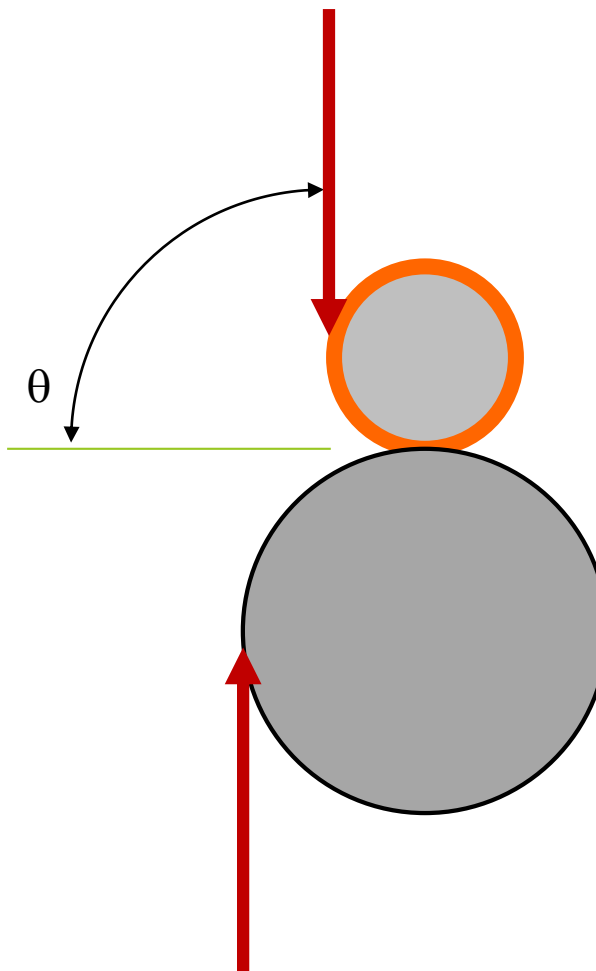


- No shape stiffening
- No pre-nip wrap friction
'isolation'
- Trouble with skew, uneven loading, deflection

Entry Web Path Options: Steel-Rubber Nips

90° Wrap on Rubber Roller = D

- Deflection gathers web = converging wrinkles
- Uneven loading shifts web laterally = shear wrinkles
- Insensitive to skew (web twist = no problem)

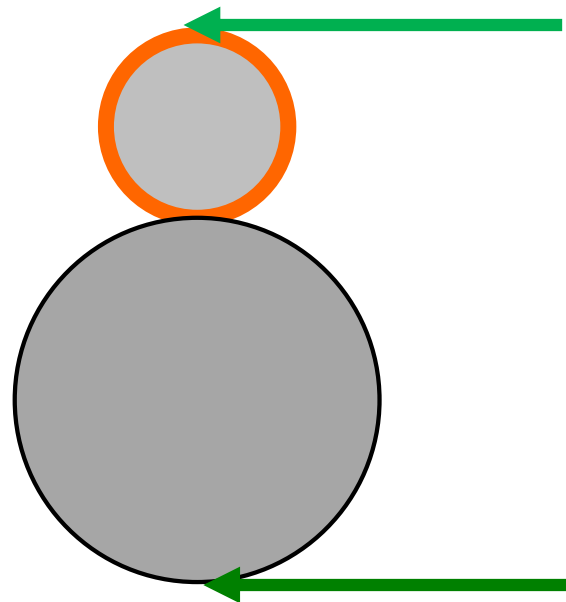


90° Wrap on Steel Roller = C-

Entry Web Path Options: Steel-Rubber Nips

- Skew shifts web laterally = shear wrinkles
- Insensitive to deflection and uneven loading
- Good for pre-heating lamination
- Increase rubber roller wear from speed differential of wrapped and nipped rubber covering.

180° Wrap on Rubber Roller = B

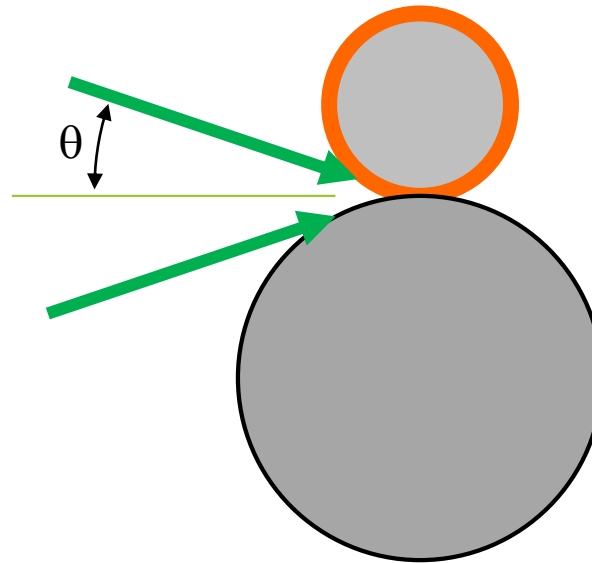


180° Wrap on Steel Roller = B+

Entry Web Path Options: Steel-Rubber Nips

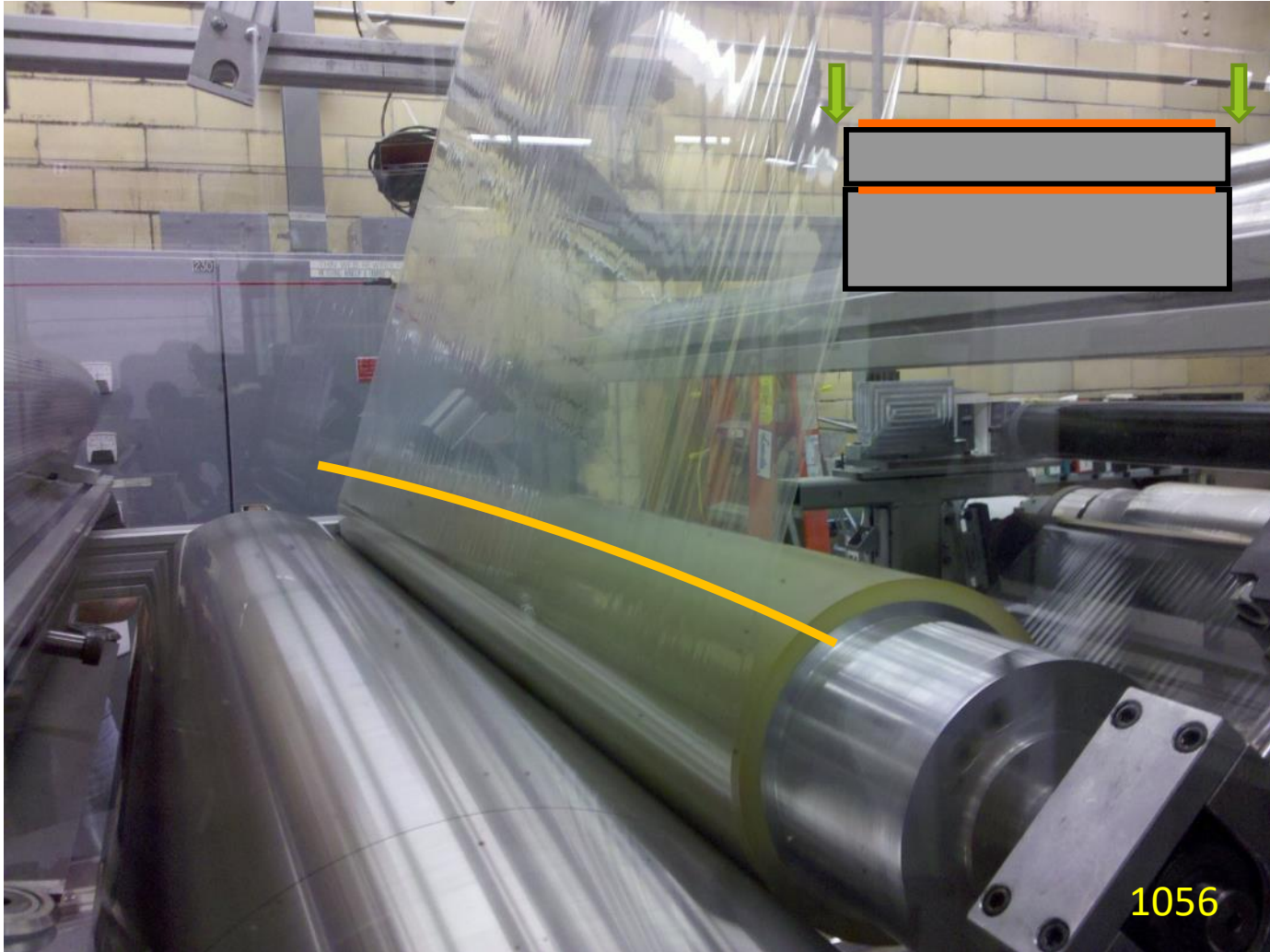
- Insensitive to deflection
- Insensitive to uneven loading misalignment
- Small pre-nip wrap angle creates shape stiffening
- Minimum rubber roller wear
- Uneven loading speed variations cause lateral shift and shear wrinkles
- Skew shifts web laterally = shear wrinkles

10-30° Wrap on Rubber Roller = A



10-30° Wrap on Steel Roller = A+

No Skew, Nip Low & Even, High Tension



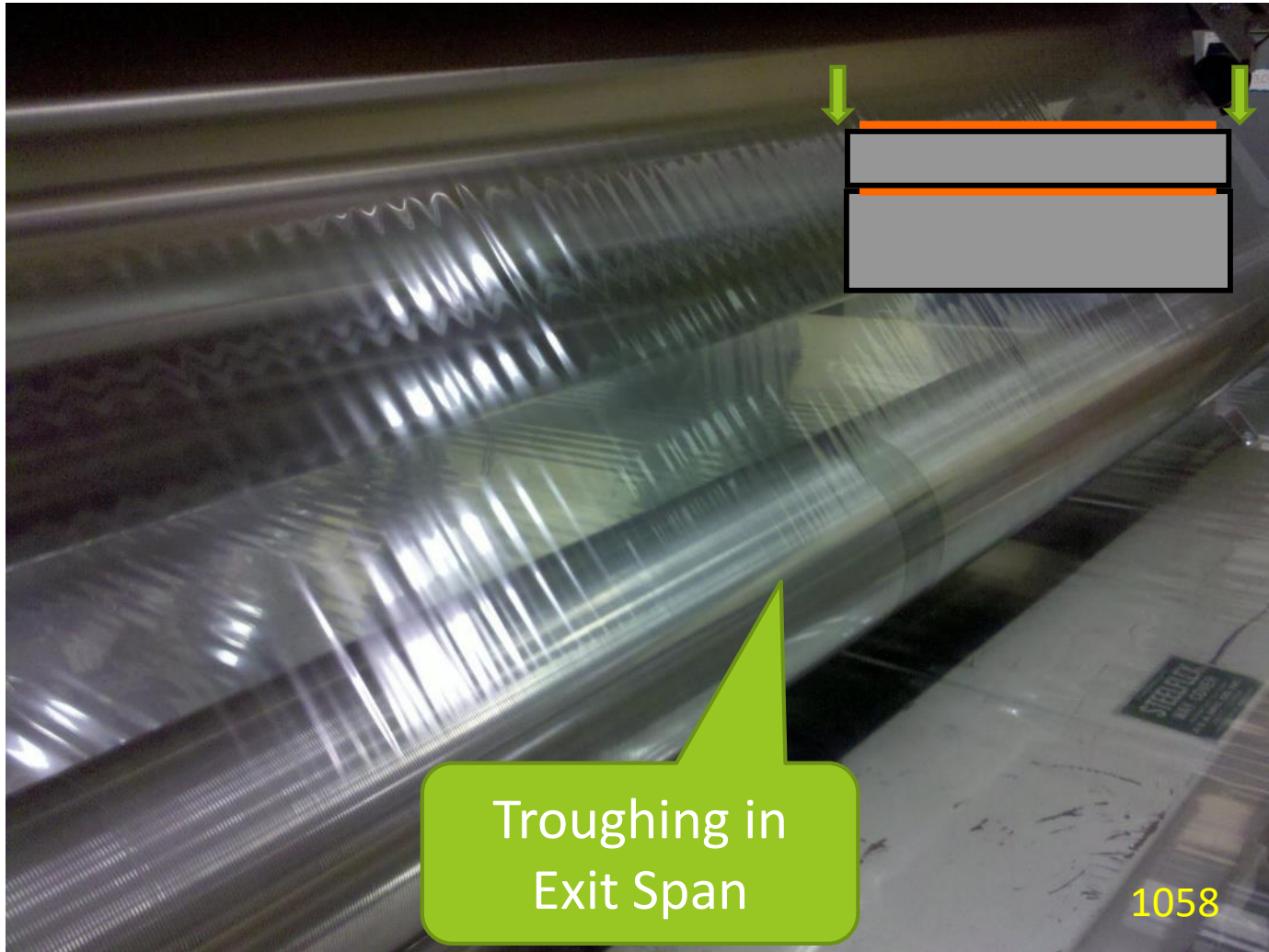
Nip Far
= 25 lbf

Nip Near
= 25 lbf

Skew
= 0

Tension
= 0.6 PLI

No Skew, Nip Low & Even, High Tension



Nip Far
= 25 lbf

Nip Near
= 25 lbf

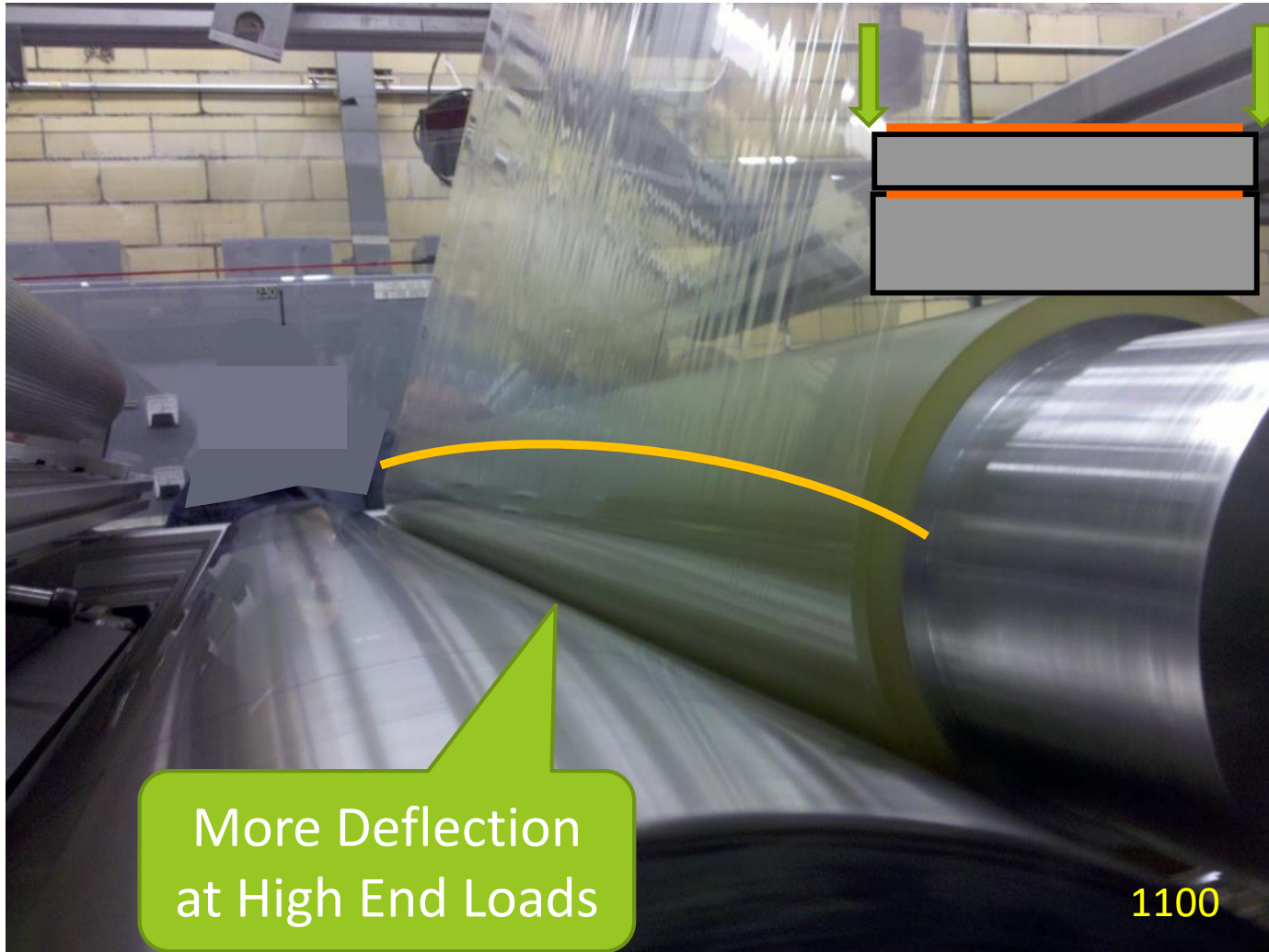
Skew
= 0

Tension
= 0.6 PLI

Troughing in
Exit Span

1058

No Skew, Nip Med. & Even, High Tension



Nip Far
= 150 lbf

Nip Near
= 150 lbf

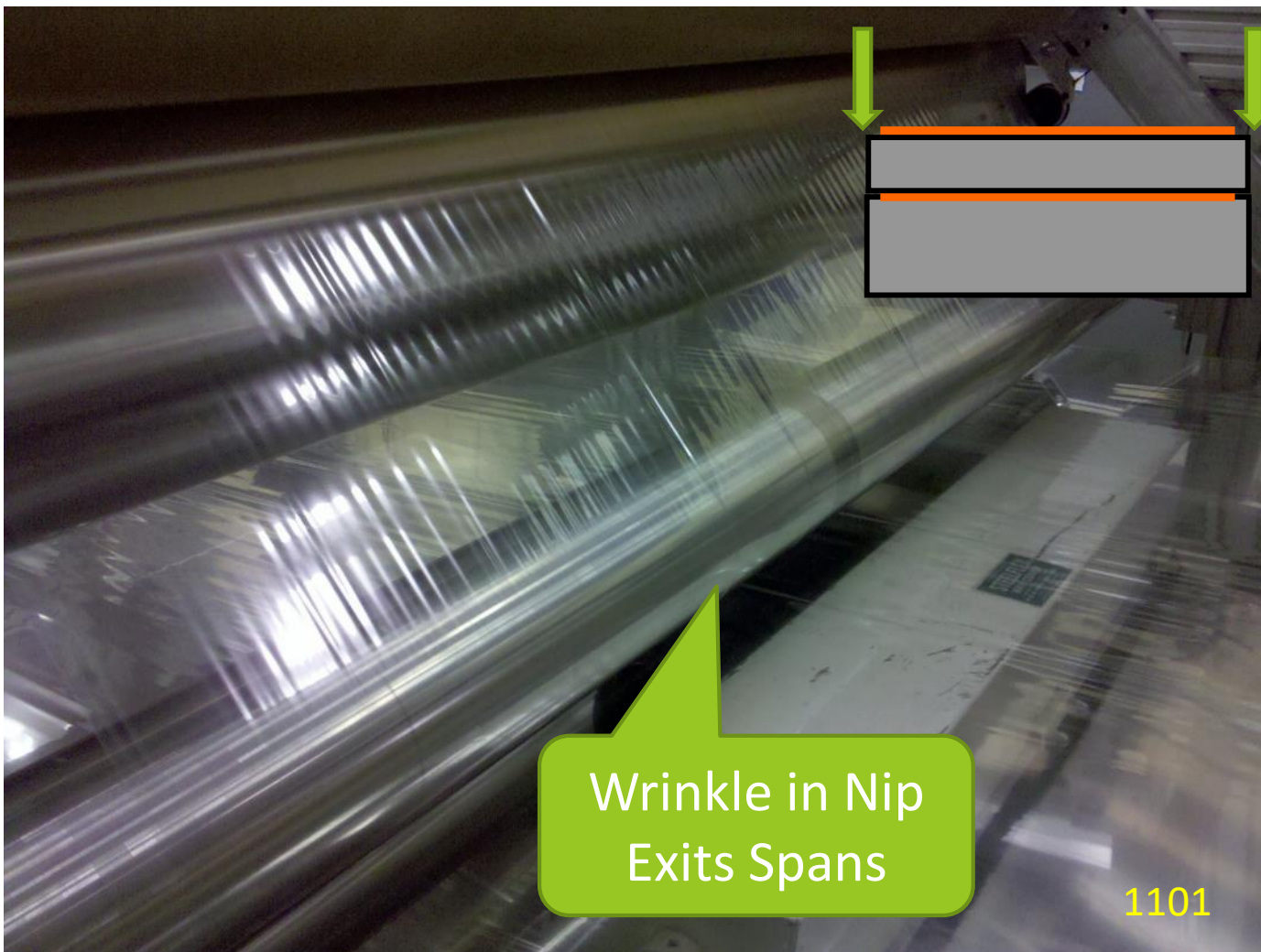
Skew
= 0

Tension
= 0.6 PLI

More Deflection
at High End Loads

1100

No Skew, Nip Med. & Even, High Tension



Nip Far
= 150 lbf

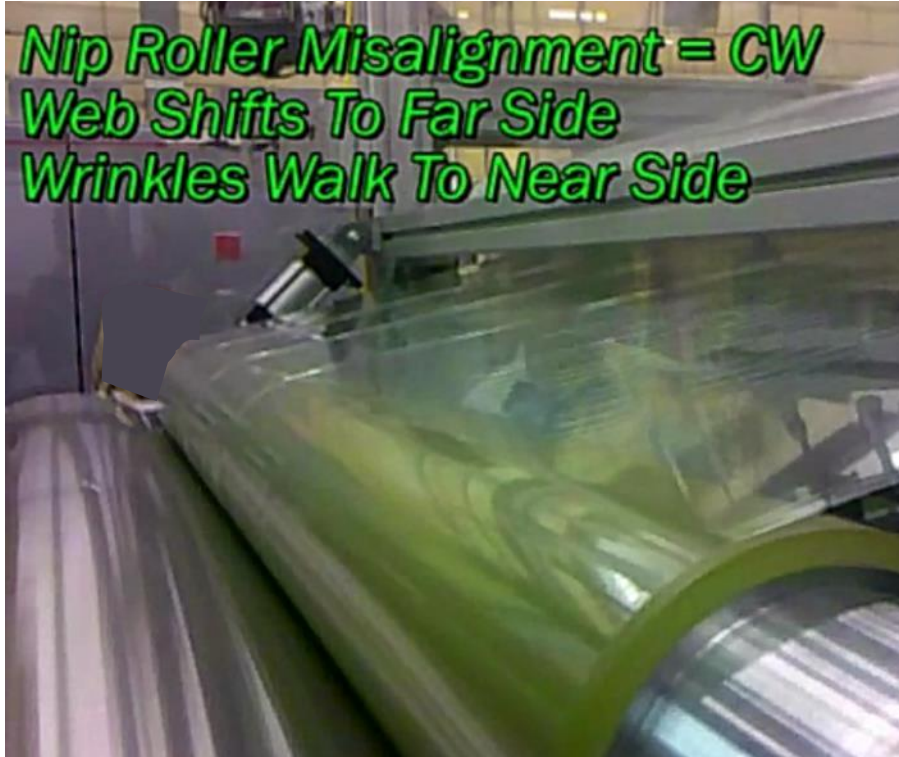
Nip Near
= 150 lbf

Skew
= 0

Tension
= 0.6 PLI

Nip Wrinkles From Skewing

Nip Roller Misalignment = CW
Web Shifts To Far Side
Wrinkles Walk To Near Side



Nip Roller Misalignment = CCW
Web Shifts To Near Side
Wrinkles Walk To Far Side



Lamination Defined

Lamination is...

The process of combining two or more substrates, creating a product of combined properties not found in any layer individually.

Laminated Products

- Flexible Packaging
- Cartons
- Heat Sealable Insulation
- IDs (passports, licenses)
- RFID Tags
- LCD Displays
- High-Tech Fabrics
- Reflective Sheeting
- Labelstock
- Trash Bags
- Flooring
- Medical Pouches
- Batteries
- Adhesives
- Book Covers, Maps
- In-Process “Carriers”

Laminate Properties

- ▶ Moisture Barrier
- ▶ Light Barrier
- ▶ Vapor Barrier
- ▶ Oxygen Barrier
- ▶ Abrasion Resistance
- ▶ Heat Sealability
- ▶ Gloss/Matte Finish
- ▶ Aesthetics
- ▶ Tear Resistance
- ▶ Break Strength
- ▶ Modulus (Stiffness)
- ▶ Release

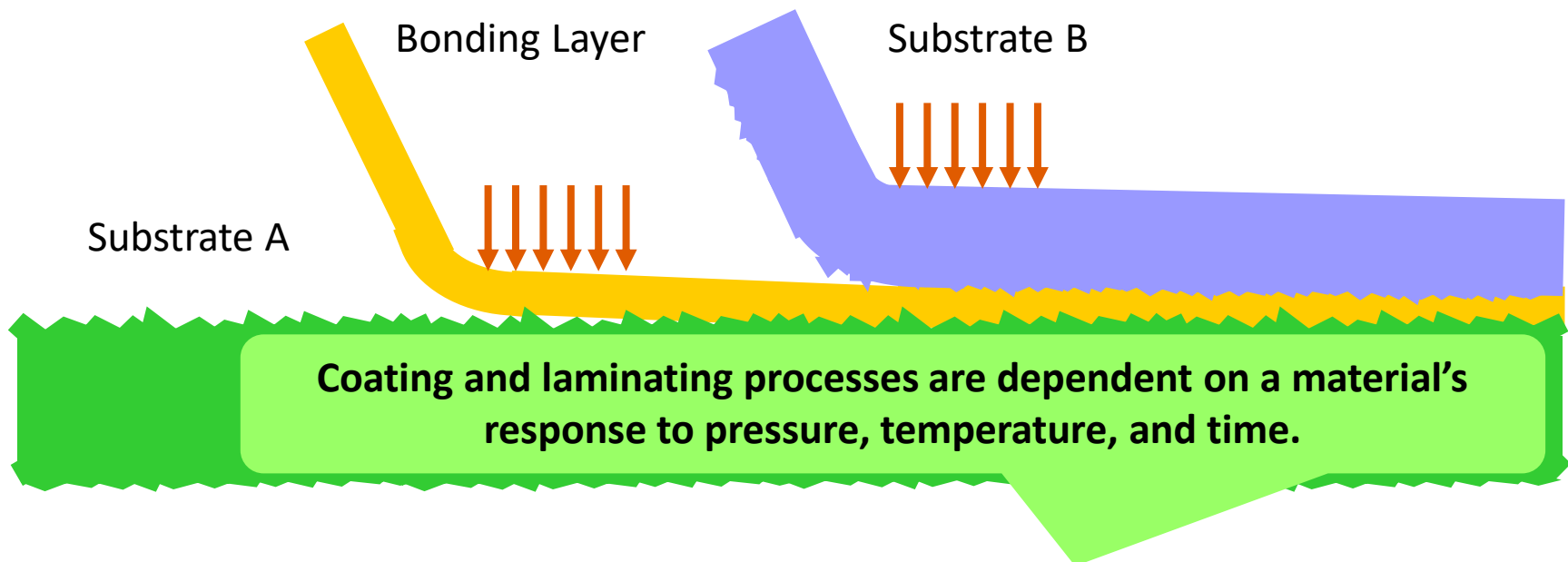
Lamination Micro-View

Coat:

Applying the bonding layer to A,
forcing intimate contact, excluding air
at A interface.

Laminate:

Applying substrate B to the bonding layer,
forcing intimate contact, excluding air at B-
interface.



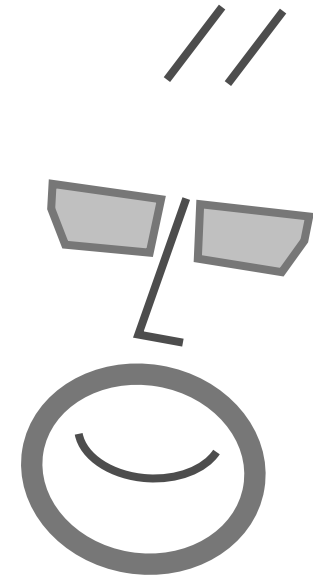
Nipping and Lamination Key Points

Tensions & Curl:

- Minimize tension-induced curl by tensioning each web to match pre-laminating strains.
- Consider thermal and hygroscopic expansion coefficients.
- Consider Poisson's ratio. If mismatched, laminate at lowest tension possible to avoid CD curl.
- Avoid post-nip wrap on nipping rollers.
- Exit tension set equal to the sum of the input tensions.
- Driven steel roller and idling or torque driven rubber roller.
- If needed, drive the compliant roller with torque control.

Tension & Laminating Qs

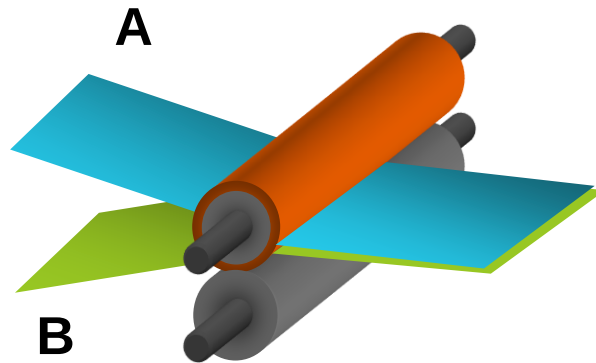
How can laminating tension
create curl?



Elastic Curl

The most common cause of curl is mismatched strains at lamination.

$$\epsilon_A > \epsilon_B$$



Flat under tension.



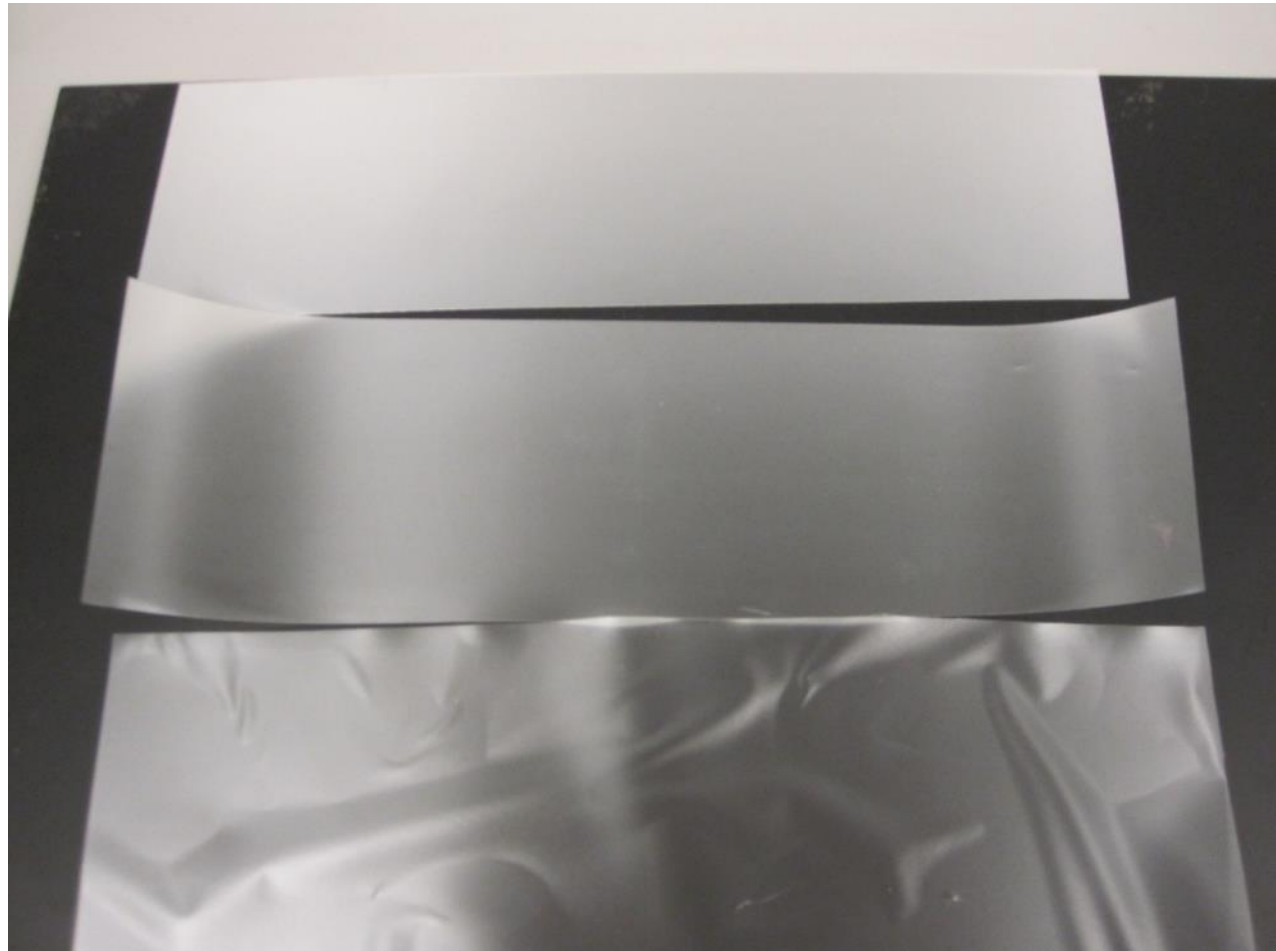
Curled when tension is removed.

Example: Laminate with Strain Mismatch

Matched strains:
Flat laminate

Thin film strain 30%
above thick film:
Curled laminate

Thin film strain 60%
above thick film:
Delaminated laminate



2C: Curled Webs

Problems:

- o *Curled web may fold over in handling, especially in long spans.*
- o *Customer requiring untensioned flatness in their end use.*

Elastic Curl

1. Materials are laminated under tension.



2. When tension is removed, materials return to unstrained dimensions.

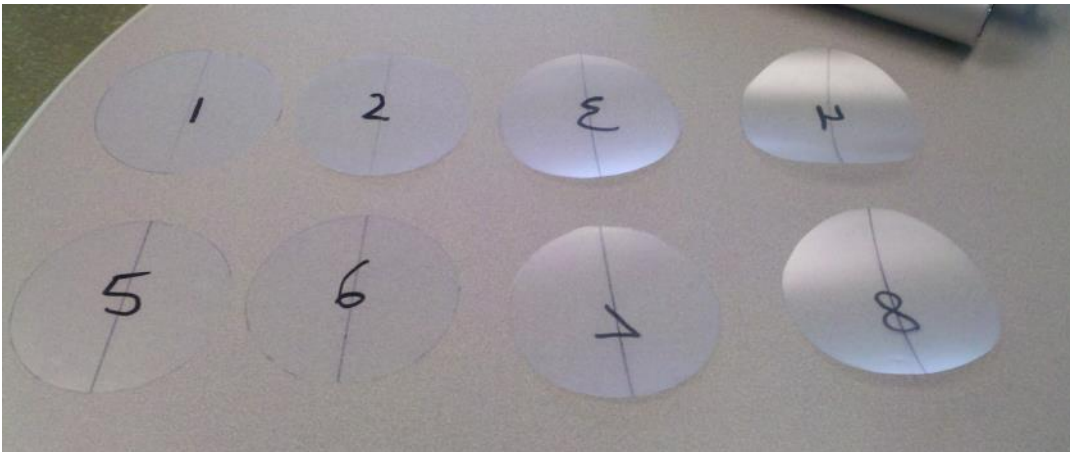


3. Non-uniform recovery creates curl.

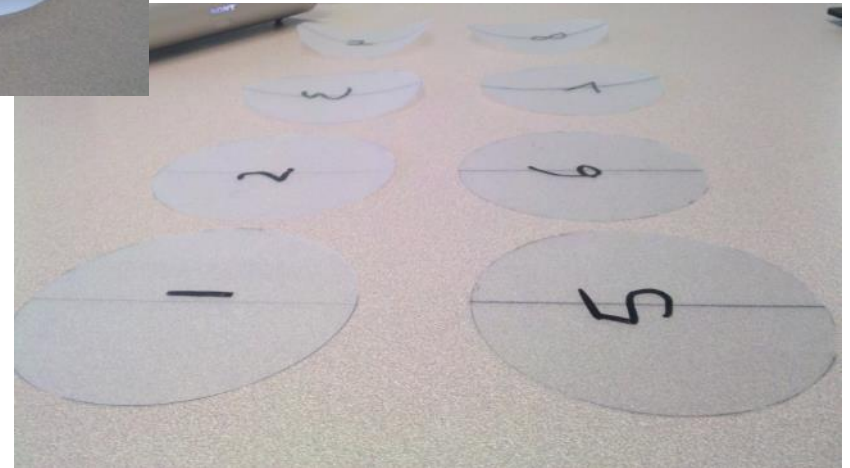


Checking Curl with Circular Samples

Increasing excessive strain on Side A



Increasing excessive strain on Side B



Calculating Curvature of Curl

$$K = \frac{6E_A E_B (h_A + h_B) h_A h_B \Delta \epsilon}{E_A^2 h_A^4 + 4E_A E_B h_A^3 h_B + 6E_A E_B h_A^2 h_B^2 + 4E_A E_B h_A h_B^3 + E_B^2 h_B^4}$$

K = 1/radius of curvature

E_A = Young's modulus of layer A

E_B = Young's modulus of layer B

h_A = thickness of layer A

h_B = thickness of layer B

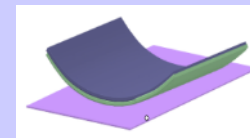
$\Delta \epsilon$ = strain differential (from tension or expansion from thermal or moisture)

Anything that causes strain mismatch can create curled laminates.

2A: Curled Webs

Root Causes

1. Strain mismatch in laminating
2. *Variations in thermal expansion or contraction*
3. *Shrinkage from drying or curing*
4. *Hygroscopic expansion or contraction*
5. *Polymer relaxation*
6. *Yielding or viscoelastic strain*



Curled
Webs

Recommendation #1

Minimize curl by tensioning each web to match pre-laminating strains.

Why? Laminates with matched strains will contract equally when untensioned, avoiding curl.



Rule #1 of Laminating

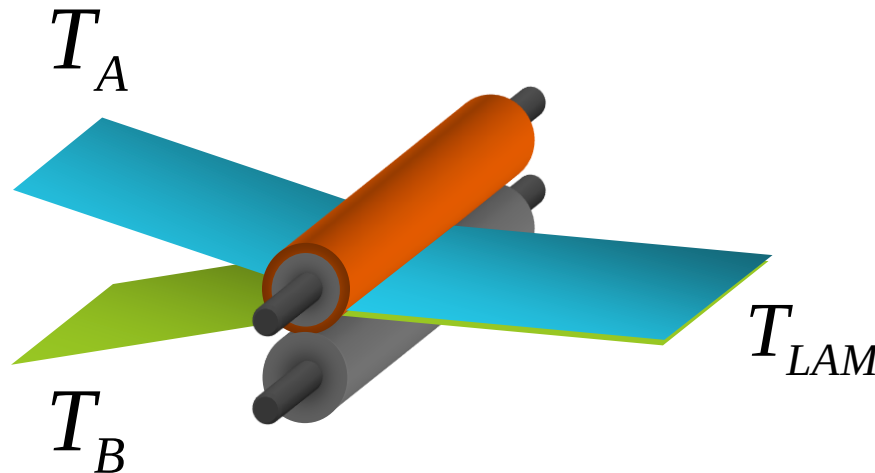
Match Strains of Incoming Webs.

To prevent elastic curl, set input tension ratio proportional to web spring constants.

$$\varepsilon_A = \varepsilon_B$$

$$\frac{T_A}{T_B} = \frac{t_A E_A}{t_B E_B}$$

$$\varepsilon_A = \frac{\sigma_A}{E_A} = \frac{T_A}{t_A E_A}$$



Rule #1 of Laminating

Example Strain Calcs:

$$\varepsilon_A = \frac{\sigma_A}{E_A} = \frac{T_A}{t_A E_A}$$

$$T_A = 0.2 \text{ PLI}$$

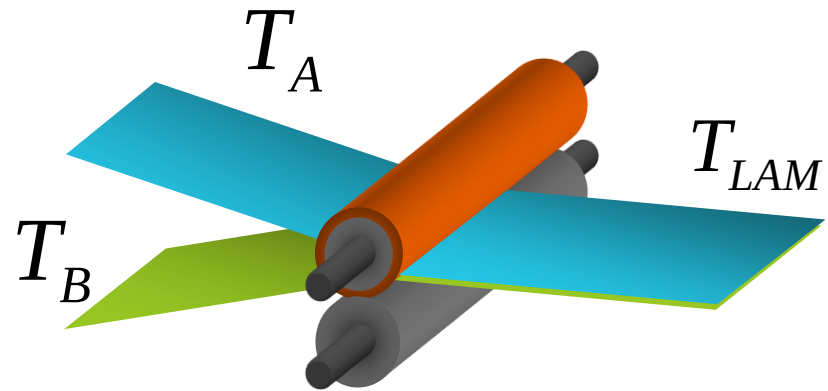
$$t_A = 0.001''$$

$$E_A = 50,000 \text{ psi (OPP)}$$

$$\text{Stress} = (0.2)/(0.001) = 200 \text{ psi}$$

$$\text{Strain} = (200)/(50000) = \underline{0.4\%}$$

$$\begin{aligned} \text{Strain / PLI} &= (0.4\%)/(0.2 \text{ PLI}) \\ &= 2\% \text{ per PLI} \end{aligned}$$



$$T_B = 1.0 \text{ PLI}$$

$$t_B = 0.0005''$$

$$E_B = 500,000 \text{ psi (PET)}$$

$$\text{Stress} = (1.0)/(0.0005'') = 2000 \text{ psi}$$

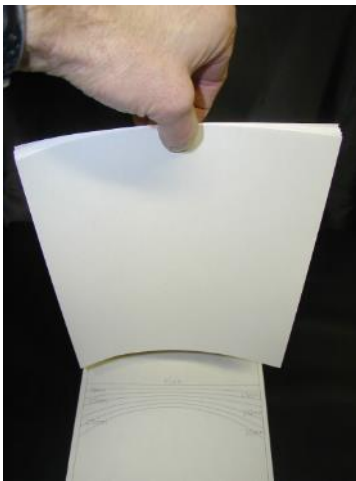
$$\text{Strain} = (2000)/(500000) = \underline{0.4\%}$$

$$\begin{aligned} \text{Strain / PLI} &= (0.4\%)/(1.0 \text{ PLI}) \\ &= 0.4\% \text{ per PLI} \end{aligned}$$

Strain Matched

Curl Measurement

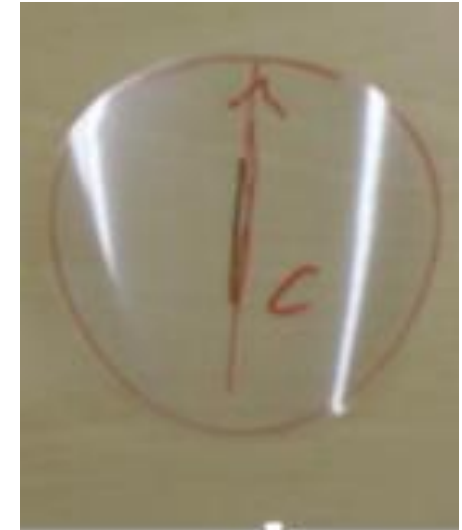
Strip or Sheet Curl



X Curl

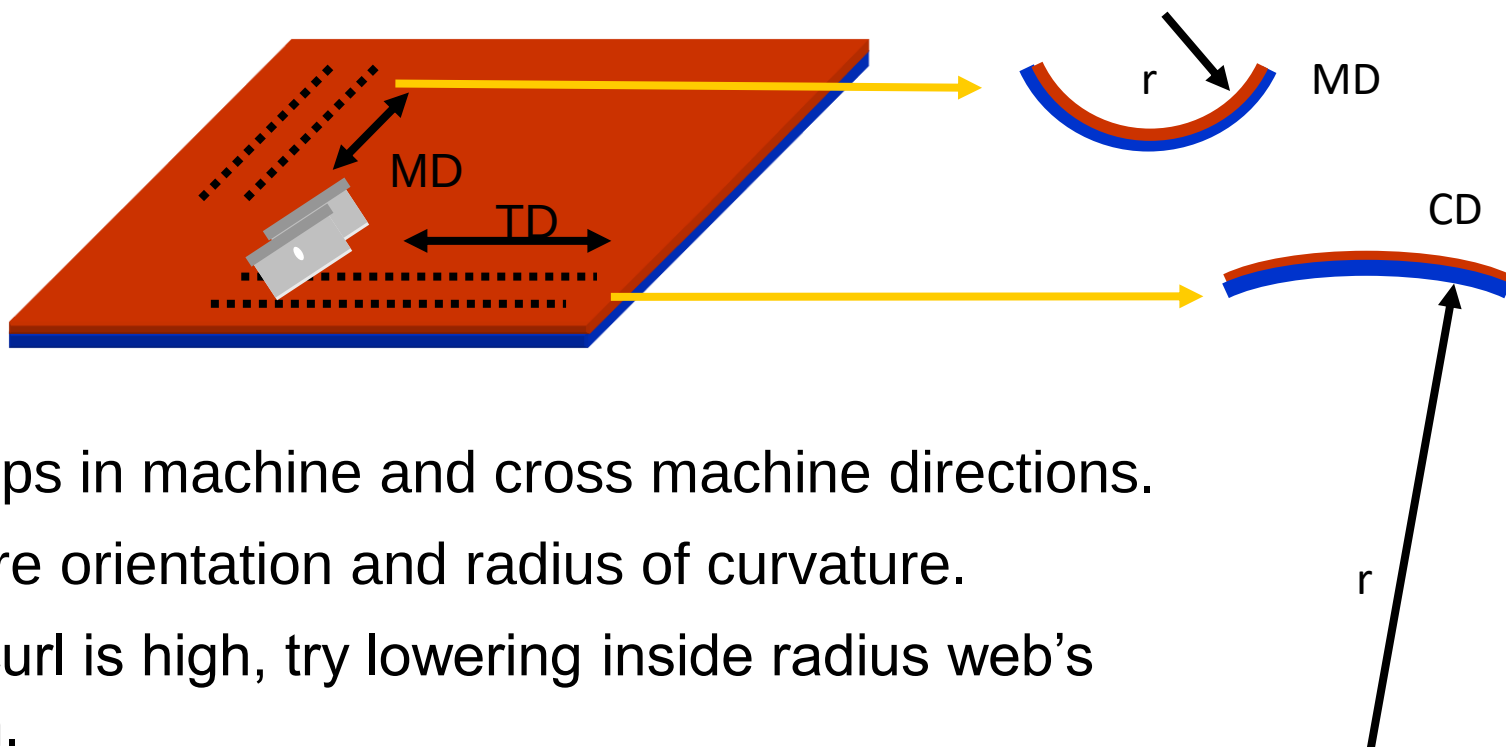


Circle Curl



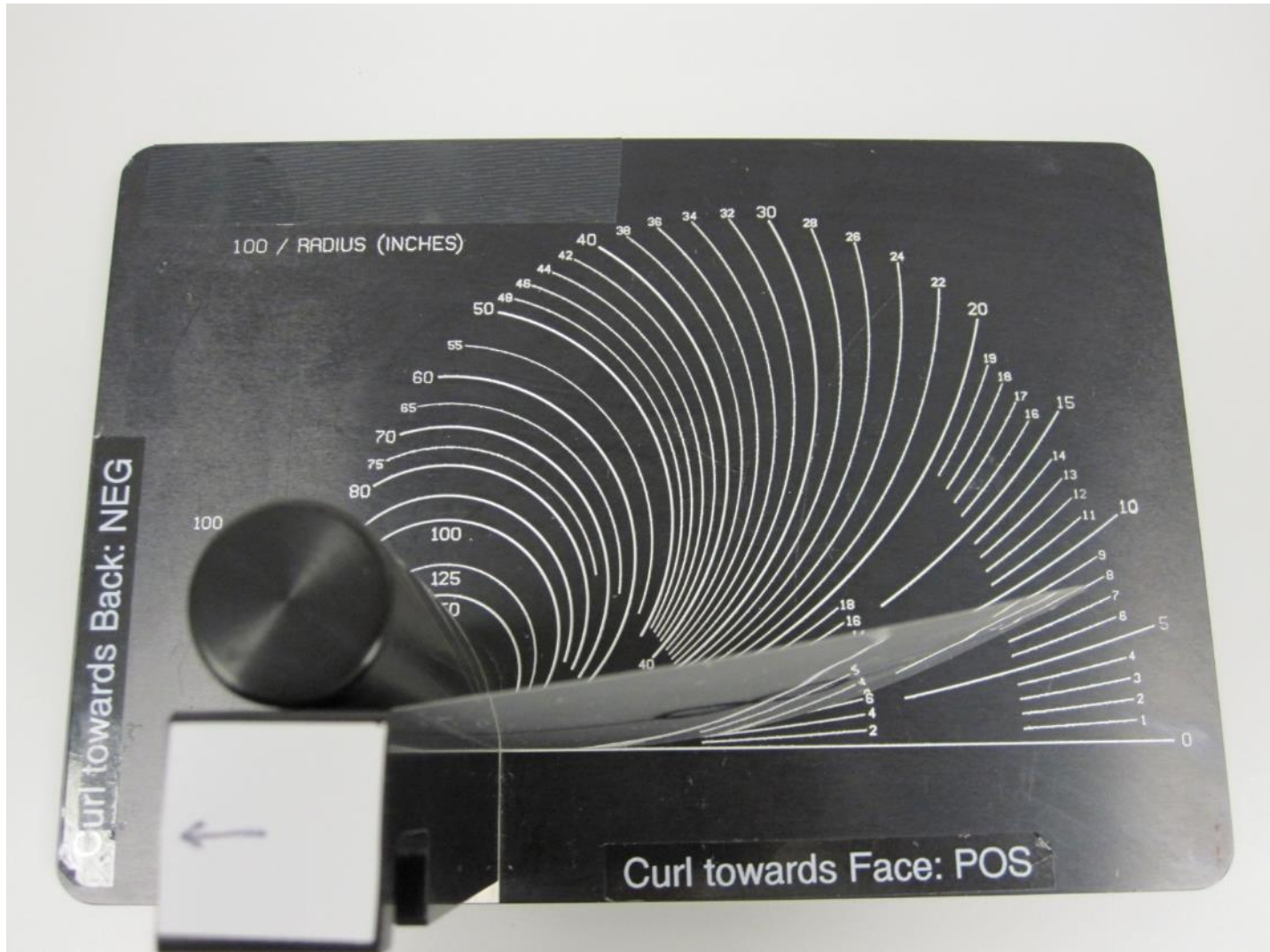
Measure Laminate Curl

Quantify curl magnitude and direction with MD and CD strips.



1. Cut strips in machine and cross machine directions.
2. Measure orientation and radius of curvature.
3. If MD curl is high, try lowering inside radius web's tension.
4. If CD curl is high, look for cause in coating or web dimensional change.

Film Curl Gauge



Checking Curl with Circular Samples

Crossweb curl samples from 'O' operator side to 'A' automated or drive side.



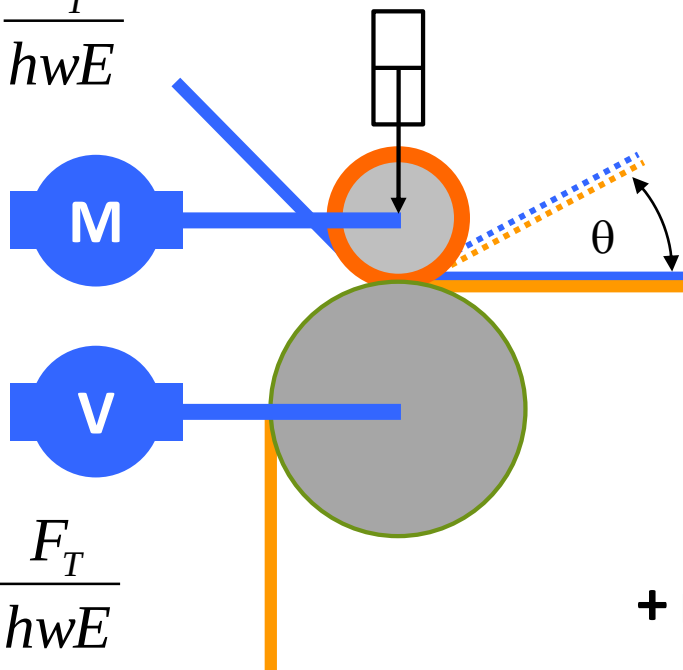
Key Variables of Laminate Curl

Tension

Moisture (paper)

Temperature

$$\varepsilon = \frac{F_T}{hwE}$$



$$\varepsilon = \frac{F_T}{hwE}$$

Tension

Moisture (paper)

Temperature

What create more negative curl?

- Lower top tension (or higher bottom tension)
- Higher input moisture in bottom, if paper/nylon (or lower top moisture)
- Higher input temperature in bottom web (or lower top web temperature)
- Higher rubber roller torque (M)*
- Higher nip load*
- Wrap bottom roller on exit side of nip* (*Rubber roller on top side. These effects are not well understood.)

+ Positive Curl



— Negative Curl



Recommendation #2

Set exit tension equal to the sum of the input tensions.

Why? To run at minimum loads without slippage.

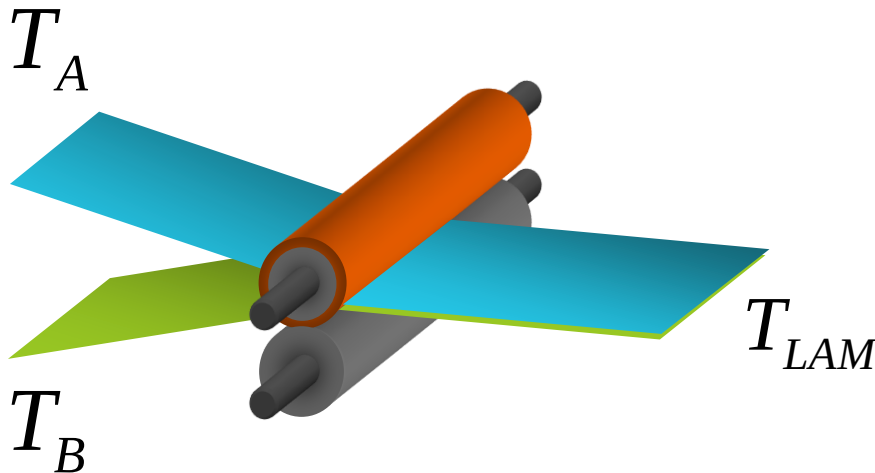


Rule #2 of Laminating

Set Laminate Tension to Sum of Input Tensions.

To reduce in-nip shear and allow minimum nip load without slippage.

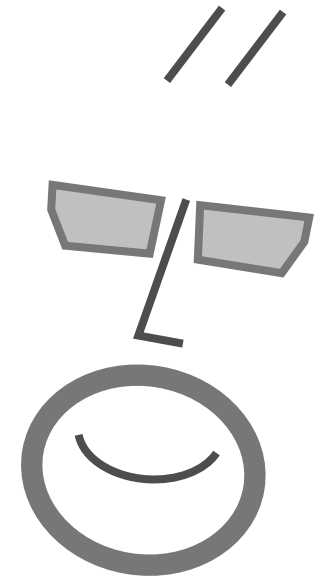
$$T_{LAM} = T_A + T_B$$
$$\Delta T_{LAM} = 0$$



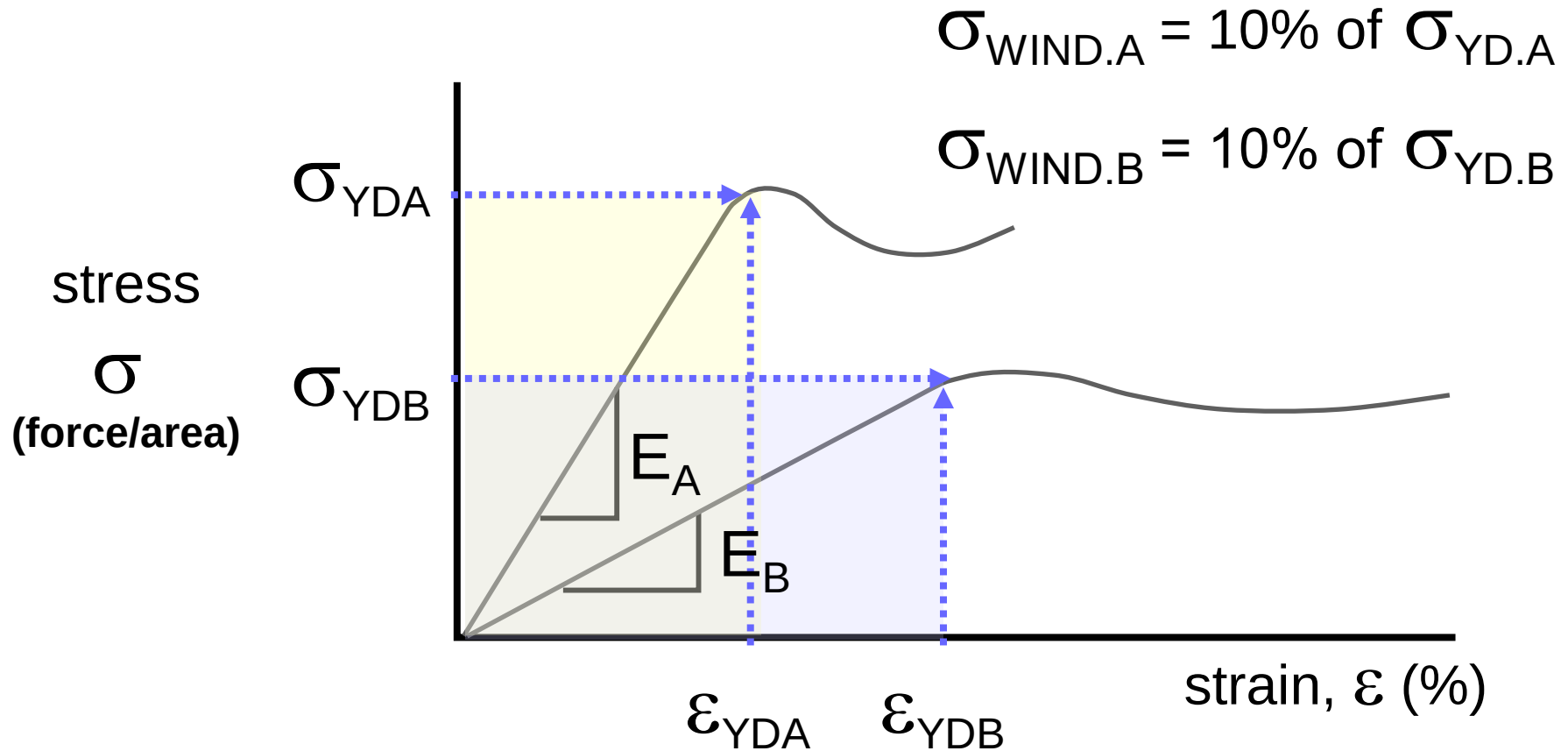
Laminate Modulus

What is the effective modulus of a multi-layer web?

$$E_{LAM} = \frac{E_A t_A + E_B t_B + E_C t_C + \dots}{t_A + t_B + t_C + \dots}$$



Web Handling Stress



Target handling strain = 10% of yield strain.

Calculating Laminate Modulus



12 micron Aluminum
75 micron Polyethylene

$$\begin{aligned}
 E_{\text{LAM}} &= (E_{\text{AL}} t_{\text{AL}}) + (E_{\text{PE}} t_{\text{PE}}) / (t_{\text{AL}} + t_{\text{PE}}) \\
 &= [(69\text{GPa})(12\mu\text{m}) + (1.0\text{GPa})(75\mu\text{m})] / (12\mu\text{m} + 75\mu\text{m}) \\
 &= (828 + 75) / (87) \\
 &= 10 \text{ GPa}
 \end{aligned}$$

The laminate modulus is 1/7th the modulus of aluminum or 10X higher modulus than polyethylene.

The aluminum layer will yield at 0.24% strain. Target laminate tension should be 10% of yield point or
 $(0.024\%)(10 \text{ GPa}) = 2.4 \text{ MPa} = 208 \text{ N/m} / (87 \mu\text{m})$